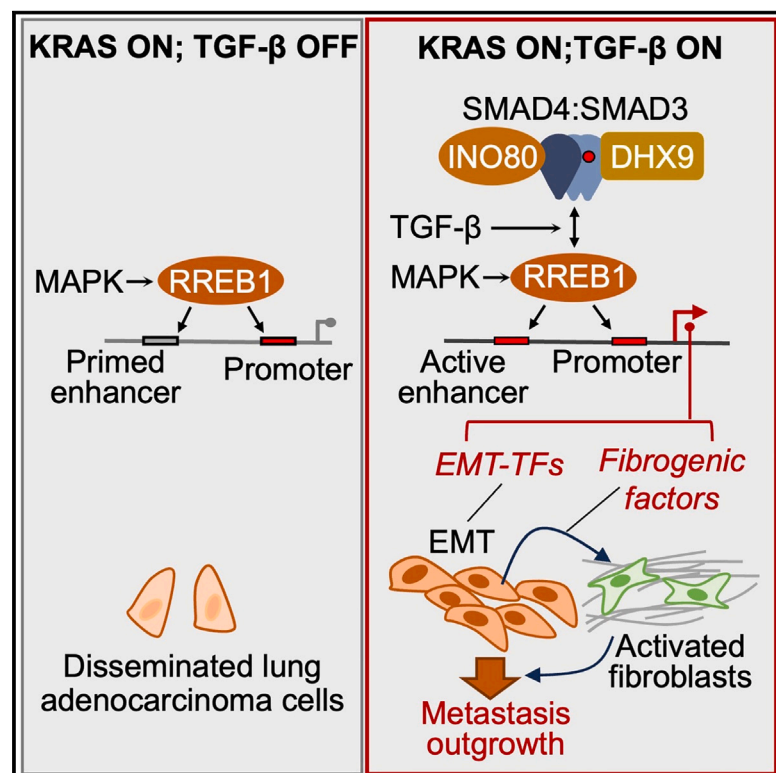


TGF- β and RAS jointly unmask primed enhancers to drive metastasis

Graphical abstract



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In brief

During carcinoma metastasis, malignant progenitors undergo a TGF- β -dependent EMT associated with fibroblast activation and extracellular matrix remodeling in the tumor microenvironment. RAS-activated RREB1 primes enhancers of EMT and fibrogenic genes in lung adenocarcinoma cells for activation by chromatin remodeling complexes that the TGF- β /SMAD pathway recruits to these enhancers. Inhibiting RREB1 disables this pro-metastatic process.

Highlights

- TGF- β and RAS drive expression of EMT and fibrogenic genes that jointly fuel metastasis
- Epigenetic determinants segregate different programs within a global TGF- β response
- RAS/MAPK-regulated RREB1 primes enhancers for activation by SMAD-recruited INO80
- Targeting RREB1 selectively inhibits LUAD metastasis



Article

TGF- β and RAS jointly unmask primed enhancers to drive metastasis

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SUMMARY

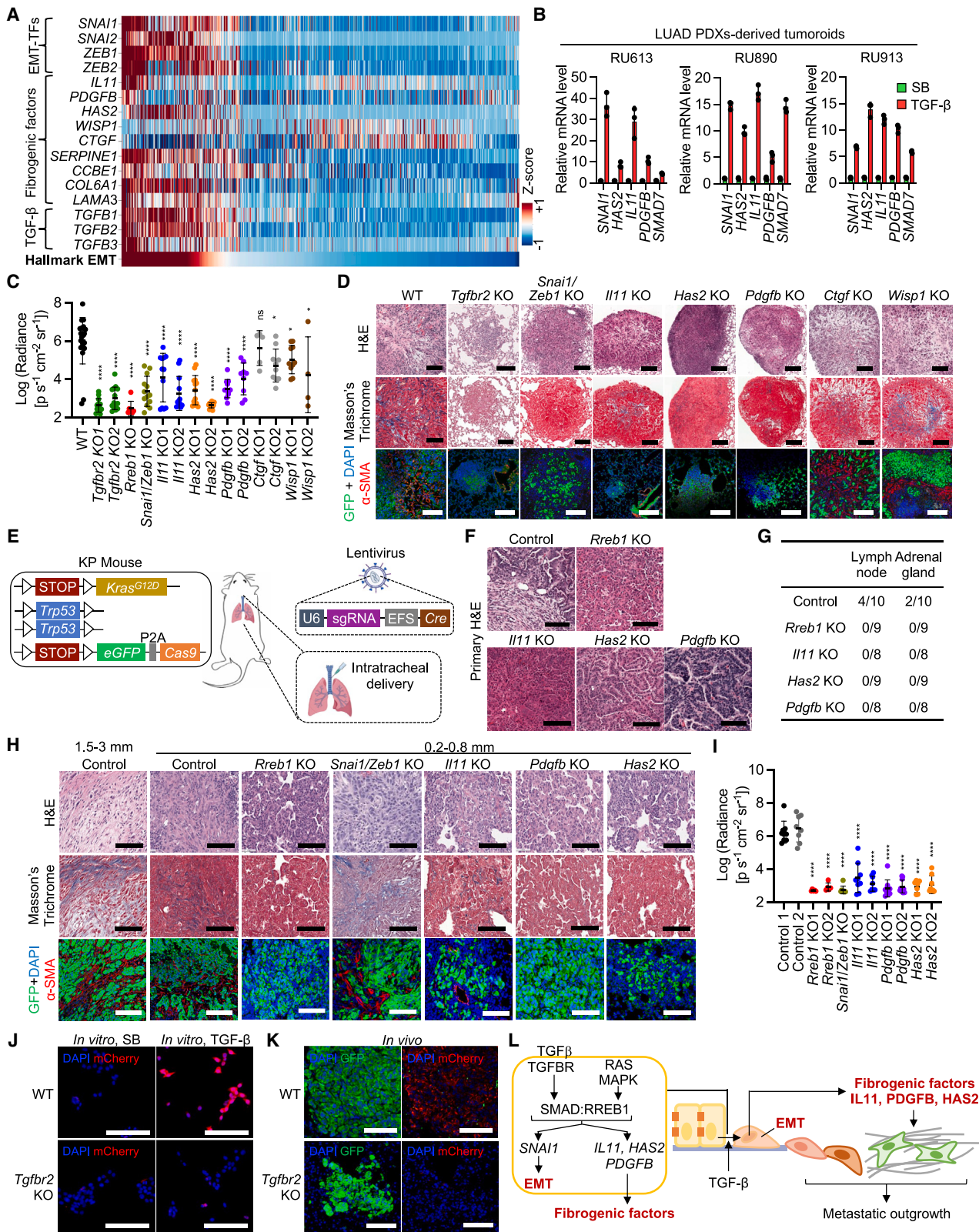
Epithelial-to-mesenchymal transitions (EMTs) and extracellular matrix (ECM) remodeling are distinct yet important processes during carcinoma invasion and metastasis. Transforming growth factor β (TGF- β) and RAS, signaling through SMAD and RAS-responsive element-binding protein 1 (RREB1), jointly trigger expression of EMT and fibrogenic factors as two discrete arms of a common transcriptional response in carcinoma cells. Here, we demonstrate that both arms come together to form a program for lung adenocarcinoma metastasis and identify chromatin determinants tying the expression of the constituent genes to TGF- β and RAS inputs. RREB1 localizes to H4K16acK20ac marks in histone H2A.Z-loaded nucleosomes at enhancers in the fibrogenic genes interleukin-11 (*IL11*), platelet-derived growth factor-B (*PDGFB*), and hyaluronan synthase 2 (*HAS2*), as well as the EMT transcription factor *SNAI1*, priming these enhancers for activation by a SMAD4-INO80 nucleosome remodeling complex in response to TGF- β . These regulatory properties segregate the fibrogenic EMT program from RAS-independent TGF- β gene responses and illuminate the operation and vulnerabilities of a bifunctional program that promotes metastatic outgrowth.

INTRODUCTION

Epithelial-to-mesenchymal transitions (EMTs) are phenotypic plasticity processes that play critical roles during development and injury repair and reemerge during carcinoma invasion and metastasis.^{1–3} Metastasis specifically depends on the ability of disseminated cancer cells to invade host tissue, a function that EMTs support. In an EMT, epithelial cells lose apicobasal polarity, gain motility, and remodel cell adhesion contacts to invade the surrounding stroma. EMTs are driven by specialized transcription factors (EMT-TFs) that repress epithelial genes and induce the expression of mesenchymal traits. EMTs are

frequently accompanied by other key processes. For example, during gastrulation, epiblast progenitors undergoing an EMT also undergo differentiation into mesoderm and endoderm.⁴ During carcinoma metastasis, malignant progenitors undergoing an EMT are frequently associated with activated fibroblasts and extracellular matrix (ECM) remodeling.^{1,5} Understanding how these processes are interconnected will shed light on the role of EMTs in normal development and diseases like cancer.

The pleiotropic cytokine transforming growth factor β (TGF- β) potentially induces EMTs during development, wound healing, fibrosis, and cancer.^{6,7} TGF- β requires synergistic inputs from the RAS pathway to trigger EMTs.^{8–13}, a synergy hinging on a



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specialized effector of RAS/mitogen-activated protein kinase (MAPK) signaling, the RAS response element-binding protein 1 (RREB1).⁸ MAPK-activated RREB1 interacts with TGF- β -activated SMAD TFs to induce the expression of EMT-TFs in embryonic and adult epithelial progenitors and in carcinoma cells.^{8,14} The SMAD-RREB1 synergy additionally couples the induction of EMTs with the expression of mesendoderm identity factors in epiblast progenitors and of fibrogenic factors in carcinoma cells, thus coordinating EMTs with context-dependent expression of other genes.⁸

Why a specific subset of TGF- β transcriptional responses comprising EMT-TFs and fibrogenic genes require RREB1 inputs remains unknown. Also unknown is whether the EMT and fibrogenic arms of the TGF- β /RAS response coalesce into a functional program required for metastatic outgrowth in carcinoma cells. Here, we address these questions in the context of lung adenocarcinoma (LUAD), a leading cause of cancer mortality globally.^{15,16} We demonstrate that both the EMT and the fibrogenic arms of the SMAD and RREB1 transcriptional programs are necessary for the growth of pulmonary metastases. We elucidate specific chromatin features required for RREB1-mediated activation of EMT-TFs and associated fibrogenic genes. Leveraging these insights, we provide proof of concept that RREB1 competitive inhibitors can be deployed to blunt the TGF- β fibrogenic EMT program and suppress LUAD metastasis.

RESULTS

Fibrogenic and EMT hallmarks in human LUAD metastases

Activating *KRAS* mutations define a major genetic subtype of human LUAD. *RAS*-mutant LUAD cell lines respond to TGF- β through RREB1-dependent activation of a stereotypical gene set encoding the EMT-TFs *SNAIL* (*SNAI1*), *SLUG* (*SNAI2*), zinc-finger E-box-binding 1 (*ZEB1*), and *ZEB2*;¹ the fibrogenic cyto-

kines interleukin-11 (*IL11*)^{17,18} and platelet-derived growth factor-B (*PDGFB*);^{19,20} the ECM modifying enzyme hyaluronan synthase 2 (*HAS2*),^{21–23} the ECM-associated proteins connective tissue growth factor (*CTGF*),²⁴ and WNT1-inducible-signaling pathway protein 1 (*WISP1*);²⁵ and other fibrogenic factors.⁸ To investigate the expression of this gene set in clinical samples, we analyzed our previously generated single-cell RNA sequencing (scRNA-seq) dataset from patient-derived LUAD metastasis cancer cells.^{26,27} We ranked individual cells based on expression of a hallmark EMT signature²⁸ and queried the expression of fibrogenic and EMT-TF genes (Figure 1A). All fibrogenic genes examined, except *CTGF* and *WISP1*, were concordantly expressed with EMT-TFs in cancer cells expressing the EMT signature. Cells expressing the fibrogenic EMT program also showed peak expression of the three TGF- β forms, *TGFB1*, *TGFB2*, and *TGFB3* (Figure 1A).

We generated tumor sphere cultures (“tumoroids” for short) from patient-derived xenografts (Figure S1A) and incubated these with TGF- β 1 (henceforth TGF- β) or the TGF- β receptor inhibitor SB505124 (SB)²⁹ to eliminate endogenous TGF- β signaling (Figures S1B and S1C). Incubation with TGF- β triggered a rapid (2 h) induction of representative EMT-TFs and fibrogenic factors in three independent tumoroids (Figure 1B). Thus, expression of a TGF- β -dependent fibrogenic EMT program is a clinically relevant feature of human LUAD metastases.

EMT and fibrogenic mediators of LUAD metastasis

RREB1 is essential for the metastatic growth of 393T3 cells,⁸ a LUAD cell line derived from the *Kras*^{G12D}; *p53*^{-/-} (KP) genetically engineered mouse model (GEMM).³⁰ We knocked out fibrogenic mediators and EMT-TFs in 393T3 cells (Figures S1D–S1F). Wild-type (WT) 393T3 cultures incubated with TGF- β showed canonical EMT traits, including downregulation of the epithelial marker E-cadherin, dispersion of cell colonies, and induction of *Snai1* (Figures S1G and S1H). Knockout of the TGF- β receptor *Tgfr2*

Figure 1. The EMT and fibrogenic branches of the TGF- β response are required for LUAD metastasis

(A) Heat map representation of the expression of hallmark EMT gene signature, TGF- β -response signature, EMT-TFs, fibrogenic factors, and TGF- β forms across patient-derived metastatic LUAD cells.^{26,27} Individual cells (columns) are ranked left to right by hallmark EMT score. For each gene, imputed expression was z-normalized across all cells. *n* = 991 cells isolated from brain, vertebral, and adrenal metastases from 5 patients.

(B) LUAD PDX-derived tumoroids grown in Matrigel for 2 days were treated with SB505124 (SB, 2.5 μ M) or TGF- β 1 (TGF- β , 100 pM) for 2 h. Data are mean $\pm$ SD; *n* = 3.

(C) Lung colonization burden in mice 21 days after tail-vein inoculation of WT, *Tgfr2*-, *Rreb1*-, *Snai1/Zeb1*-, *Il11*-, *Pdgfb*-, *Has2*-, *Ctgf*-, or *Wisp1*-knockout (KO) 393T3 cells in syngeneic immunocompetent mice. Colonization was quantified by *ex vivo* firefly luciferase bioluminescence imaging (BLI). Data are mean $\pm$ SD; *n* = 5–20 mice per group; one-way analysis of variance (ANOVA). Relative to WT, ns: not significant, **p* < 0.05, *****p* < 0.0001.

(D) Representative hematoxylin and eosin (H&E) staining, Masson's trichrome staining, and α -smooth muscle actin (α -SMA) immunofluorescence (IF) of lung colonized tumors from (C). Scale bars, 100 μ m.

(E) Schematic representation of the generation of compound KP LUAD GEMMs.

(F) Representative H&E staining of lung tumors from these GEMMs. Scale bars, 100 μ m.

(G) Incidence of GFP⁺ cell colonies in mediastinal lymph nodes and adrenal glands in GEMMs.

(H) Representative H&E staining, Masson's trichrome staining, and α -SMA IF of pulmonary metastases 56 days after tail-vein inoculation of control, *Rreb1*-, *Snai1/Zeb1*-, *Il11*-, *Pdgfb*-, or *Has2*-KO KP LUAD tumoroids in immunocompetent mice. The range of tumor diameters is indicated at the top. Scale bars, 100 μ m.

(I) Lung colonization load from Figure 1H was quantified by firefly luciferase bioluminescence imaging (BLI). Data are mean $\pm$ SD; *n* = 8–10 mice per group; one-way ANOVA; *****p* < 0.0001.

(J) Representative GFP IF, mCherry IF, and DAPI staining of WT and *Tgfr2*-KO 393T3 cells transduced with an inducible SMAD reporter construct and incubated with SB or TGF- β for 24 h. Scale bars, 100 μ m.

(K) Representative GFP IF, mCherry IF, and DAPI staining of lung metastatic nodules in doxycycline diet-fed mice that were inoculated with 393T3 cells harboring an inducible SMAD reporter construct. Scale bars, 100 μ m.

(L) Graphical summary of the fibrogenic and EMT branches of the RAS-dependent TGF- β response in LUAD cells. Both arms and each of their constituent genes are required for the outgrowth of pulmonary metastases.

See also Figures S1 and S2.

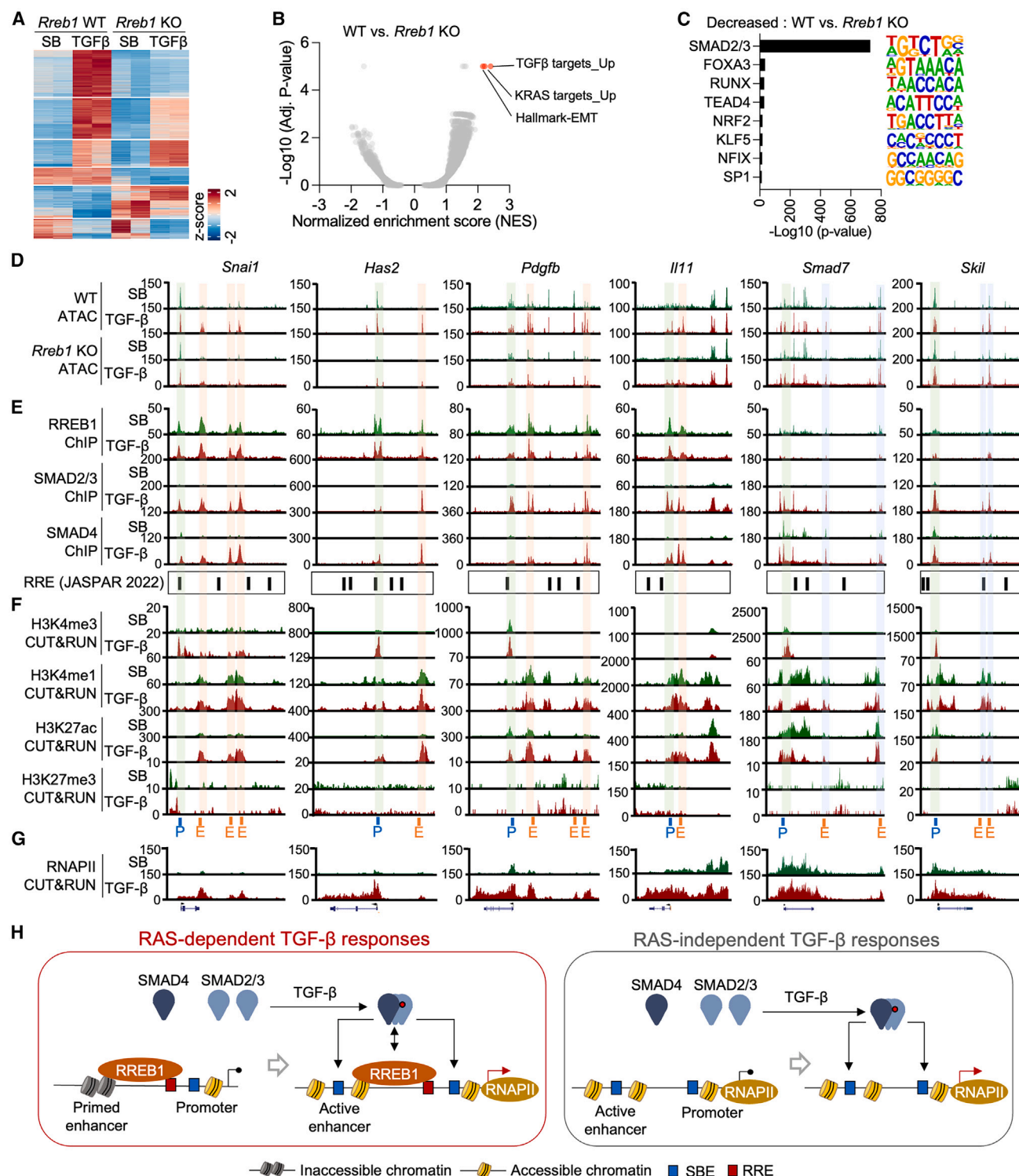


Figure 2. RREB1 primes enhancers for activation by TGF- β

(A) Heat map representation of ATAC-seq peaks in WT and *Rreb1*-KO 393T3 cells treated with SB or TGF- β for 1.5 h. ATAC-seq peaks were identified by DESeq2 analyses ($\log_2$ -transformed fold change ≥ 1 ; false discovery rate [FDR] ≤ 0.05).

(B) Volcano plot of gene signatures differentially represented in TGF- β -dependent ATAC-seq peaks from WT versus *Rreb1*-KO 393T3 cells.

(C) Transcription factor-binding motifs differentially enriched in TGF- β -dependent ATAC-seq peaks from WT versus *Rreb1*-KO 393T3 cells, as identified by DESeq2 analyses.

(legend continued on next page)

or a combined knockout⁸ of *Snai1* and *Zeb1* prevented the induction of EMT by TGF- β , whereas knockout of *Il11*, *Pdgfb*, *Has2*, *Wisp1*, or *Ctgf* did not (Figure S1G). Knockout of *Snai1/Zeb1*, *Il11*, *Pdgfb*, *Has2*, *Wisp1*, or *Ctgf* did not prevent the induction of the other genes by TGF- β , except for an effect of *Has2* knockout on *Wisp1* (Figure S1H). Notably, 393T3 cells lacking *Rreb1*, *Il11*, *Pdgfb*, *Has2*, *Tgfb2*, or both *Snai1* and *Zeb1* were impaired in pulmonary metastasis formation upon intravenous inoculation into syngeneic immunocompetent mice (Figures 1C and S1I). Masson's trichrome staining of stromal collagen and α -smooth muscle actin immunofluorescence (α -SMA IF, a marker of myofibroblasts) demonstrated intra-tumoral fibrosis in the metastatic colonies formed by WT 393T3 cells but not in size-matched small lesions formed by knockout cells (Figures 1D and S1J). Knockouts of *Ctgf* and *Wisp1* had limited effects on intra-tumoral fibrosis or metastasis (Figures 1C, 1D, and S1J).

To assess the functional role of these genes in primary tumor formation, we initiated lung tumors in *Kras*^{LSL-G12D}; *Trp53*^{flx/flx}; *Rosa26*^{LSL-Cas9-EGFP/+} KP mice³¹ by intratracheal delivery of a vector encoding Cre recombinase and sgRNAs against *Rreb1*, *Il11*, *Pdgfb*, or *Has2* (Figure 1E). These compound KP LUAD mice (Figures S2A–S2C) did not significantly differ from control KP mice in primary tumor incidence or burden, except for KP-*Rreb1* knockout tumors, which developed more slowly (Figures S2D and S2E). The histology of *Rreb1*, *Il11*, *Pdgfb*, and *Has2* knockout tumors was more acinar and papillary than that of control tumors (Figures 1F and S2F). Autochthonous metastases were observed in mediastinal lymph nodes and/or adrenal glands in 4 out of 10 tumor-bearing KP mice, while no metastases were detected in 34 mice bearing tumors deficient in *Rreb1*, *Il11*, *Pdgfb*, or *Has2* ($p = 0.0015$, Fisher's exact test) (Figures 1G and S2H).

Next, we generated tumoroids from this panel of compound KP lung tumors (Figures S2G, S2I, and S2J). Using a TGF- β /SMAD-dependent mCherry reporter²⁷ (Figure S2K), we validated the need for SB to block endogenous TGF- β signaling and set a baseline in these tumoroids (Figure S2L). On incubation with TGF- β , control tumoroids and tumoroids deficient in *Il11*, *Pdgfb*, or *Has2* showed E-cadherin downregulation, dispersion, and *Snai1* expression (Figures S2M and S2N). Tumoroids deficient in *Rreb1* or both *Snai1* and *Zeb1* (Figure S2G) failed to undergo EMT with TGF- β (Figures S2M and S2N). Deletion of *Snai1/Zeb1*, *Il11*, *Pdgfb*, or *Has2* did not prevent TGF- β induction of the other genes in this set (Figure S2N). KP tumoroid cells formed pulmonary metastases in immunocompetent mice, whereas tumoroids lacking *Rreb1*, *Il11*, *Pdgfb*, *Has2*, or a combination of *Snai1* and *Zeb1* showed limited metastatic activity (Figures 1H, 1I, and S2O).

Intra-tumoral collagen deposition and SMA+ cells were abundant in pulmonary metastases formed by KP tumoroids but

largely absent in size-matched small colonies formed by the knockout tumoroids (Figures 1H and S2P). Collagen-rich areas in 393T3 or KP tumoroid metastases contained high levels of *Il11*, *Pdgfb*, and *Has2* transcripts compared with poorly fibrotic areas, as determined by RNA-fluorescence *in situ* hybridization (FISH) (Figures S2Q and S2R). TGF- β signaling activity was present in 393T3 lung metastatic colonies, as confirmed using the TGF- β /SMAD reporter (Figure 1J). Pulmonary metastases formed by these cells showed high mCherry expression compared with colonies formed by *Tgfb2*-knockout counterparts (Figure 1K). Collectively, these results indicated that the EMT and fibrogenic arms of the TGF- β response in LUAD are essential components of a program driving LUAD metastasis (Figure 1L).

TGF- β - and RREB1-dependent chromatin accessibility

Snai1, *Has2*, *Il11*, and *Pdgfb* differ from other TGF- β target genes in that their induction by TGF- β -activated SMADs requires RREB1. To determine the basis for this requirement, we measured global chromatin accessibility in WT and *Rreb1*-knockout 393T3 cells with assay for transposase-accessible chromatin using sequencing (ATAC-seq). Exposure of WT cells to TGF- β led to widespread gains in chromatin accessibility across the genome (Figure 2A). Nearly half of these gains were absent in *Rreb1*-knockout cells. Most RREB1-dependent gains occurred near genes related to EMT, TGF- β signaling, and RAS signaling (Figure 2B). SMAD-binding elements were the most significantly enriched TF motifs in these regions (Figure 2C). In ATAC-seq tracks over *Snai1*, *Has2*, *Pdgfb*, and *Il11* loci, all RREB1- and TGF- β -dependent accessibility peaks were located distant to the transcription start site (TSS) (Figure 2D). By contrast, two representative RAS-independent TGF- β target genes, *Smad7* and *Skil*,⁸ lacked RREB1- and TGF- β -dependent chromatin accessibility peaks (Figure 2D). In sum, RREB1 and SMAD TFs cooperatively increase chromatin accessibility at specific regions in fibrogenic EMT loci.

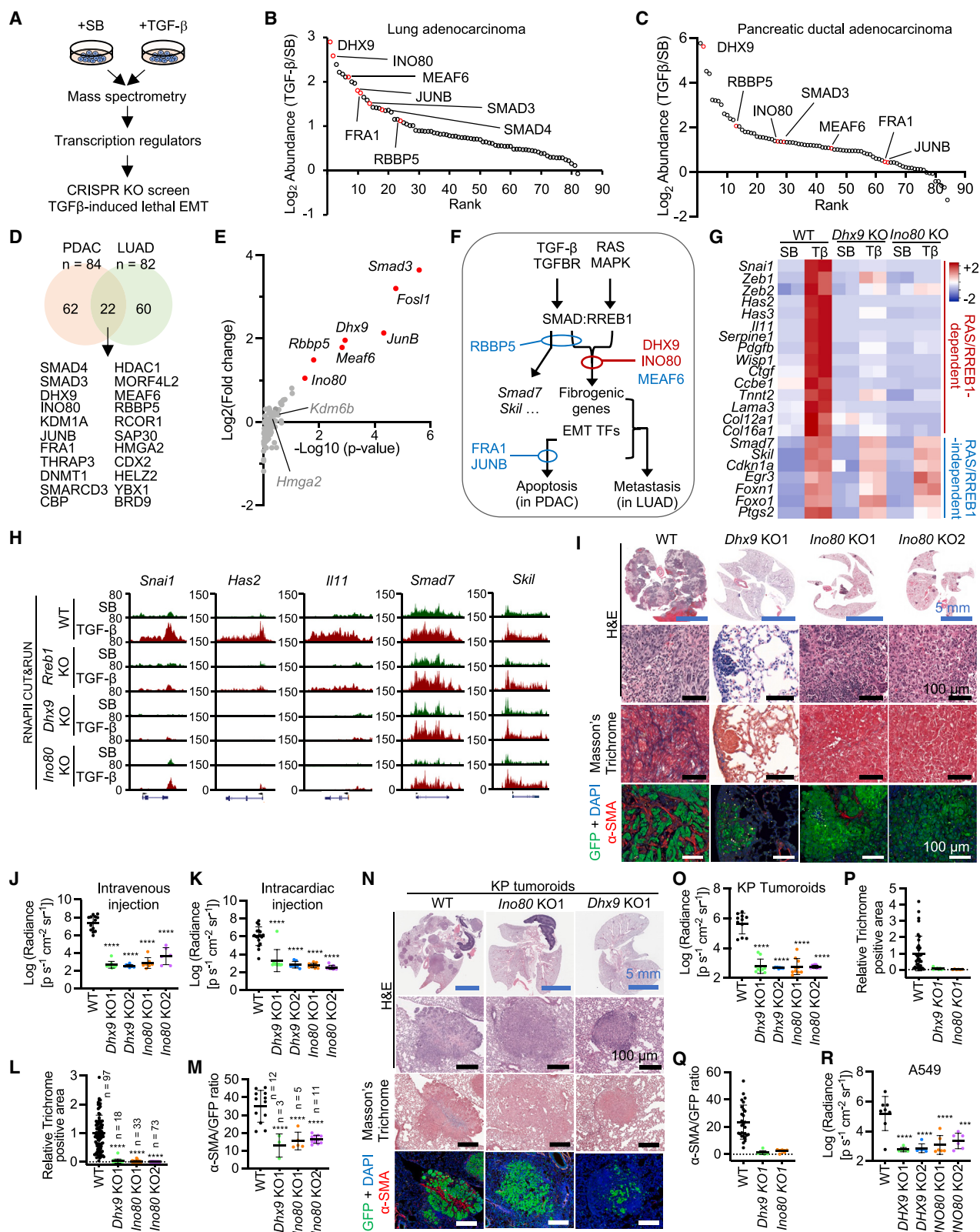
RREB1 primes enhancers for activation by TGF- β

Using chromatin immunoprecipitation and sequencing (ChIP-seq), we determined that RREB1 was already bound throughout the genome prior to TGF- β stimulation in 393T3 cells (Figures S3A and S3B). In the TGF- β pathway, SMAD2 and SMAD3 are phosphorylated by the TGF- β receptor to form heterotrimeric complexes with SMAD4, which then bind to target enhancers and promoters (refer to Figure 2H).^{6,32} SMAD2 and SMAD3 are known to recruit histone acetyltransferases p300 and CBP^{33,34}; however, the specific function of SMAD4 in transcription remains unknown.⁶ Approximately 75% of the sites and 92% of the annotated genes occupied at baseline by RREB1 were subsequently occupied by SMAD2/3 and SMAD4 in response to TGF- β genome-wide (Figures S3A–S3C) and at

(D–G) Tracks of ATAC-seq (D), HA-RREB1, SMAD2/3, and SMAD4 ChIP-seq (E), H3K4me3, H3K4me1, H3K27ac, and H3K27me3 CUT&RUN (F), and RNA polymerase II (RNAPII) CUT&RUN (G) in the indicated gene loci in 393T3 cells expressing HA-RREB1 incubated with SB- or TGF- β -treated for 1.5 h. *Rreb1*-KO cells were included in (D). The location of RAS response elements (RRE) was determined based on RREB1 JASPAR motifs. P, promoter; E, enhancer.

(H) Schematic representation of distinct enhancer and promoter states in RAS-dependent versus RAS-independent TGF- β target genes. SBEs, SMAD-binding elements.

See also Figure S3.



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fibrogenic and EMT-TF gene loci (Figure 2E). The RREB1-bound regions fell into two classes. One class included regions of open chromatin located near a TSS; these regions were proximal to a consensus RAS response element (RRE), the DNA sequence motif bound by RREB1 (Figures 2E and S3D). The other class included regions where the chromatin became accessible in response to TGF- β . These regions lacked recognizable RRE motifs, suggesting that RREB1 contacts the chromatin in these regions to prime it for the TGF- β response. No RREB1 binding was detected in *Smad7* or *Skil* (Figure 2E).

We profiled histone 3 (H3) modifications using cleavage under targets and release using nuclease (CUT&RUN) assays. H3K4me3, a marker of active promoters,^{35,36} was enriched near the TSS of *Snai1*, *Has2*, *Pdgfb*, *Smad7*, and *Skil* in response to TGF- β (Figure 2F). By contrast, the RRE-free RREB1-binding regions of *Snai1*, *Has2*, *Il11*, and *Pdgfb* contained H3K4me1 under basal conditions and gained H3K27ac in response to TGF- β (Figure 2F), marking these regions as RREB1-primed enhancers that become activated in response to TGF- β . The *Smad7* and *Skil* loci contained active enhancers (Figure 2F). This aligns with a previous stratification of TGF- β -responsive enhancers into primed (H3K4me1^{high};H3K27ac^{low};H3K27me3^{low}) and active enhancers (H3K4me1^{high};H3K27ac^{high};H3K27me3^{low}).³⁷ Knockout of *Rreb1* inhibited the TGF- β -dependent accumulation of H3K27ac and SMADs in primed enhancers (Figures S3E–S3G) but did not prevent SMAD2/3 binding to the promoter region (Figure S3F). RNA polymerase II (RNAPII) CUT&RUN assays showed that TGF- β stimulated the loading of RNAPII on *Snai1*, *Has2*, *Il11*, and *Pdgfb*, whereas RNAPII was preloaded on *Smad7* and *Skil* (Figure 2G). These results suggested that RREB1-mediated priming of fibrogenic and EMT gene enhancers is required for transcriptional activation of these genes by TGF- β (Figure 2H).

Identification of essential RREB1 cofactors

We integrated proteomic and CRISPR knockout screens to identify RREB1 cofactors mediating the activation of fibrogenic EMT (Figure 3A). RREB1 exists in multiple splice variants containing up to 15 C2H2-type zinc-finger (ZF) domains.⁸ A murine RREB1 isoform including the first 11 ZF (RREB1 residues 1–1,291; RREB1[ZF1-11]) restores RAS-dependent TGF- β gene responses in *Rreb1*-knockout cells.⁸ We thus transduced 393T3 LUAD cells with a HA epitope-tagged RREB1(ZF1-11) vector followed by cell incubation with SB or TGF- β . Mass spectrometry (MS) analysis of immunoprecipitated RREB1(ZF1-11) complexes identified 82 RREB1-associated proteins, 28 of which were enriched >2-fold in samples from TGF- β -treated cells compared with SB-treated cells (Figure 3B; Table S1).

Another context where coordinated induction of this set of EMT-TFs and fibrogenic genes by the TGF- β /SMAD and RAS/RREB1 pathways plays an active role is in pancreatic ductal adenocarcinoma (PDAC).¹⁰ A crucial difference between PDAC and LUAD cells is that in PDAC, this response leads to apoptosis due to the disruption of a vital transcriptional network by SNAIL,^{8,10} whereas EMT is not coupled to apoptosis in LUAD cells.^{6,14} The coupling of TGF- β -dependent EMT to apoptosis in PDAC cells provided a robust functional “up assay” to screen for mediators of this response.^{8,10} PDAC cells derived from genetically engineered *Pdx1-Cre*; *Kras*^{LSL-G12D}; *Cdkn2a*^{fllox/fllox}; *Smad4*^{fllox/fllox} mice, restoration of SMAD4 expression¹⁰ leads to TGF- β - and RAS/RREB1-mediated induction of a fibrogenic EMT followed by apoptosis.^{8,10} After transduction of HA-RREB1(ZF1-11), incubation with SB or TGF- β , and MS analysis of immunoprecipitated HA-RREB1(ZF1-11) complexes, 84 RREB1-associated proteins were identified, of which 47 were enriched >2-fold in samples from TGF- β -treated cells, and 22 overlapped with the

Figure 3. Identification of INO80 and DHX9 as essential RREB1 cofactors

(A) Schematic representation of RREB1-targeted proteomics coupled with a focused CRISPR knockout screen to identify essential mediators of TGF- β -dependent EMTs.

(B and C) 82 (B) and 84 (C) transcription-related proteins identified in HA-RREB1 immune complexes (refer to Table S1) in 393T3 LUAD cells (B) and SMAD4-restored PDAC cells (C) were plotted according to the effect of TGF- β on their abundance in these complexes. RREB1-associated proteins scoring in the CRISPR knockout screen (refer to E) are indicated.

(D) Venn diagram showing the overlap of HA-RREB1-interacting proteins from (B) and (C).

(E) sgRNAs enriched in SMAD4-restored mouse PDAC cells that survived a TGF- β -induced EMT. Genes corresponding to the top enriched sgRNAs are indicated.

(F) Schematic summary of the TGF- β signaling cascade showing the steps that require the indicated factors as determined by loss-of-function analyses. Red, factors specifically required for RAS-dependent induction of fibrogenic and EMT-TF genes in PDAC and LUAD cells; blue, all other factors tested.

(G) Heat map representation of RAS-dependent and -independent TGF- β -responsive genes from RNA-seq analysis of WT, *Dhx9*-, and *Ino80*-KO 393T3 cells after 1.5 h with TGF- β . *n* = 2.

(H) RNAPII CUT&RUN tracks in the *Snai1*, *Has2*, *Il11*, *Smad7*, and *Skil* loci in SB- or TGF- β -treated WT, *Rreb1*-, *Dhx9*-, and *Ino80*-KO 393T3 cells.

(I) Representative H&E staining, Masson's trichrome staining, and α -SMA and GFP IF of colonized lung tissues from mice after intravenous inoculation of WT, *Dhx9*-, or *Ino80*-KO 393T3 cells. *Dhx9*-KO2 393T3 cells generated no detectable metastatic colonies.

(J and K) Metastatic tumor burden as quantified by BLI 21 days after intravenous (J) or intracardiac inoculation (K) of WT, *Dhx9*-, or *Ino80*-KO 393T3 cells. Data are mean $\pm$ SD; *n* = 6–14; one-way ANOVA; *****p* < 0.0001.

(L and M) Quantification of trichrome staining (L) and α -SMA staining (M) in lung metastatic colonies. *n* for each group is indicated in graph. *Dhx9*-KO2 393T3 cells generated no detectable metastatic colonies.

(N) Representative H&E staining, Masson's trichrome staining, and α -SMA IF of lung tissues from mice 56 days after tail-vein inoculation of WT, *Dhx9*-, or *Ino80*-KO KP tumoroids in immunocompetent mice.

(O) Lung metastasis burden in mice from Figure 3N quantified by firefly luciferase BLI. Data are mean $\pm$ SD; *n* = 9–12 mice per group; one-way ANOVA; *****p* < 0.0001.

(P and Q) Quantification of trichrome staining (P) and α -SMA staining (Q) in lung metastatic colonies.

(R) Lung colonization burden in athymic mice 63 days after tail-vein inoculation of WT, *DHX9*-, or *INO80*-KO human LUAD A549 cells, as quantified by firefly luciferase BLI. Data are mean $\pm$ SD; *n* = 6–7 mice per group; one-way ANOVA; *****p* < 0.0001.

See also Figure S4 and Tables S1 and S2.

RREB1-associated proteins identified in 393T3 cells (Figures 3C and 3D; Table S1).

We designed and screened a focused CRISPR-Cas9 knockout library targeting the 84 RREB1-associated proteins in PDAC (6 single guide RNAs, sgRNAs, per target) (Table S2). As a positive control, we included sgRNAs targeting *Smad3*.¹⁰ We determined library representation by deep sequencing of sgRNAs amplified from genomic DNA harvested early (T0) and three population doublings after transduction (T3). To quantify the impact of gene targeting in this process, we assessed gene scores (GSs) by calculating the median log₂ fold change in sgRNA abundance between T3 and T0 (Table S2). *Smad3*-targeting sgRNAs were the top hits in the screen, followed by sgRNAs against *Fos1*, *JunB*, *Dhx9*, *Meaf6*, *Rbbp5*, and *Ino80* (Figure 3E). The role of each of these genes in TGF- β -induced apoptosis was confirmed by independent gene knockout experiments (Figure S4A).¹⁰ SMAD3, DHX9, INO80, and RBBP5 were among the proteins whose association with RREB1 was significantly enriched in TGF- β -treated cells (Figures 3B and 3C). Previously reported EMT regulators, such as HMGA2 and KDM6B,^{38,39} also emerged as RREB1-associated proteins but failed to score in our functional screen, in agreement with the weak effect of their knockout on TGF- β -dependent induction of *Snai1* (Figures S4B and S4C).

Individual knockout of *Smad3*, *Dhx9*, *Meaf6*, *Rbbp5*, or *Ino80* in the SMAD4-restored PDAC cells blunted the induction of *Snai1*, *Has2*, and *Il11* by TGF- β (Figure S4D). *JunB* and *Fos1* knockouts showed limited effects on the induction of fibrogenic EMT genes by TGF- β , suggesting that JUNB and FRA1 (Fos-related antigen 1, encoded by *Fos1*) function downstream of EMT (summarized in Figure 3F). We also excluded RBBP5 from further analysis because its knockout also inhibited the induction of *Smad7*, an RAS-independent TGF- β effect (Figure S4D). *Smad4*-proficient PDAC tumors arising in *Pdx1-Cre*; *Kras*^{G12D/+}; *Trp53*^{R172H/wt} mice, which develop with an intact SMAD4 and decouple TGF- β signaling from apoptosis,⁴⁰ showed a weak induction of fibrogenic and EMT genes by TGF- β (Figure S4E). Thus, we decided to focus on DHX9 and INO80 in LUAD cells.

Knockout of *Dhx9* and *Ino80* did not affect the proliferation of 393T3 cells (Figures S4F and S4H) but impaired the TGF- β -dependent expression of *Snai1*, *Has2*, and *Il11*, while *Meaf6* knockout showed a limited effect (Figure S4I). More broadly, RNA-seq analysis of *Dhx9* or *Ino80* knockout cells showed a specific loss of a common set of TGF- β gene responses (Figures S4J and S4K), including the entire RREB1-dependent TGF- β gene expression program (Figure 3G).⁸ In total, of 98 TGF- β -responsive genes (log₂ fold change > 4) in 393T3 cells, 41 required both DHX9 and INO80, 16 only required INO80, and 3 required only DHX9 (Figure S4L). CUT&RUN assays in 393T3 cells showed that the knockouts of *Rreb1*, *Dhx9*, or *Ino80* inhibited TGF- β -dependent recruitment of RNAPII to the *Snai1*, *Has2*, and *Il11* promoters (Figure 3H).

DHX9 and INO80 in fibrogenic EMT responses and LUAD metastasis

Dhx9 and *Ino80* knockouts inhibited lung metastases by 393T3 cells inoculated intravenously (Figures 3I and 3J) or intracardially

(Figure 3K). The small lesions formed by *Ino80*-knockout and *Dhx9*-knockout cells showed reduction in collagen deposition (Figure 3L) and SMA⁺ cells (Figure 3M), compared with size-matched control lesions, thus recapitulating the phenotype of LUAD cells that were deficient in *Rreb1* or fibrogenic EMT components. Knockout of *Dhx9* and *Ino80* in KP tumoroids (Figure S4M) did not affect tumoroid formation (Figures S4N and S4O) but impeded the induction of fibrogenic EMT gene responses to TGF- β (Figure S4P) and inhibited pulmonary metastasis in immunocompetent mice (Figures 3N–3Q). We also knocked out *TGFBR2*, *RREB1*, *DHX9*, and *INO80* in the KRAS-mutant human LUAD A549 cell line⁸ (Figure S4Q) and in tumoroids grown from patient-derived KRAS-mutant LUAD xenograft RU631 (Figure S4R). Knockout of *DHX9*, *INO80*, or *RREB1* abolished the TGF- β -mediated induction of fibrogenic EMT gene responses but not the induction of *SMAD7* expression, whereas knockout of *TGFBR2* abolished all these responses (Figure S4S). Knockout of *DHX9* or *INO80* in A549 cells inhibited the formation of pulmonary metastasis in athymic mice (Figures 3R and S4T). Thus, INO80 and DHX9 phenocopy RREB1 as mediators of TGF- β responses and metastasis.

SMAD interactions with DHX9 and INO80

TGF- β induced the binding of DHX9, INO80, and SMADs to regions with pre-bound RREB1 genome-wide (Figure 4A) and at primed enhancers in *Snai1*, *Has2*, *Il11*, and *Pdgfb* (Figures 2E and 4B), as determined by ChIP-seq analysis. *Rreb1* knockout diminished the TGF- β -induced interaction of DHX9 and INO80 with these regions (Figures 4B and S5A). The TGF- β -induced recruitment of DHX9 to *Snai1* and *Has2* enhancers required INO80, whereas recruitment of INO80 did not require DHX9 (Figure S5B). Proximity ligation assays and immunoprecipitation experiments showed an association of RREB1 with SMAD2/3, SMAD4, DHX9, and INO80 in the nucleus within 30 min of TGF- β addition (Figures 4C–4F).

We investigated whether SMAD proteins play a role in recruiting DHX9 and INO80 to RREB1-primed enhancers. In the absence of TGF- β treatment, DHX9 was in a complex with SMAD3, while INO80 was in a complex with immunoprecipitated SMAD4, and these interactions were not increased by incubation of the cells with TGF- β (Figure 4G). SMAD3 was required for the interaction of DHX9 with RREB1 (Figure 4H). SMAD4 was required for a TGF- β -induced interaction of INO80 with RREB1 (Figure 4I). SMAD3 and SMAD4 were required for the TGF- β -induced interaction of DHX9 and INO80, respectively, with RREB1-primed enhancers (Figures 4J and 4K). A basal level of interaction was observed between INO80 and DHX9 (Figure 4G) and between INO80 and RREB1 (Figure 4I).

SMAD TFs consist of an N-terminal (MH1) DNA-binding domain connected by a flexible linker to a C-terminal (MH2) domain.^{41,42} SMAD2 and SMAD3 share extensive sequence identity but differ in the allosteric regulation of their interaction with DNA.⁴³ Although SMAD2 interacted with SMAD4 in response to TGF- β in 393T3 cells, SMAD2 was largely excluded from RREB1 complexes with SMAD3, SMAD4, DHX9, and INO80 (Figures 4E–4H). Knockout of *Smad3* or knockdown of *Smad4* (Figure S5C) eliminated the fibrogenic and EMT-TF gene responses to TGF- β (Figures S5D and S5E), whereas

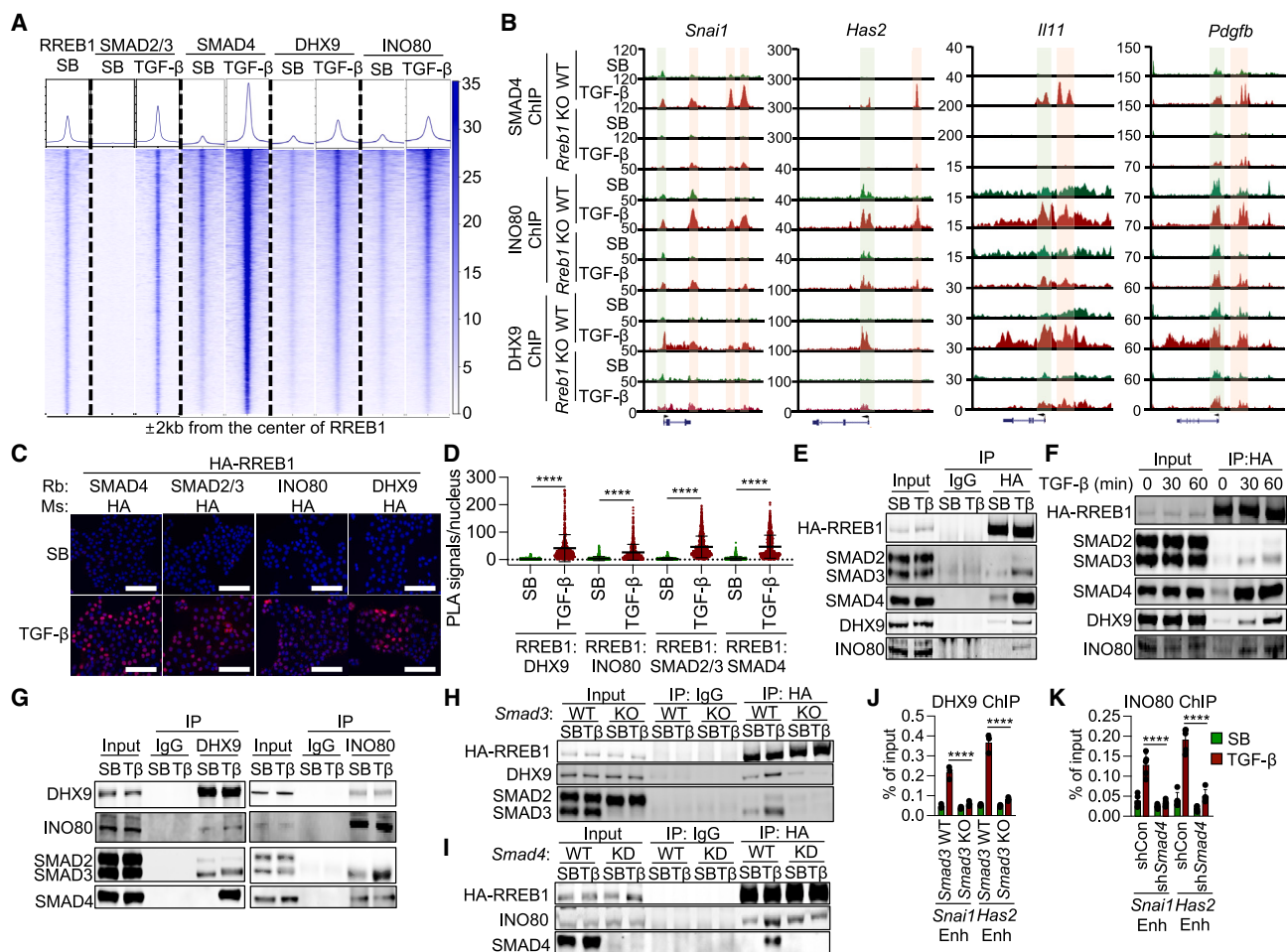


Figure 4. SMAD-mediated recruitment of INO80 and DHX9 to RREB1-primed enhancers

(A) Metaplots and tornado plots representing chromatin occupancy by SMAD2/3, SMAD4, DHX9, and INO80 in genomic regions ± 2 kb from the center of RREB1 ChIP-seq peaks in 393T3 cells treated with SB or TGF- β for 1.5 h.

(B) SMAD4, INO80, and DHX9 ChIP-seq tracks on *Snai1*, *Has2*, *Il11*, and *Pdgfb* loci in WT and *Rreb1*-KO 393T3 cells. Promoter (green) and primed enhancer regions (pink) are highlighted (refer to Figure 2D).

(C) Proximity ligation analysis showing TGF- β -dependent interactions of HA-RREB1 with DHX9, INO80, SMAD2/3, and SMAD4 in 393T3 cells. Scale bars, 100 μ m.

(D) Quantification of nuclear PLA signal in Figure 4C. One-way ANOVA; **** $p < 0.0001$.

(E–G) 393T3 cells expressing HA-RREB1 were incubated with SB or TGF- β (T β) for 1.5 h (E and G) or for the indicated time periods (F) and immunoprecipitated (IP) with the indicated antibodies. The immune complexes were subjected to western immunoblotting with antibodies indicated at left.

(H and I) WT and *Smad3*-KO (H) or *Smad4*-knockdown (KD) 393T3 cells (I) were incubated with SB or TGF- β for 1.5 h and immunoprecipitated (IP) with indicated antibodies. The immune complexes were subjected to western immunoblotting with antibodies indicated at left.

(J) ChIP-PCR analysis of DHX9 binding to enhancers of *Snai1* and *Has2* in WT or *Smad3*-KO 393T3 cells. Data are mean $\pm$ SD; $n = 3$; two-way ANOVA; **** $p < 0.0001$.

(K) ChIP-PCR analysis of INO80 binding to enhancers of *Snai1*, *Has2*, and *Smad7* in 393T3 cells expressing control or *Smad4*-targeting short-hairpin RNA (shRNA). Data are mean $\pm$ SD; $n = 3$; two-way ANOVA; **** $p < 0.0001$.

See also Figure S5.

knockdown of *Smad2* did not (Figures S5C and S5F). These results indicated that preexisting SMAD3-DHX9 and SMAD4-INO80 complexes converge on RREB1-primed enhancers in response to TGF- β .

SMAD3-DHX9 recruits CBP to primed fibrogenic EMT enhancers

DHX9, or RNA helicase A, comprises a helicase domain that unwinds three-stranded DNA:RNA R-loops,^{44,45} a dsRNA-

binding domain (RBD), a DNA-binding domain (RGG), and a minimal transactivation domain (MTAD).⁴⁶ Although resolution of R-loops is important for transcription,^{47,48} R-loop mapping using DNA-RNA immunoprecipitation and cDNA conversion coupled with high-throughput sequencing (DRIPc-seq)⁴⁹ detected DNA-RNA hybrids in *Snai1*, *Has2*, *Il11*, *Pdgfb*, *Smad7*, or *Skil* only after TGF- β addition (Figure S5G). Moreover, DHX9-(K417R) harboring a helicase-inactivating mutation⁵⁰ (Figure S5H) rescued the induction of *Snai1* and *Has2*

by TGF- β in *Dhx9*-knockout cells, as did WT DHX9 (Figure S5I).

Domain mapping experiments showed that the SMAD3 MH2 domain mediates the SMAD3-DHX9 interaction, and the DHX9 RBD is not required for this interaction (Figures S5J–S5L). However, an RBD deletion construct (DHX9- Δ RBD) failed to rescue the induction of *Snai1* and *Has2* by TGF- β in *Dhx9*-knockout 393T3 (Figure S5M). The RBD interacts with the coactivator acetyltransferase CREB-binding protein (CBP).⁵¹ Indeed, knockout of *Dhx9* inhibited a TGF- β -induced interaction between CBP and SMAD3 (Figure S5N). Moreover, TGF- β induced the recruitment of CBP to the RREB1-primed enhancers of *Snai1*, *Has2*, *Il11*, and *Pdgfb*, and this was inhibited in *Dhx9*-knockout cells (Figure S5O). The TGF- β -dependent gain of H3K27 acetylation in primed enhancers also required DHX9 (Figure S5P). By contrast, the recruitment of CBP to the *Smad7* enhancer was relatively weak and did not require DHX9 (Figure S5O), in line with the limited gain of H3K27ac observed in *Smad7* enhancers (refer to Figure 2F). These results suggest that SMAD3-bound DHX9 recruits CBP to activate RREB1-primed enhancers in response to TGF- β .

Essential role of INO80 recruitment by SMAD4

One structural difference between the SMAD4^{52,53} versus SMAD2 and SMAD3 is in a region connecting the linker and MH2 domains.^{54,55} This region contributes a parallel strand to the MH2 domain as well as a flexible loop (Figure 5A). There is also an extended α -helix followed by a long loop that packs against the C-terminal tail (Figure 5A). Although the existing X-ray crystal structures of heterotrimeric SMAD complexes reveal only the rigid cores of the MH2 domains and their interfaces,⁵⁶ additional structural elements have been observed in structures of SMAD4 MH2 monomers.^{52,53} Modeling the SMAD4 MH2 domain in a heterotrimeric complex with SMAD3 showed that these additional elements are not part of the SMAD trimer interface but remain accessible (Figure 5B). We generated “M1,” a mutant SMAD4 construct with the three turns of the extended α -helix and its loop (SMAD4 residues 454–493) replaced with the GFEAVY sequence (residues 359–364) of SMAD3 and also with the last 8 C-terminal residues removed (Figure 5A). We also generated the “M2” mutant by substituting 15 residues in the flexible loop (SMAD4 residues 294–310) with a triplicated GGGGS sequence (Figure 5A).

The WT, M1, and M2 proteins were expressed at similar levels in *Smad4* knockdown 393T3 cells. The M1 mutant completely lost the ability to form a complex with INO80 under basal conditions; by contrast, the M2 mutant retained this ability compared with WT SMAD4 (Figure 5C). These mutants retained the ability to associate with SMAD3 in response to TGF- β , although M1 was partly impaired in this activity (Figure 5D). These results suggested a role of the protruding structure formed by the extended α -helix/loop and C-terminal tail of SMAD4 in binding INO80. Re-expression of WT SMAD4, but not the M1 mutant, in *Smad4*-knockdown 393T3 cells restored the induction of fibrogenic EMT genes by TGF- β (Figure 5E) and metastatic competency (Figures 5F and 5G), demonstrating the essential role of INO80 recruitment by SMAD4 in the activation of pro-metastatic EMT-TF and fibrogenic genes by TGF- β .

INO80 clears H2A.Z nucleosomes at RREB1-primed enhancers

INO80 is the core ATPase subunit of an ATP-dependent chromatin remodeling complex with roles in DNA transcription, replication, and repair.⁵⁷ INO80 modifies the position and composition of nucleosomes at promoters and enhancers. The complex cradles the nucleosome through multiple DNA and histone contacts for nucleosome sliding and may additionally exchange histone H2A for the variant H2A.Z in target nucleosomes. H2A.Z has been shown to mediate repression.^{58–62}

Knockout of *Ino80* blunted the genome-wide TGF- β -induced gain in chromatin accessibility (Figure 6A), including at the promoters and RREB1-primed enhancers of fibrogenic and EMT genes in 393T3 cells (Figure 6B). Expression of INO80 rescued the induction of *Snai1* and *Has2* by TGF- β in *Ino80*-knockout cells, whereas expression of a catalytic domain deletion construct (INO80- Δ SNF) did not (Figures 6C, S6A, and S6B). H2A.Z ChIP-seq analysis showed peak H2A.Z accumulation at RREB1-bound sites genome-wide (Figures 6D and S6C). Cells incubated with TGF- β excluded H2A.Z from the center of the RREB1-binding peaks without affecting H2A (Figures 6D and S6D). H2A.Z was present in the RREB1-primed enhancers of *Snai1*, *Has2*, *Il11*, and *Pdgfb*, and incubation with TGF- β rapidly (<2 h) cleared H2A.Z-loaded nucleosomes from these enhancers (Figure 6E). This effect was dependent on RREB1 and INO80 (Figure 6F). We assessed nucleosome occupancy by means of micrococcal nuclease digestion with deep sequencing (MNase-seq), whereby decreased occupancy leads to increase accessibility to the nuclease in between neighboring nucleosomes. While nucleosome periodicity remained largely unchanged along gene bodies and intergenic regions, we observed clearing of nucleosomes at RREB1-primed fibrogenic and EMT gene enhancers upon cell stimulation with TGF- β (Figure 6E).

H2A.Z inhibits RREB1-primed enhancers

Next, we investigated the role of H2A.Z in enforcing the inhibited state of RREB1-primed enhancers. To avoid global effects of a permanent depletion of H2A.Z in cells, we transiently depleted H2A.Z using conditional CRISPR interference (CRISPRi) under doxycycline control in 393T3 cells (Figure S6E). Downregulation of *H2afz* (encoding H2A.Z) with two independent gRNAs rescued the induction of *Snai1*, *Has2*, *Il11*, and *Pdgfb* by TGF- β in *Ino80* knockout cells (Figure S6F). Downregulation of *H2afz* additionally rescued the lung metastatic competency of these *Ino80*-deficient cells (Figures 6G and 6H). By contrast, *H2afz* downregulation only marginally rescued these gene responses in *Rreb1*-knockout cells (Figure S6G), consistent with RREB1 fulfilling additional requirements such as recruitment of SMADs and DHX9.

We conducted MNase-qPCR assays to evaluate the effect of *H2afz* knockdown on the nucleosome profiles at the *Snai1* and *Has2* primed enhancers. TGF- β treatment induced nucleosome-free regions at these enhancers, an effect that was dampened by knockout of *Ino80* (Figure S6H). Downregulation of *H2afz* rescued the ability of TGF- β to induce chromatin accessibility and nucleosome mobilization at the enhancers in *Ino80*-knockout cells, as assessed through MNase-qPCR (Figure S6H). These results suggest that H2A.Z-containing nucleosomes

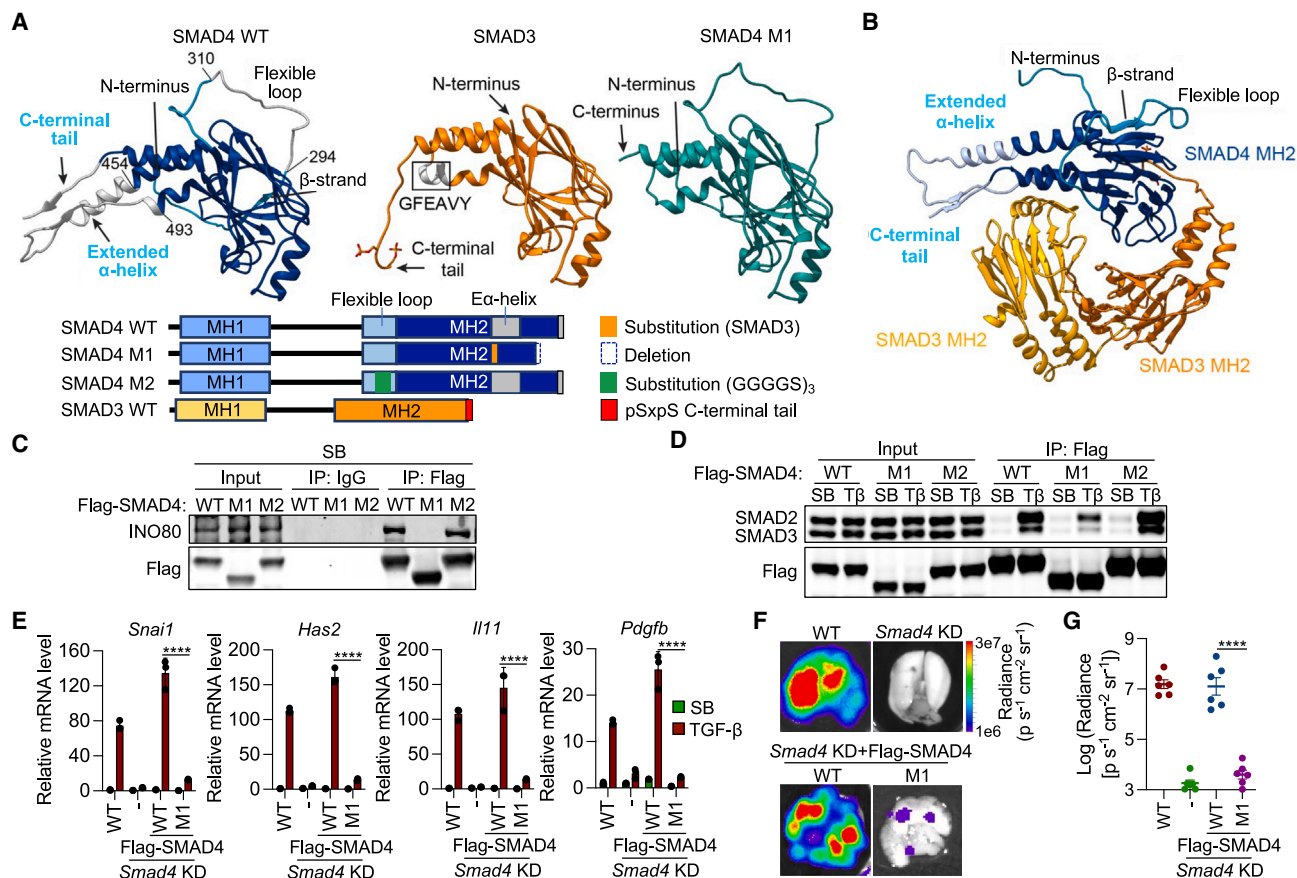


Figure 5. Essential role of SMAD4-mediated recruitment of INO80 in LUAD metastasis

(A) Comparison of WT and mutant MH2 structures of SMAD3 (PDB: 1U7F) and SMAD4 (PED00194 entries) with the differences highlighted. The SMAD4 M1 mutant model (mint green) was generated with ColabFoldv1.5. Structural elements of interest are indicated. Bottom, schematic representation of the full-length SMAD3 and SMAD4 domain structures, indicating the regions that were modified in the M1 and M2 mutant constructs.

(B) Heterotrimer including one SMAD4 MH2 domain (blue) and two SMAD3 MH2 domains (yellow and orange). The structurally distinct regions of SMAD4 are color coded and labeled. The SMAD4 MH2 domain contains the extended α -helix and C-terminal tail visible in solution (PED00194 entries) but absent in the X-ray data.

(C and D) *Smad4*-KD 393T3 cells expressing WT or M1, or M2 SMAD4 constructs were incubated with SB or TGF- β for 1.5 h as indicated, immunoprecipitated (IP), and immune complexes were subjected to western immunoblotting with antibodies indicated at left.

(E) *Snai1*, *Has2*, *Il11*, *Pdgfb*, and *Smad7* mRNA levels in WT and *Smad4*-KD 393T3 cells expressing empty vector (–), SMAD4 (WT), or SMAD4 M1 vectors, and treated with SB or TGF- β for 1.5 h. $n = 3$; two-way ANOVA; **** $p < 0.0001$.

(F) Representative *ex vivo* BLI images of lungs from mice 21 days after tail-vein inoculation of the cells described in (E). Mice were fed with doxycycline diet.

(G) Lung metastasis burden in mice from the same experiment, as quantified by *ex vivo* firefly luciferase BLI. Data are mean $\pm$ SD; $n = 5$ mice per group; one-way ANOVA; **** $p < 0.0001$.

suppress chromatin accessibility at fibrogenic and EMT-TF enhancers at baseline, and RREB1 primes these nucleosomes for clearing by the SMAD4-dependent recruitment of INO80.

Inhibition of RREB1 suppresses metastasis

Our findings raised the possibility that perturbing RREB1 function would blunt fibrogenic EMT gene responses and suppress LUAD metastasis. To define the functional RREB1 domains required for these processes, we then transduced 393T3 cells with vectors encoding different HA epitope-tagged RREB1 fragments (Figure 7A). RREB1(ZF1-11) containing the first 11 ZF domains bound to the primed RRE-free enhancers as well as to the RRE-containing promoters of *Snai1* and *Has2* in 393T3 cells (Fig-

ure S7A; refer to Figure 2E). RREB1 fragment composed of only ZFs 1 to 5 (RREB1[ZF1-5]) also bound to these enhancer and promoter regions, whereas another mutant containing ZFs 6 to 11 (RREB1[ZF6-11]) with an added nuclear localization signal did not bind to these regions (Figure S7A). Electrophoretic mobility shift assays (EMSAs) using oligonucleotide probes corresponding to the RRE-containing regions of *Snai1* and *Has2* demonstrated the ability of HA-RREB1(ZF1-5) to directly bind to these probes (Figure S7B). Altogether, these results suggested that RREB1 binds to RREs through the ZF1-5 region.

Given the interaction of RREB1(ZF1-11) and RREB1(ZF1-5) with RRE-free closed enhancers, we tested whether RREB1 has an intrinsic affinity for specific histone modifications.

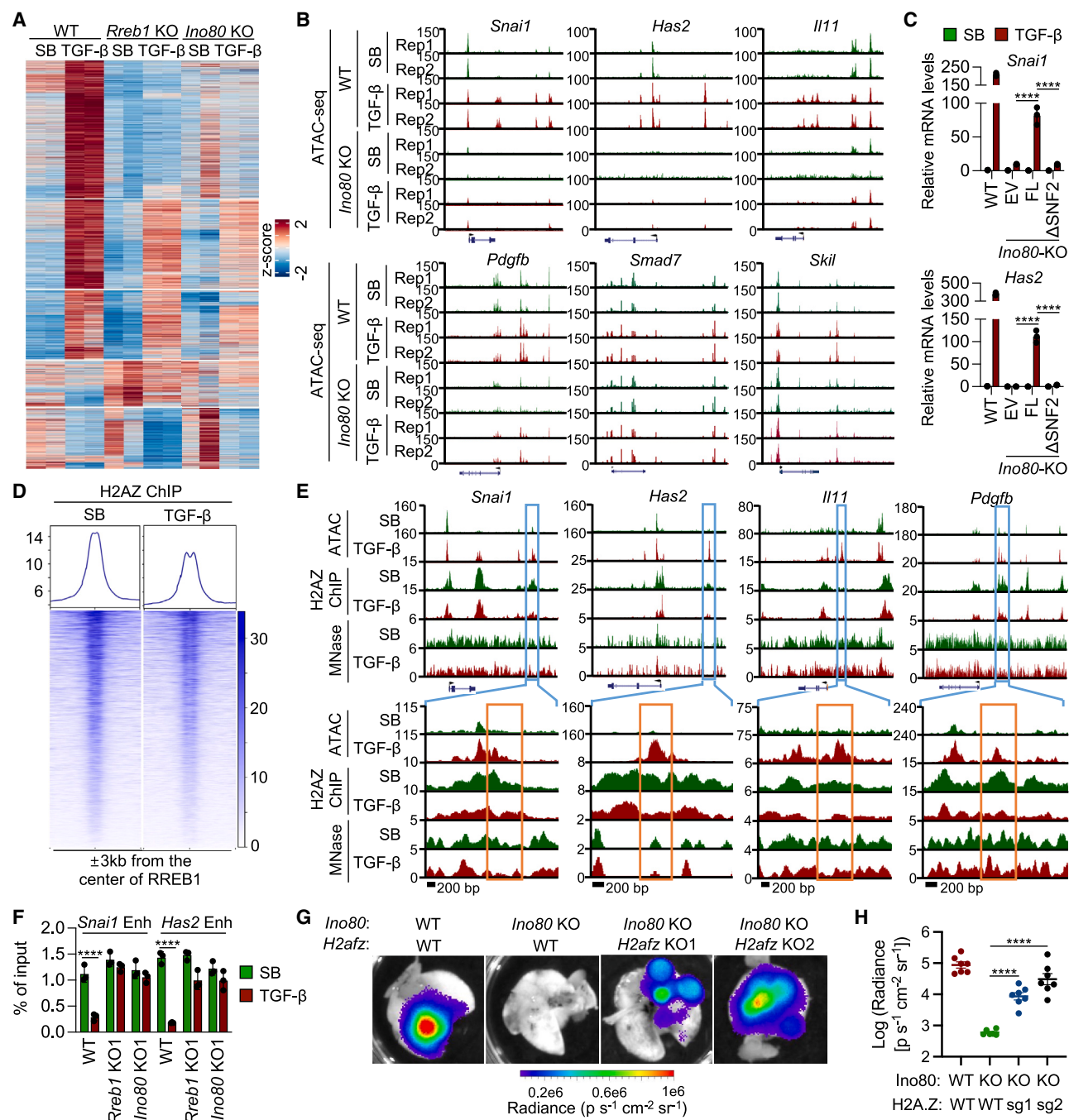


Figure 6. INO80 clears inhibitory H2A.Z nucleosomes from RREB1-primed enhancers

(A) Heat map representation of TGF- β -regulated ATAC-seq peaks. DESeq2 analyses ($\log_2$ -transformed fold change ≥ 1 ; false discovery rate [FDR] ≤ 0.05) in WT, *Rreb1* KO, and *Ino80* KO 393T3 cells treated with SB or TGF- β for 1.5 h.

(B) ATAC-seq tracks in loci of interest in WT and *Ino80*-KO 393T3 cells treated with SB or TGF- β .

(C) Effect of TGF- β on *Snai1* and *Has2* mRNA levels in WT cells and *Ino80*-KO 393T3 cells expressing the indicated INO80 mutants. Data are mean $\pm$ SD; $n = 3$; two-way ANOVA; **** $p < 0.0001$. EV, empty vector; FL, full-length INO80; Δ SNF, INO80- Δ RBD.

(D) Metaplots and tornado plots representing chromatin occupancy by histone H2A.Z in genomic regions ± 3 kb from the center of RREB1 ChIP-seq peaks in 393T3 cells treated with SB or TGF- β for 1.5 h.

(E) ATAC-seq, H2A.Z ChIP-seq, and MNase-seq tracks in *Snai1*, *Has2*, *Il11*, and *Pdgfb* loci of 393T3 cells treated with SB or TGF- β for 1.5 h.

(F) ChIP-PCR analysis of H2A.Z enrichment at the enhancers of *Snai1* and *Has2* in WT, *Rreb1* KO, and *Ino80* KO 393T3 cells. Data are mean $\pm$ SD; $n = 3$; two-way ANOVA; **** $p < 0.0001$. Enh, enhancer.

(legend continued on next page)

Recombinant RREB1(ZF1-5) showed a marked preference for H4 tail peptides containing K16ac or K20ac alone or combined (Figures 7B and S7C; Table S3). H4K16ac and H4K20ac ChIP-seq peaks matched the position of RREB1 at the enhancers and promoters of *Snai1*, *Il11*, *Pdgfb*, and *Has2* in 393T3 cells (Figure 7C), suggesting that the RREB1 interacts with H4K16acK20ac dual marks and H2A.Z-enriched nucleosomes. RREB1(ZF1-5) retained the ability to bind SMADs and INO80 in TGF- β -treated cells but lacked the ability to interact with DHX9 (Figure 7D).

Expression of RREB1(ZF1-5) in *Rreb1*-knockout 393T3 cells did not rescue the induction of fibrogenic and EMT-TF genes by TGF- β (Figure S7D) or the metastatic activity of these cells (Figure S7E), whereas RREB1(ZF1-11) restored these functions. In line with the inhibitory effect of RREB1(ZF1-5) on the binding of RREB1(ZF1-11) to dsDNA RRE-containing probes (Figure S7F), the expression of RREB1(ZF1-5) in *Rreb1*-knockout 393T3 cells blunted the induction of gene responses and associated metastatic activity (Figures S7D and S7E). Overexpression of RREB1(ZF1-5) in WT cells significantly inhibited the induction of fibrogenic EMT genes by TGF- β (Figure S7G). RREB1(ZF1-5) also inhibited the induction of fibrogenic and EMT gene responses in human LUAD A549 cells and RU631 patient-derived tumoroids (Figures S7H and S7I). Therefore, RREB1(ZF1-5) could be used as a competitive inhibitor of endogenous RREB1.

To enhance the potential of RREB1(ZF1-5) as a molecular tool to inhibit endogenous RREB1 in metastatic cells, we introduced a MAPK phosphomimetic mutation (S161D) to generate the RREB1(ZF1-5;S161D) variant (Figure 7E).⁸ Overexpression of this engineered variant in 393T3 cells was sufficient to impair lung metastasis and intra-tumoral fibrosis (Figures 7F–7H and S7J). Collectively, these establish an attractive proof-of-concept strategy for therapeutic disruption of RREB1 function to impede this malignant process and, potentially, other processes involving TGF- β -dependent EMT and fibrogenesis.

DISCUSSION

Two arms of the fibrogenic EMT program jointly promote LUAD metastasis

This work shows that the EMT and the fibrogenic arms of the TGF- β response in LUAD cells are essential for the outgrowth of pulmonary metastases and come together to form a program that drives this malignant process (Figure 7I). EMT-TFs and fibrogenic genes are expressed concordantly yet independently of each other in LUAD metastatic cells exposed to TGF- β . The expression of SNAIL and ZEB1 was necessary but not sufficient for metastatic outgrowth in LUAD mouse models and patient-derived tumoroids and required the concerted expression of IL11, PDGFB, and HAS2. These three fibrogenic factors were dispensable for EMT but essential for metastasis. Although the

EMT and fibrogenic arms do not depend on each other for expression, their constituent genes are jointly regulated by a common set of TGF- β and RAS inputs and functionally converge as essential parts of a program driving metastasis. Based on these properties, the fibrogenic EMT genes can be defined as a synexpression group.⁶³

Determinants of activation of the fibrogenic EMT program

We have elucidated the molecular basis for the requirement of RREB1 in the activation of fibrogenic and EMT genes by the TGF- β pathway in carcinoma cells. *Snai1*, *Il11*, *Pdgfb*, and *Has2* share a set of chromatin features that render their response to TGF- β dependent on RAS-activated RREB1. These features include an RREB1-binding RRE motif located in accessible chromatin near the promoter region and enhancer elements occupied by nucleosomes containing the histone variant H2A.Z and the H4K16acK20ac dual histone mark. RREB1 binds to the RRE near the promoter and contacts the H4K16acK20ac-containing nucleosomes, priming these enhancers for SMAD-mediated activation. Upon TGF- β stimulation, SMAD4-INO80 and SMAD3-DHX9 complexes bind to the RREB1-primed enhancers. The INO80 nucleosome remodeler mediates clearing of H2A.Z-containing nucleosomes at fibrogenic and EMT gene enhancers, whereas DHX9 recruits the histone acetylase CBP and RNAPII for activation of these genes (Figure 7I). These properties segregate the fibrogenic EMT program from RAS-independent TGF- β target genes, which have active enhancers and preloaded RNAPII in the basal state and do not need INO80 or DHX9 for activation by TGF- β .

Role of SMAD4 in INO80-mediated chromatin remodeling

SMAD4 mediates TGF- β regulation of numerous developmental, immune, and homeostatic processes, as well as tumor suppression in gastrointestinal cancers and metastasis progression in LUAD and other cancers.^{6,10,32,64} SMAD4 forms transcriptionally active complexes with receptor-phosphorylated SMADs.^{41,65} Most, though not all, gene responses to TGF- β require SMAD4, and this requirement extends to all other members of the TGF- β superfamily.⁶ In spite of this central position, the specific role of SMAD4 in the TGF- β pathway has remained elusive for decades. Our evidence shows that SMAD4 binds INO80 through distinctive structural elements of its MH2 domain and recruits it to RREB1-primed enhancers for clearing H2A.Z-containing nucleosomes. To our knowledge, the recruitment of INO80 for nucleosome clearing at primed enhancers is the first identified function of SMAD4 in TGF- β signaling. SMAD4-recruited INO80 modulates H2A.Z-containing nucleosomes subtly, perhaps to increase chromatin accessibility at these enhancers

(G) *Ino80*-KO 393T3 cells were transduced with two independent sgRNAs against *H2afz* for CRISPR interference and further engineered to express doxycycline (Dox)-inducible Cas9. Representative *ex vivo* BLI images of lungs from mice 21 days after tail-vein inoculation of the cells as indicated. Mice were fed with doxycycline diet.

(H) Lung metastasis burden in these mice as quantified by *ex vivo* firefly luciferase BLI. Data are mean $\pm$ SD; $n = 6$ –7 mice per group; one-way ANOVA; **** $p < 0.0001$.

See also Figure S6.

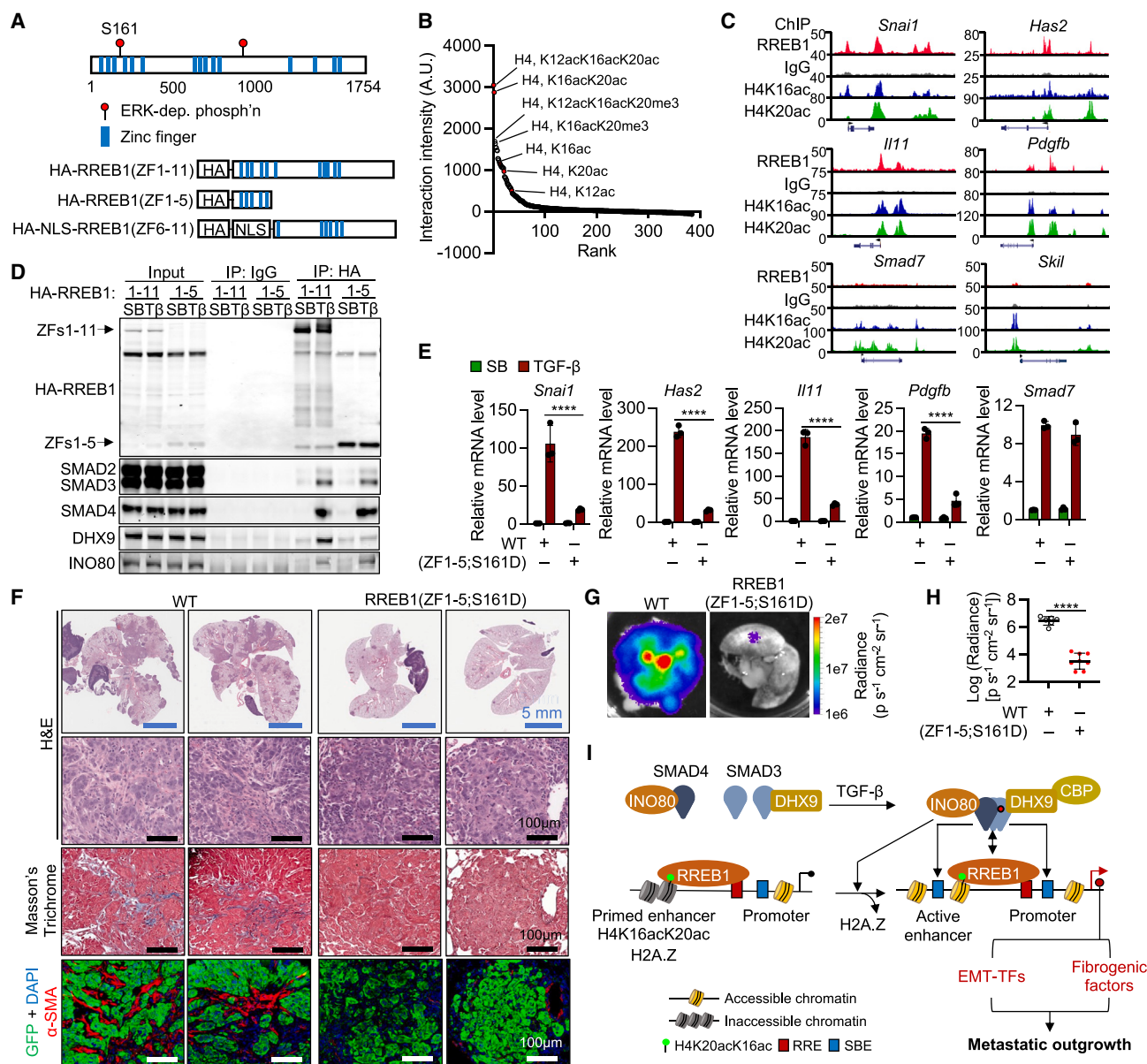


Figure 7. Inhibition of RREB1 suppresses metastasis

(A) Schematic domain structure of RREB1, indicating the location of zinc fingers annotated in Uniprot and conserved ERK phosphorylation sites.⁸

(B) Quantified signal intensity of the binding of recombinant RREB1(ZF1-5) to a histone modification peptide array as shown in Figure S7C. H4K16acK20ac-containing peptides are highlighted in red.

(C) IgG, H4K16ac, and H4K20ac ChIP-seq tracks of *Snai1*, *Has2*, *Il11*, *Pdgfb*, *Smad7*, and *Skil* in WT 393T3 cells.

(D) 393T3 cells expressing HA-RREB1(ZF1-11) or HA-RREB1(ZF1-5) were incubated with SB or TGF- β for 1.5 h and immunoprecipitated (IP) with the indicated antibodies. Immune complexes were subjected to western immunoblotting with antibodies indicated at left.

(E) *Snai1*, *Has2*, *Il11*, *Pdgfb*, and *Smad7* mRNA levels in 393T3 cells expressing vector alone WT or HA-RREB1(ZF1-5;S161D) and incubated with SB or TGF- β for 1.5 h. Data are mean $\pm$ SD; $n = 3$; two-way ANOVA; **** $p < 0.0001$.

(F) Representative H&E staining, Masson's trichrome staining, and α -SMA IF of lungs from mice 21 days after tail-vein inoculation with WT or HA-RREB1(ZF1-5;S161D)-expressing 393T3 cells. Two sections per cell type are shown.

(G and H) Representative *ex vivo* BLI images (**G**) and lung metastatic burden (**H**) of mice in this experiment, as quantified by firefly luciferase BLI. Data are mean $\pm$ SD; $n = 7$ mice per group; two-tailed unpaired t test; **** $p < 0.0001$.

(I) Schematic summary of the roles of INO80 and DHX9 in the induction of fibrogenic and EMT-TF genes by TGF- β as two branches of a regulated program that jointly support tumor invasion, fibroblast activation, and ECM deposition for the outgrowth of pulmonary metastases.

See also Figure S7 and Table S3.

only transiently to allow enhancer return to the inhibited state after TGF- β stimulation ends.

Limitations of the study

This work focused on LUAD pulmonary metastasis, a major clinical complication of LUAD. However, metastatic LUAD also affects the bones, brain, liver, and adrenal glands, which differ extensively in stromal composition. The TGF- β fibrogenic EMT program may also be important in injury repair and the pathogenesis of fibrotic diseases,⁶ implications that remain to be investigated. TGF- β is widely pleiotropic, thwarting decades-long efforts to target this pathway for the treatment of major diseases of dysregulated TGF- β signaling, including fibrosis, cancer, and others.⁶ TGF- β ligands and receptors are nodes of broad pleiotropy. By contrast, RREB1 is dedicated to RAS-dependent TGF- β fibrogenic EMTs, providing a potential target for the selective suppression of pathogenic forms of these programs. Our proof of concept that rationally designed RREB1 inhibitors can be deployed to inhibit this program and suppress metastasis opens the possibility of testing this approach against diseases in which RREB1 enables TGF- β -driven pathogenesis.

RESOURCE AVAILABILITY

Lead Contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Joan Massagué (Massagué@mskcc.org).

Materials availability

This study did not generate new unique reagents. Further information and requests for any reagents generated in this study, including plasmids and cancer cell lines, are available upon reasonable request from the [lead contact](#).

Data and code availability

- Raw sequencing reads and processed files for RNA-seq, ChIP-seq, CUT&RUN, ATAC-seq, MNase-seq, and DRIP-seq have been deposited in the Gene Expression Omnibus database (GEO) under the SuperSeries accession number GSE256020 and are publicly available as of the date of publication.
- All raw and processed scRNA-seq data generated by Laughney et al.²⁶ and used in this paper were accessible from NCI's Gene Expression Omnibus database through accession number GSE123904. Accession numbers are listed in the [key resources table](#). Code used for scRNA-seq analysis, derived from <https://zenodo.org/records/7618821> and used by Hu et al.,²⁷ are deposited to GitHub: https://github.com/digvijayky/Lee_et_al_Cell_2024. The DOI is listed in the [key resources table](#).
- Original Western blot images and qRT-PCR Ct values have been deposited at Mendeley (<https://10.17632/k83v27djfh.1>) and are publicly available as of the date of publication. The DOI is listed in the [key resources table](#).
- All software programs used for analyses are publicly available and listed in the [key resources table](#).
- Any additional information required to reanalyze data reported in this paper are available from the [lead contact](#) upon request.

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AUTHOR CONTRIBUTIONS

J.H.L. and J.M. conceived studies, designed experiments, interpreted results, and wrote the manuscript; J.H.L. performed experiments and analyzed data; F.J.S.-R. and S.W.L. designed and analyzed CRISPR knockout screens; L.H. provided technical assistance; H.B. designed and analyzed mass spectrometry; R.P.K. and J.H.L. analyzed epigenetic profiling data; D.V.K.Y. analyzed scRNA-seq data; F.X.C. and M.J.M. designed and constructed SMAD4 mutants; C.T. performed EMSA experiments; E.S. and L.L. generated and tested 393T3 knockout cell lines; J.E.C., C.S., and T.T. generated KP LUAD mice; J.S.P. generated PDX tumoroids; PDX models were derived by C.M.R.; and F.J.S.-R., M.J.M., R.P.K., and D.V.K.Y. helped with manuscript preparation.

DECLARATION OF INTERESTS

J.M. owns company stock in Scholar Rock. C.M.R. has consulted regarding oncology drug development with Amgen, Astra Zeneca, Chugai, Daiichi Sankyo, Hoffman-La Roche, and Jazz Pharmaceuticals; C.M.R. serves on the scientific advisory boards of Auron, Bridge Medicines, DISCO, Earli, and Harpoon Therapeutics. S.W.L. is a consultant and holds equity in Blueprint Medicines, ORIC Pharmaceuticals, Mirimus, PMV Pharmaceuticals, Faeth Therapeutics, and Senecea Therapeutics, and is a consultant for Fate Therapeutics. T.T. is a consultant and holds equity in Lime Therapeutics; the Tammela Lab receives research support from Ono Pharmaceuticals Co., Ltd. (unrelated to this work). T.T.'s spouse is an employee of Recursion Pharmaceuticals. R.P.K. is a co-founder of and consultant for Econic Biosciences.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Anti-rat HA High Affinity	Sigma Aldrich	Cat #: 11867423001 RRID: AB_390918
Anti-rabbit HA-tag Antibody	Cell Signaling Technology	Cat #: 3724S RRID: AB_1549585
Anti-mouse HA-tag Antibody	Cell Signaling Technology	Cat #: 2367S RRID: AB_10691311
Anti-rabbit SMAD2/3 Antibody	Cell Signaling Technology	Cat #: 8685S RRID: AB_10889933
Anti-rabbit SMAD4 Antibody	Cell Signaling Technology	Cat #: 46535S RRID: AB_2736998
Normal rabbit IgG	Cell Signaling Technology	Cat #: 2729S RRID: AB_1031062
Normal mouse IgG	Sigma Aldrich	Cat #: 12-371 RRID: AB_145840
Anti-rabbit DHX9 Antibody	Abcam	Cat #: ab26271 RRID: AB_777725
Anti-rabbit DHX9 Antibody	Proteintech	Cat #: 17721-1-AP RRID: AB_2092506
Anti-rabbit INO80 Antibody	Proteintech	Cat #: 18810-1-AP RRID: AB_10598463
Anti-rabbit H2A.Z Antibody	Abcam	Cat #: ab4174 RRID: AB_304345
Anti-rabbit CBP Antibody	Cell Signaling Technology	Cat #: 7389S RRID: AB_2616020
Anti-mouse Flag-tag Antibody	Sigma Aldrich	Cat #: F1804 RRID: AB_262044
Anti-mouse RREB1 Antibody	Santa Cruz Biotechnology	Cat #: sc-515901 RRID: AB_3083806
Anti-mouse FRA1 Antibody	Santa Cruz Biotechnology	Cat #: sc-28310 RRID: AB_627632
Anti-rabbit JUNB Antibody	Cell Signaling Technology	Cat #: 3753S RRID: AB_2130002
Anti-rabbit RBBP5 Antibody	Bethyl Laboratories	Cat #: A300-109A RRID: AB_210551
Anti-rabbit KDM6B Antibody	Cell Signaling Technology	Cat #: 3457S RRID: AB_1549620
Anti-rabbit HMGA2 Antibody	Cell Signaling Technology	Cat #: 5269S RRID: AB_10694917
Anti-mouse TGFB2 Antibody	Proteintech	Cat #: 66636-1-Ig RRID: AB_2881995
Anti-rabbit SNAIL Antibody	Cell Signaling Technology	Cat #: 3879S RRID: AB_2255011
Anti-rabbit ZEB1 Antibody	Proteintech	Cat #: 21544-1-AP RRID: AB_10734325
Anti-rabbit HAS2 Antibody	Thermo Fisher Scientific	Cat #: PA5-25593 RRID: AB_2543093
Anti-rabbit GAPDH Antibody	Cell Signaling Technology	Cat #: 2118S RRID: AB_561053

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
Anti-mouse α -Tubulin Antibody	Sigma Aldrich	Cat #: T6074 RRID: AB_477582
Anti-rabbit H3K4me3 Antibody	Epiccypher	Cat #: 13-0041 RRID: AB_3076423
Anti-rabbit H3K4me1 Antibody	Epiccypher	Cat #: 13-0057 RRID: AB_3076424
Anti-rabbit H3K27Ac Antibody	Active Motif	Cat #: 39133 RRID: AB_2561016
Anti-rabbit H3K27me3 Antibody	Active Motif	Cat #: 39157 RRID: AB_2561020
Anti-rabbit H4K16ac Antibody	Abcam	Cat #: ab109463 RRID: AB_10858987
Anti-rabbit H4K20ac Antibody	RevMAb Biosciences	Cat #: 31-1084-00 RRID: AB_2716398
Anti-rabbit Polymerase II Antibody	Active Motif	Cat #: 39097 RRID: AB_2732926
Anti-DNA-RNA Hybrid Antibody, clone S9.6	Millipore Sigma	Cat #: MABE1095 RRID: AB_2861387
Anti-rabbit E-Cadherin antibody	Cell Signaling Technology	Cat #: 3195S RRID: AB_2291471
Anti-chicken Green Fluorescent Protein Antibody	Aves Labs	Cat #: GFP-1010 RRID: AB_2307313
Anti-chicken mCherry Antibody	Abcam	Cat #: ab205402 RRID: AB_2722769
IRDye® 800CW Goat anti-Rabbit IgG Secondary Antibody	LI-COR Biosciences	Cat #: 926-32211 RRID: AB_621843
IRDye® 800CW Goat anti-Mouse IgG Secondary Antibody	LI-COR Biosciences	Cat #: 926-32210 RRID: AB_621842
IRDye® 680RD Goat Anti-Rabbit IgG Secondary Antibody	LI-COR Biosciences	Cat #: 926-68071 RRID: AB_10956166
IRDye® 680RD Goat Anti-Mouse IgG Secondary Antibody	LI-COR Biosciences	Cat #: 926-68070 RRID: AB_10956588
Alexa Fluor 488 tyramide	Life Technologies	Cat #: B40953
CF 594 tyramid	Biotium	Cat #: 92174
Alexa Fluor 647 tyramide	Life Technologies	Cat #: B40958
Alexa-Fluor 488 Goat anti-Chicken	Thermo Fisher Scientific	Cat #: A11039 RRID: AB_2534096
Alexa-Fluor 488 Goat anti-Rabbit	Thermo Fisher Scientific	Cat #: A11008 RRID: AB_143165
Alexa-Fluor 546 Goat anti-Rabbit	Thermo Fisher Scientific	Cat #: A11035 RRID: AB_2534093
Alexa-Fluor 546 Goat anti-Mouse	Thermo Fisher Scientific	Cat #: A11030 RRID: AB_2737024
Alexa-Fluor 647 Goat anti-Chicken	Thermo Fisher Scientific	Cat #: A21449 RRID: AB_2535866
Alexa-Fluor 647 Goat anti-Mouse	Thermo Fisher Scientific	Cat #: A21235 RRID: AB_2535804
Alexa-Fluor 647 Goat anti-Rabbit	Thermo Fisher Scientific	Cat #: A21244 RRID: AB_2535812
Bacterial and virus strains		
NEB® 5-alpha Competent E. coli	New England Biolabs	Cat #: C2987H
Endura electrocompetent cells	Fisher Scientific	Cat #: 50-104-7945
E.coli B834 (DE3)	Merck KGaA	Cat #: 69041

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
Chemicals, peptides, and recombinant proteins		
Dulbecco's Modified Eagle's high glucose medium	Media Preparation Core, MSKCC	Powder Cat #: 52100047
Roswell Park Memorial Institute 1640 medium	Media Preparation Core, MSKCC	Powder Cat #: 31800105
Advanced DMEM/F-12	Thermo Fisher	Cat#: 12634-028
Dulbecco's Phosphate-Buffered Saline, no calcium, no magnesium	Media Preparation Core, MSKCC	Powder Cat #: 21600044
Fetal Bovine Serum	Sigma Aldrich	Cat #: F2442
L-glutamine	Thermo Fisher Scientific	Cat #: 25030081
GlutaMAX	Thermo Fisher Scientific	Cat #: 35050061
Penicillin-Streptomycin	Thermo Fisher Scientific	Cat #: 15140163
Amphotericin B	Gemini Bio-Products	Cat #: 400104
HEPES	Sigma Aldrich	Cat #: H0887
Gentamicin	Thermo Fisher Scientific	Cat #: 15710-072
Puromycin dihydrochloride	Sigma Aldrich	Cat #: P9620
G418	Thermo Fisher Scientific	Cat #: 10131035
Blasticidin	Invivogen	Cat #: ant-bl-05
Hygromycin B	Thermo Fisher	Cat #: 10687-010
Polybrene	Santa Cruz Biotechnology	Cat #: sc-134220
Opti-MEM	Thermo Fisher Scientific	Cat #: 31985062
Lipofectamine 2000	Thermo Fisher Scientific	Cat #: 11668019
MK2206	ChemieTek	Cat #: CT-MK2206
SB-505124	Sigma Aldrich	Cat #: S4696
TGF- β 1	R&D Systems	Cat #: 7754-BH
Lenti-X Concentrator	Takara Bio	Cat #: 631231
D-luciferin, Potassium Salt	GoldBio	Cat #: LUCK-10G
Sucrose	Fisher Scientific	Cat #: S53
Glycerol	Thermo Fisher Scientific	Cat #: BP2291
β -Mercaptoethanol	Sigma Aldrich	Cat #: M3148100ML
DAPI (4',6-Diamidino-2-Phenylindole Dilactate)	Thermo Fisher Scientific	Cat #: D3571
Hoechst 33342	Thermo Fisher Scientific	Cat #: H3570
Dispase II	Sigma Aldrich	Cat #: D4693
Collagenase type IV	Thermo Fisher Scientific	Cat #: 17104019
DNase I	STEMCELL Technologies	Cat #: 07900
BD Pharm Lyse™ Lysing Buffer	BD Biosciences	Cat #: 555899
Corning™ Matrigel™ GFR Membrane Matrix	Fisher Scientific	Cat #: CB-40230C
BsmBI-V2	New England Biolabs	Cat #: R0739S
BbsI-HF	New England Biolabs	Cat #: R3539S
Esp3I	New England Biolabs	Cat #: R0734S
AsiSI	New England Biolabs	Cat #: R0630S
MluI-HF	New England Biolabs	Cat #: R3198S
BsrGI-HF	New England Biolabs	Cat #: R3575S
EcoRI-HF	New England Biolabs	Cat #: R3101S
HindIII	New England Biolabs	Cat #: R3104S
SspI-HF	New England Biolabs	Cat #: R3132S
XbaI	New England Biolabs	Cat #: R0145S
BamHI-HF	New England Biolabs	Cat #: R3136S

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
Bovine serum albumin	New England Biolabs	Cat #: B9001
Klenow Fragment (3-5' exo-)	New England Biolabs	Cat #: M0212S
NEBNext End Repair Module	New England Biolabs	Cat #: E6050S
E.coli DNA Ligase	New England Biolabs	Cat #: M0205S
DNase I	New England Biolabs	Cat #: M0303S
Beta-Nicotinamide adenine dinucleotide	New England Biolabs	Cat #: B9007S
RNaseH	New England Biolabs	Cat#: M0297S
dUTP Solution	New England Biolabs	Cat #: N0459S
Deoxynucleotide (dNTP) Solution Set	New England Biolabs	Cat #: N0446S
DNA Polymerase I (E.coli)	New England Biolabs	Cat #: M0209S
T4 DNA ligase	New England Biolabs	Cat #: M0202S
Salt-T4® DNA Ligase	New England Biolabs	Cat #: M0467S
Gibson Assembly Master Mix	New England Biolabs	Cat #: E2611L
Q5 High-Fidelity 2x Master Mix	New England Biolabs	Cat #: M0492S
IScript reverse transcription supermix	BioRad	Cat #: 1708840
Chameleon™ Duo Pre-stained Protein Ladder	LI-COR Biosciences	Cat #: 928-60000
Odyssey Blocking Buffer	LI-COR Biosciences	Cat #: 927-60001
LB-carbenicillin plates	Teknova	Cat #: L1010
1 M Magnesium dichloride (MgCl ₂)	Thermo Fisher Scientific	Cat #: AM9530G
2 M Potassium chloride (KCl)	Thermo Fisher Scientific	Cat #: AM9640G
1 M Calcium chloride (CaCl ₂)	Sigma Aldrich	Cat #: 21115-100ML
Glycine	Fisher Scientific	Cat #: G48-212
Nonidet P-40 (NP-40)	Sigma Aldrich	Cat #: 56741
1 M Tris-HCl pH 8.0	Sigma Aldrich	Cat #: 93283
1M Tris-HCl pH 7.5	Thermo Fisher Scientific	Cat #: 15567027
1 M Tris-HCl pH 6.9	Boston BioProducts	Cat #: BM-311
Triton™ X-100	Fisher Scientific	Cat #: AC215682500
Tween® 20	Sigma Aldrich	Cat #: P1379-1L
N,N-Dimethylformamide, anhydrous (DMF)	Sigma Aldrich	Cat #: 227056-100mL
Formaldehyde solution	Sigma Aldrich	Cat #: F8775-500ML
16% Paraformaldehyde Aqueous Solution	Fisher Scientific	Cat #: 50-980-488
Nuclease-Free Water	Thermo Fisher Scientific	Cat #: AM9930
0.5 M EDTA	Thermo Fisher Scientific	Cat #: AM9260G
10% SDS	Thermo Fisher Scientific	Cat #: 15553027
5M NaCl	Thermo Fisher Scientific	Cat #: AM9759
1M HEPES/KOH pH 7.5	Boston BioProducts	Cat #: BBH-75-K
1M HEPES/KOH pH 8.0	Thermo Fisher Scientific	Cat #: J63578-AP
Sodium phosphate monobasic solution	Sigma Aldrich	Cat #: 74092-100ML
Sodium phosphate dibasic solution	Sigma Aldrich	Cat #: 94046-100ML
10 % Na-deoxycholate	Bioworld	Cat #: 40430018-1
Sodium bicarbonate (NaHCO ₃)	Sigma Aldrich	Cat #: S5761
Phenol-chloroform isoamyl alcohol	Thermo Fisher Scientific	Cat #: 15593031
cOmplete™, Mini, EDTA-free Protease Inhibitor Cocktail	Roche	Cat #: 11836170001
Halt™ Phosphatase Inhibitor Cocktail	Thermo Fisher Scientific	Cat #: 78427
UltraPure™ BSA	Thermo Fisher Scientific	Cat #: AM2616
RNaseA	Thermo Fisher Scientific	Cat #: EN0531
Proteinase K	Thermo Fisher Scientific	Cat #: EO0491
Spermidine	Sigma Aldrich	Cat #: S0266-1G

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
Digitonin	Sigma Aldrich	Cat #: CHR103
Nextera TDE1	Illumina	Cat #: FC-121-1030
TruSeq Single indexes	Illumina	Cat #: 20015960
NEBNext High-Fidelity 2x PCR Master Mix	New England Biolabs	Cat #: M0541
GlycoBlue Coprecipitant	Thermo Fisher Scientific	Cat #: AM9515
Micrococcal Nuclease Solution	Thermo Fisher Scientific	Cat #: 88216
Phase lock gel tubes	VWR	Cat #: 10847-800
QuickExtract™ DNA Extraction Solution	Lucigen	Cat #: QE09050
N-glycosylase	Thermo Fisher Scientific	Cat #: N8080096
Poly[d(I-C)]	Sigma Aldrich	Cat #: 10108812001
NuPAGE LDS Sample Buffer (4X)	Thermo Fisher Scientific	Cat #: NP0007
NuPAGE Sample Reducing Agent	Thermo Fisher Scientific	Cat #: NP0009
Antigen Unmasking Solution, Citrate-Based	Vector Laboratories	Cat #: H-3300
Histo-Clear	National Diagnostics	Cat #: HS-200
Hydrogen peroxide solution	Sigma Aldrich	Cat #: 216763-500ML
Normal donkey serum	Sigma Aldrich	Cat #: D9663
Normal goat serum	Life Technologies	Cat #: 50062Z
Normal Horse Serum	Vector Laboratories	Cat #: S-2012
ImmPRESS HRP Anti-Rabbit Ig	Vector Laboratories	Cat #: MP-7401
ImmPACT DAB Peroxidase	Vector Laboratories	Cat #: SK-4105
Vectashield mounting medium	Vector Laboratories	Cat #: H-1000-10
ProLong diamond antifade mountant	Life Technology	Cat #: P36970
Calbiochem™ MOWIOL™ 4-88 Reagent	Fisher Scientific	Cat #: 47-590-4100GM
Bond Dewax Solution	Leica	Cat #: AR9222
EDTA-based epitope retrieval ER2 solution	Leica	Cat #: AR9640
Terrific Broth	CondaLab	Cat #:1246
Isopropyl β-D-1-thiogalactopyranoside	NZYTEch, Lda	Cat #: MB02603
Tris HCl	Cymit Quimica, SL	Cat #: AN-AG00HEKS
NaCl	Merck KGaA	Cat #: S9625
Imidazole	Merck KGaA	Cat #: A1073.1000
TCEP tris(2-carboxyethyl)phosphine HCl	ChemoSapiens S. L.	Cat #: AB121644
TWEEN 20	Panreac Quimica, S.L.U.	Cat #: A4974.0500
Lysozyme	Merck KGaA	Cat #:L6876
DNase I	Merck KGaA	Cat #: 11284932001
Glycerol	Merck KGaA	Cat #: 49767
40% Acrylamide/Bis Solution, 19:1	Bio-Rad	Cat #: 1610144
3C protease	In-house	N/A
Duolink® In Situ Detection Reagents Red	Sigma Aldrich	Cat #: DUO92008
Duolink® In Situ PLA® Probe Anti-Rabbit MINUS	Sigma Aldrich	Cat #: DUO92001
Duolink® In Situ PLA® Probe Anti-Mouse PLUS	Sigma Aldrich	Cat #: DUO92001
Critical commercial assays		
RNeasy Mini Kit	QIAGEN	Cat #: 74106
QIAshredder	QIAGEN	Cat #: 76956
RNase-Free DNase Set	QIAGEN	Cat #: 79256
QIAquick PCR Purification Kit	QIAGEN	Cat #: 28106
MinElute PCR Purification Kit	QIAGEN	Cat #: 28004
CUTANA DNA purification kit	Epicpypher	Cat #: 14-0050

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
QIAquick Gel Extraction Kit	QIAGEN	Cat #: 28706
DNeasy Blood and Tissue Kit	QIAGEN	Cat #: 69504
QuickLyse Miniprep Kit	QIAGEN	Cat #: 27406
QIAGEN Plasmid Plus Maxi Kit	QIAGEN	Cat #: 12965
ZymoPURE II Plasmid Maxiprep Kits	Zymo Research	Cat #: D4203
Transcriptor First Strand cDNA Synthesis Kit	Roche	Cat #: 04897030001
Mouse/Rat PDGF-BB Quantikine ELISA Kit	R&D Systems	Cat #: MBB00
DuoSet ELISA Ancillary Reagent Kit 2	R&D Systems	Cat #: DY008B
Mouse IL-11 DuoSet ELISA	R&D Systems	Cat #: DY418
Caspase-Glo® 3/7 Assay	Promega	Cat #: G8092
CellTiter-Glo® 2.0 Cell Viability Assay	Promega	Cat #: G9243
Pierce™ BCA Protein Assay Kit	Thermo Fisher Scientific	Cat #: 23227
TruSeq Stranded mRNA LT Kit	Illumina	Cat #: RS-122-2102
NovaSeq 6000 S4 Reagent Kit	Illumina	Cat #: 20028315
KAPA HTP Library Preparation Kit	Kapa Biosystems	Cat #: KK8234
NEBNext Ultra RNA Library Prep Kit	New England Biolabs	Cat #: E7530S
CUTANA™ ChIC/CUT&RUN Kit	Epiccypher	Cat #: 14-1048
Bioanalyzer High-Sensitivity DNA Analysis kit	Agilent	Cat #: 5067-4626
MODified™ Histone Peptide Array	Active Motif	Cat #: 13005
RNAscope LS Multiplex Reagent Kit	Advanced Cell Diagnostics Bio	Cat #: 322800
NEBNext Ultra RNA Library Prep Kit	New England Biolabs	Cat #: E7530S

Deposited data

Raw files for RNA-seq	This paper	GEO: GSE256020
Raw and processed data files for ChIP-seq and CUT&RUN	This paper	GEO: GSE256020
Raw and processed data files for DRIP-seq	This paper	GEO: GSE256020
Raw and processed data files for MNase-seq	This paper	GEO: GSE256020
Raw and processed data files for ATAC-seq	This paper	GEO: GSE256020
Raw and processed data files for scRNA-seq	Laughney et al. ²⁶ and Hu et al. ²⁷	GEO: GSE123904
Original Western blots and qRT-PCR Ct values	Mendeley Data	https://10.17632/k83v27djfh.1
Code used for scRNA-seq analysis	GitHub	https://github.com/digvijayky/Lee_et_al_Cell_2024

Experimental models: Cell lines

393T3	Li et al. ⁶⁹	This paper
mouse KSIC pancreatic ductal adenocarcinoma cell line	Bardeesy et al. ⁶⁴	This paper
KP LUAD tumoroids	In-house	This paper
PDX-derived tumoroids	In-house	This paper
A549	ATCC	ATCC #: CCL-185
HEK293T	ATCC	ATCC #: CRL-3216

Experimental models: Organisms/strains

Mouse: B6129SF1/J	The Jackson Laboratory	Strain #: 101043
Mouse: 129S1/SvImJ	The Jackson Laboratory	Strain #: 002448

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
Mouse: Athymic Nude-Crl:NU(NCr)- <i>Foxn1</i> ^{nu}	Charles River Laboratories	Strain #: 490
Mouse: B6J.129(B6N)-Gt(ROSA) 26Sor ^{tm1(CAG-cas9*,-EGFP)Fzh/J}	The Jackson Laboratory	Strain #: 026175
Oligonucleotides		
sgRNA sequences for CRISPR screens: Table S2	This paper	Custom
Primers for CRISPR screen: Table S2	This paper	Custom
sgRNA oligos: Table S4	This paper	Custom
qRT-PCR primers: Table S4	This paper	Custom
m <i>Pdgfb</i> -C1 probe	Advanced Cell Diagnostics Bio	Cat #: 424658
m <i>Has2</i> -C2 probe	Advanced Cell Diagnostics Bio	Cat #: 465178-C2
m <i>Il11</i> -C3	Advanced Cell Diagnostics Bio	Cat #: 552468-C3
<i>Snai1</i> EMSA: Table S4	CondaLab	Custom
<i>Has2</i> EMSA: Table S4	CondaLab	Custom
<i>Smad2</i> shRNA #1	Sigma Aldrich	TRCN0000089333
<i>Smad2</i> shRNA #2	Sigma Aldrich	TRCN0000089334
<i>Smad4</i> shRNA	Lowe lab	CAAAGATGAATTGGATTCTTT
Recombinant DNA		
pEntr/pLenti-HA-Rreb1-puro	Su et al. ⁸	N/A
pEntr/pLenti-HA-Rreb1(ZF1-5)-puro (1-315)	In-house	N/A
pEntr/pLenti-HA-Rreb1(ZF6-11)-puro (316-1291)	In-house	N/A
pEntr/pLenti-HA-Rreb1(ZF1-5) S161D-puro	In-house	N/A
pOPINF/RREB1/ZF1-5	In-house/ThermoFisher	N/A
Guide-it CRISPR/Cas9 vector	Takara	Cat #: 632602
pLenti CMV rtTA3 Hygro (w785-1) vector > 12 SMAD binding elements (SBE, 5'-AGCCAGACA-3') with minimal promoter	In-house	Addgene #26730
pLenti CMVtight eGFP Puro (w771-1) > mCherry replacing eGFP	In-house	Addgene #26431
U6-sgRNA-EFS-Cas9-2A-Cre (pUSEC)	Sánchez-Rivera et al. ⁷⁰	Addgene #60820
pLVXT-Flag-SMAD4 WT-puro	In-house	N/A
pLVXT-Flag-SMAD4 M1-puro: Replace SMAD4_454A-493D with SMAD3_359G- 364Y & delete SMAD4_5451-552D	In-house	N/A
pLVXT-Flag-SMAD4 M2-puro: Replace SMAD4_294M-310F with 15-residue GS- linker	In-house	N/A
pLVXT-Flag-DHX9 WT-puro	In-house	N/A
pLVXT-Flag-DHX9 K417R-puro	In-house	N/A
pLVXT-Flag-DHX9 ΔRBD-puro (Δ1-330)	In-house	N/A
pLVXT-Flag-DHX9 ΔRM-puro (Δ1-404)	In-house	N/A
pLVXT-Flag-DHX9 ΔRGG-puro (Δ1151-1270)	In-house	N/A
pCS2-Flag-SMAD3 WT	In-house	N/A
pCS2-Flag-SMAD3 NL	In-house	N/A
pCS2-Flag-SMAD3 LC	In-house	N/A
pLVXT-INO80 WT-C-Myc-DDK-Puro	In-house	N/A

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
pLVXT-INO80 ΔSNF-C-Myc-DDK-Puro (Δ273-1225)	In-house	N/A
Software and algorithms		
Living Image	PerkinElmer	Version 4.5
ImageJ	NIH	Version 2
FIJI	NIH	Version 2.3.0/1.53q
Prism	GraphPad	Version 9
SnapGene Viewer	SnapGene	Version 2
CaseViewer	3DHISTECH	Version 2.4
RStudio	Rstudio	Version 1.2.5029
Integrative Genomics Viewer (IGV)	Broad Institute	Version 2.16.1
STAR	Dobin et al. ⁷⁷	Version 2.7.10b
deepTools	Ramírez et al. ⁷⁸	Version 3.3.0
BEDTools	Quinlan et al. ⁷⁹	Version 2.29.2
GSEA	Mootha et al. ⁸⁰	Version 4.0.3
DESeq2	Bioconductor	Version 1.40.2
TrimGalore	Bolger et al. ⁸¹	Version 0.4.5
Bowtie2	Langmead et al. ⁸²	Version 2.3.4.1
SAMtools	Li et al. ⁸³	Version 1.8
MACS2	Zhang et al. ⁸⁴	Version 2.1
HOMER	Heinz et al. ⁸⁵	Version 4.5
Picard Tools	N/A	Version 2.16.0
FeatureCounts	Liao et al. ⁸⁶	Version 1.6.1
Heatmap	Seaborn	Version 0.13.2
Python	Python	Version 3.8.5
Scanpy	Wolf et al. ⁸⁷	Version 1.9.6
Pandas	https://zenodo.org/records/10537285	Version 2.0.3
Numpy	Harris et al. ⁸⁸	Version 1.22.4
Scipy	Virtanen et al. ⁸⁹	Version 1.10.1
H5py	https://zenodo.org/records/4584676	Version 3.10.0
Matplotlib	https://zenodo.org/records/10152802	Version 3.7.4
ColabFold	Mirdita et al. ⁷⁴	v1.5.5
UCSF Chimera	Pettersen et al. ⁷⁶	v1.13.1
Other		
Mouse doxycycline diet 625 mg/kg	Envigo	TD.07383
Clamp Lamp Light with 8.5 Inch Aluminum Reflector	Simple Deluxe	Cat #: B08MZKQNP4
ProLong Diamond Antifade mountant	Thermo Fisher Scientific	Cat #: P36961
Corning Falcon Standard Tissue Culture Dishes	Fisher Scientific	Cat #: 08772E
0.2μM Nitrocellulose membrane	N/A	N/A
AMPure XP beads	Beckman Coulter	Cat #: A63880
Dynabeads® Protein G for Immunoprecipitation	Life Technologies	Cat #: 10004D
Dynabeads M-280 Streptavidin	Thermo Fisher Scientific	Cat #: 11205D
HiTrap HP 5 mL	Cytiva	Cat #: 17-5248-01
NGC™ Quest 10 Plus Chromatography System	BioRad	Cat #: 7880003
HiLoad™ 16/600 Superdex™ 75 pg	GE Healthcare Life Science	Cat #: GE28-9893-33

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Animals

All animal experiments were conducted in accordance with protocols approved by the MSKCC Institutional Animal Care and Use Committee (IACUC) and in compliance with the relevant ethical regulations regarding animal research. Mice in all experiments were monitored by the investigators and veterinary staff at the Research Animal Resource Center, MSKCC, with food and water provided ad libitum. B6129SF1/J (strain #: 101043), 129S1/SvImJ (strain #: 002448) mouse strains were obtained from The Jackson Laboratory. Athymic nude mice (*Foxn1*^{nu}, strain #: 490) were obtained from Charles River Laboratories. For all cancer cell injection studies, female mice were used between 5–7 weeks of age. For genetically engineered mouse models of lung adenocarcinoma, mice were bred in the C57BL/6 x SV129 mixed background from previously published strains: *Kras*^{LSL-G12D},⁶⁶ *Trp53*^{flox/flox},⁶⁷ and *Rosa26*^{LSL-Cas9-P2A-EGFP}⁶⁸ to generate KP-Cas9 mice.

Cell lines

The mouse KSIC pancreatic ductal adenocarcinoma cell line was provided by N. Bardeesy.⁶⁴ The mouse lung epithelial cell line 393T3 was provided by T. Jacks.⁶⁹ The human LUAD cell line A549 and embryonic kidney cell line HEK293T were purchased from ATCC. All cell lines were tested negative for mycoplasma contamination prior to use in experiments. KSIC and 393T3 cells were genotyped by PCR amplification. All cell lines and organoid lines were maintained at 37°C and 5% CO₂ were cultured in Dulbecco's Modified Eagle's (DME) high glucose medium (Media Preparation Core, MSKCC, Cat #: 52100047), supplemented with 10% fetal bovine serum (FBS) (Sigma Aldrich, Cat #: F2442), 2 mM L-glutamine (Thermo Fisher Scientific, Cat #: 25030081) and 100 units/mL of penicillin/streptomycin (Thermo Fisher Scientific, Cat #: 15140163). To block endogenous TGF-β, cells were treated with SB-505124 as a baseline condition in TGF-β treatment experiments.

METHOD DETAILS

Animal studies

For tail vein injection, 5×10⁴ 393T3 cells, 1.5×10⁵ KP LUAD tumoroids, or 1×10⁶ A549 cells were delivered in 100 μL phosphate-buffered saline solution (PBS). For tail vein injection of KP LUAD tumoroids, we used F1 from B6J.129(B6N)-Gt(*ROSA*)26Sor^{tm1(CAG-cas9*,-EGFP)Fz}/J (strain #: 026175) from the Jackson Laboratory x 129S1/SvImJ. For intracardiac injection, 2×10⁴ 393T3 cells in 100 μL PBS were delivered into the left ventricle of the B6129SF1/J mice with a 26G × 3/8" needle attached to tuberculin syringe (VWR, Cat #: BD309625). Metastatic colonization was monitored by bioluminescence imaging (BLI) using retroorbital injection of D-luciferin (150 mg/kg) and an IVIS Spectrum Xenogen instrument (PerkinElmer). For ex vivo bioluminescence imaging, mice were injected with D-luciferin, and dissected organs were analyzed by BLI. The average radiance [p/s/cm²/sr] of the organ explants was determined. Data were analyzed using Living Image software v4.5 (PerkinElmer).

Genetically engineered mouse model of lung adenocarcinoma were induced in KP-Cas9 mice by intratracheal delivery of 2,500 transducing units (TU) of lentiviral U6-sgRNA-EFS-Cre (pSECC),⁷⁰ for lentiviral vector delivery of Cre recombinase and sgRNAs against genes of interest under the control of AT2 cell-specific surfactant protein-C (SPC) promoter. Tumors were harvested at 31–35 weeks post lentiviral transduction.

For the isolation of primary tumor cells, LUAD tumor cells were isolated from micro-dissected large LUAD tumors. Following euthanasia, mice were perfused with PBS through the left ventricle of the heart. Microdissected tumors were dissociated by a mixture of dispase II (Sigma Aldrich, Cat #: D4693, final concentration 0.6 U/mL), collagenase type IV (Thermo Fisher Scientific, Cat #: 17104019, final concentration 0.166 U/mL), and DNase I (STEMCELL Technologies, Cat #: 07900, final concentration 10 U/mL) in DMEM containing Gentamicin (Gibco, Cat #: 15710064, final concentration 2 μg/mL) at 37°C for 30 min. The dissociated cells were filtered using a 100 μm strainer and spun at 300 g for 5 min at room temperature. The supernatant was removed by aspiration and red blood cells were lysed by BD Pharm Lyse™ Lysing Buffer (BD Biosciences, Cat #: 555899). Cells were then washed with media and pelleted at 300 g for 5 min at 4°C. The supernatant was removed and GFP+ cells were sorted by Aria III cytometer (BD Biosciences). Sorted GFP+ KP LUAD cells were plated on Matrigel. KP LUAD cells were mixed in 50% Matrigel (Corning, Cat #: CB-40230C) and 50% LUAD media (Advanced DMEM/F12, Gibco, Cat #: 12634-028), 2% of FBS, 2mM GlutaMAX (Gibco, Cat #: 35050061), 10 mM HEPES, 2 μg/mL of gentamicin, and 100 units/mL of penicillin/streptomycin.

Single-cell RNA sequencing data analysis

We used our previously reported scRNA-seq dataset from human LUAD specimens (GSE123904).²⁶ The normalized, imputed gene expression generated by Laughney et al.²⁶ was used as the input. Metastatic tumour cells were ranked by average lung epithelial development score and assigned to different stages.²⁶ The dataset is further filtered to only include metastasis samples and tumour cells corresponding to developmental stages II and III, as previously performed by Hu et al.²⁷ Metastatic tumour cells were then ranked by Hallmark EMT signature score.²⁸ Z-normalized, imputed expression of genes or gene signatures was plotted on heat maps using the seaborn.

LUAD PDX-derived tumoroids

PDX models were generated under an MSKCC Institutional Review Board approved clinical protocol. LUAD PDXs (Ru631, Ru890, and Ru913) were previously established by the Rudin lab, and isolated single PDX cells were provided as a frozen stock. Cells were subsequently thawed in a water bath and resuspended in RPMI containing 10% FBS and 100 U/ml Penicillin/Streptomycin (Thermo Fisher Scientific). PDX-derived tumoroids were generated by culturing cells in culture medium mixed with Matrigel (Corning) at 1:1 ratio on a 35-mm glass-bottomed dish (P35G-1.5-14-C, MatTek). Briefly, the glass-bottom area of dishes was initially coated with undiluted Matrigel solution to prevent cell attachment to glass surface. Cells were then seeded on top of the Matrigel coating at 5×10^4 cells in cold, diluted Matrigel solution. The mixture was incubated at 37°C for 30 min to promote gelation, and additional culture medium was added gently to the side of cells/Matrigel domes. Cells were grown for 5–7 days or as stated in the description to generate mature tumoroids. When collecting cells, the Matrigel dome harboring tumoroids was transferred to a 15-mL conical tube and incubated in the Cell Recovery Solution (Corning) for at least 30 min at 4°C while on a rotator to digest Matrigel. Tumoroids were furthermore dissociated into single cells using TrypLE for 15 min at 37°C, neutralized in culture media, and centrifuged before they were resuspended in fresh medium. Samples were then strained through a 35 μ m filter to remove any clustered cells and counted before additional experiments.

CRISPR screen

To generate stable Cas9-expressing SMAD4-restored PDAC cells, lentiCas9-Blast (Addgene, 52962) was used. To express sgRNAs, pUSEPB (U6-sgRNA-EFS-Puro-P2A-TurboBFP) was used. In brief, pUSEPB vector was linearized with BsmBI (NEB, Cat #: R0739S) or Esp3I (NEB, Cat #: R0734S) and ligated with BsmBI and Esp3I-compatible annealed and phosphorylated oligonucleotides encoding sgRNAs, using high-concentration T4 DNA ligase (NEB). sgRNA sequences used are listed in Table S2.

sgRNA sequences (six per gene) targeting genes of interest were designed using a combination of the Broad Institute sgRNA Designer tool and the Vienna Bioactivity CRISPR score. sgRNAs were divided into small pools of libraries, and oligonucleotide pools were synthesized by Agilent Technologies. The libraries were then cloned into the pUSEPB vector using an adapted version of a protocol designed to ensure a robust library representation exceeding 10,000-fold.⁷¹ In brief, each library underwent selective amplification using uniquely barcoded forward and reverse primers that append cloning adapters at the 5' and 3' ends of the sgRNA insert, purified using the QIAquick PCR Purification Kit (Qiagen, Cat # 28106), and ligated into BsmBI/Esp3I-digested and dephosphorylated pUSEPB vector, using high-concentration T4 DNA ligase (NEB). Electroporation of Endura electrocompetent cells (Fisher Scientific, Cat #: 50-104-7945) was then employed to introduce a total of 1.2 μ g of ligated pUSEPB-CRISPR Library plasmid DNA. Competent cells were recovered for 1 h at 37 °C, plated across four 15 cm LB-carbenicillin plates (Teknova, Cat #: L1010), and incubated at 37 °C for 16 h. Quantification of bacterial colonies per sub-pool was achieved through serial dilution plates to ensure a robust library representation exceeding 10,000-fold. The following day, bacterial colonies were scraped and briefly expanded for 4 h at 37 °C in 500 mL of LB-carbenicillin. Plasmid DNA was then isolated using the Plasmid Plus Maxi Kit (Qiagen, Cat #: 12965). A validated control sgRNA targeting a neutral region in mouse chromosome 8 was cloned into the pUSEPB backbone and spiked into each of these libraries at a defined fraction to achieve equimolarity between sgRNAs in the library and the control. To assess sgRNA distribution, the sgRNA target region was amplified using primers that append Illumina sequencing adapters on the 5' and 3' ends of the amplicon, as well as a random nucleotide stagger and unique demultiplexing barcode on the 5' end. Library amplicons were size-selected on a 2.5% agarose gel, purified using the QIAquick Gel Extraction Kit (Qiagen, Cat #: 28706), and sequenced on an Illumina NextSeq instrument (75 nt single-end reads).

To ensure that most cells harbor a single sgRNA integration event, we determined the volume of viral supernatant that would achieve a multiplicity of infection of ~ 0.3 upon spinfection of a population of Cas9-expressing cancer cells. In brief, cells were plated at 1×10^6 per well in 12-well plates along with increasing volumes of master pool viral supernatant and 8 μ g/mL polybrene (Santa Cruz Biotechnology, Cat #: sc-134220). Cells were then centrifuged at 1,500 rpm for 2 h at 37 °C and incubated at 37 °C overnight. Viral infection efficiency was determined by the percentage of TurboBFP+ cells assessed by flow cytometry on an LSRFortessa (BD Biosciences) instrument 72 h post infection. The volume of viral supernatant that achieved 20% infection rate was utilized in the screen. To ensure a representation of 1000X at the transduction step, the appropriate number of cells in 12-well plates was infected with lentiviral supernatant. Twenty-four hours after infection, cells were pooled into two 150 mm tissue culture plates (Corning) per infection replicate ($n = 3$) and selected with 10 μ g/mL of puromycin (Sigma Aldrich, Cat #: P9620) for 3 days. Subsequently, 5×10^5 puromycin-selected cells were pelleted and stored at -20 °C (cumulative population doubling T0, input population). 5×10^5 cells were passaged and plated on 150 mm tissue culture plates. The following day, cells were treated with MK2206 (ChemieTek, Cat #: CT-MK2206, 2.5 μ M) and 2.5 μ M SB-505124 (Sigma Aldrich, Cat #: S4696-5MG) or 100 pM TGF- β 1 (R&D Systems, Cat #: 7754-BH). When cells reached 70% confluency, 5×10^5 cells were pelleted and plated on 150 mm tissue culture plates until cumulative population doublings reached 12–15.

Genomic DNA (gDNA) was extracted from cells using the DNeasy Blood and Tissue Kit (Qiagen, Cat #: 69504) per manufacturer's instructions. Pellets containing 5×10^5 cells were processed in parallel and resulting gDNA was resuspended in 100 μ L of 10 mM Tris-HCl with 0.5 mM EDTA, pH 8.0.

The library was amplified from gDNA by a modified two-step PCR version of the protocol.⁷¹ All in vivo gDNA was sampled over two-step PCR reactions. In brief, an initial 'enrichment' PCR was performed, whereby the integrated sgRNA cassettes were amplified from gDNA (PCR1), followed by a second PCR to append Illumina sequencing adapters on the 5' and 3' ends of the amplicon, as

well as a random nucleotide stagger and unique demultiplexing barcode on the 5' end (PCR2). Each PCR1 reaction contained 25 μ l of Q5 High-Fidelity 2 $\times$ Master Mix (NEB, Cat #: M0492S), 2.5 μ l of Nuc PCR1 Fwd Primer (10 μ M), 2.5 μ l of Nuc PCR1 Rev Primer (10 μ M), and a maximum of 5 μ g of gDNA in 20 μ l of water. PCR1 amplicons were purified using the QIAquick PCR Purification Kit (Qiagen) and used as template for PCR2 reactions. Each PCR2 reaction contained 25 μ l of Q5 High-Fidelity 2 $\times$ Master Mix (NEB), 2.5 μ l of a unique Nuc PCR2 Fwd Primer (10 μ M), 2.5 μ l of Nuc PCR2 Rev Primer (10 μ M), and 300 ng of PCR1 product in 20 μ l of water. Two PCR2 reactions were run per PCR1 product. Library amplicons were size-selected on a 2.5% agarose gel, purified using the QIAquick Gel Extraction Kit (Qiagen) followed by normalization, pooling, purification using AMPure XP beads (Beckman Coulter, Cat #: 63880), and sequencing on an Illumina NextSeq500 instrument (75 nt single-end reads). All primer sequences are available in [Table S2](#). PCR settings for PCR1 and PCR2 were: initial denaturation at 98 °C for 30 s; then 98 °C for 10 s, 65 °C for 30 s, 72 °C for 30 s for 24 cycles; followed by extension at 72 °C for 2 min.

FASTQ files were processed and trimmed to retrieve sgRNA target sequences followed by alignment to the reference sgRNA library file. Raw sequencing read counts were quantified for each sgRNA, and samples were pooled at the organ level for downstream analyses (reads from the same organ across multiple mice were pooled). Reads were normalized to the total read counts per sample, and input samples were used as references to calculate $\log_2$ fold change values per sgRNA using a combination of MAGeCK (v 0.5.9.4) and custom R (v 4.0.5) scripts. Enriched fold change ($\log_2$ FC > 0) values were used to average across all samples to generate average fold changes. For each library, the most enriched sgRNA for each gene was identified, and the corresponding genes were rank-ordered based on average fold change enrichments, as previously described.⁷² See [Table S2](#) for all screening data.

CRISPR-mediated genetic knockouts

CRISPR-mediated knockouts were generated by cloning sgRNAs into the 830 Guide-it CRISPR/Cas9 vector (Red, Takara, Cat #: 632602) by using digestion of BsmBI sites, transfecting the construct into cells, then isolating and expanding the cells with knockouts from single cell colonies. Sequences of synthesized sgRNA oligos for generating CRISPR-Cas9 mediated knockout are listed in [Table S4](#).

Short-hairpin RNA (shRNA) knockdown

shRNAs targeting mouse *Smad4* were provided by S. Lowe. shRNAs targeting mouse *Smad2* were purchased (Sigma Aldrich, TRCN0000089333 and TRCN0000089334).

Mass spectrometry

For each experiment, 20 plates (15 cm) with 70% confluent cells were used. Cells were treated with vehicle (DMSO) or SB-505124 for 6 h to eliminate autocrine and serum-derived TGF- β signaling from the baseline, or with 100 pM TGF- β for 1.5 hours. The cells were washed with cold PBS, scrapped and incubated in cell membrane lysis buffer (10 mM HEPES pH 8.0, 1.5mM MgCl₂, 10 mM KCl) for 10 min in ice, and then Nonidet P-40 was added to a final concentration of 1%. The cells were homogenously mixed by inverting. The nuclei were collected by centrifugation for 10 min at 3,500 g. The nuclei were resuspended in binding buffer (0.1% Nonidet P-40, 150 mM NaCl, 20 mM Tris pH 8.0, 10% glycerol). The nuclei were lysed by sonication and centrifuged at 12,000 g for 10 min. The nuclear extracts were collected as supernatants and incubated with HA antibody (Sigma Aldrich Aldrich SKU: 11867423001) overnight at 4°C in a rotator. Dynabeads™ Protein G were then added to capture protein complexes bound to HA antibody. The beads were washed 5 times with the binding buffer, and twice with PBS. The beads were flash frozen in liquid nitrogen and stored at -80°C until further use.

Samples were digested in a solution of 25 ng/ μ L trypsin in 50 mM ammonium bicarbonate overnight at 37°C with mixing at 850 rpm in a thermomixer (Eppendorf). The samples were then dried down in a vacuum centrifuge. Peptides were desalted with C18 resin-packed stage-tips. For LC-MS/MS analysis, desalted peptides were dissolved in 3% acetonitrile/0.1% formic acid and injected onto a C18 capillary column on a nano ACQUITY UPLC system (Water) which was coupled to the Orbitrap Fusion Lumos mass spectrometer (Thermo Scientific). Peptides were eluted with a linear 90 min gradient of 0.5%-50% buffer B (0.1% (v/v) formic acid, 100% acetonitrile) at a flow rate of 300 nL/min. After each gradient, the column was washed with 90% buffer B for 5 min and re-equilibrated with 99.5% buffer A (0.1% formic acid, 100% HPLC-grade water). MS data were acquired with an automatic switch between a full scan in profile mode and data-dependent MS/MS scans (MaxN method) in centroid mode with 3 s cycle time. Target value for the full scan MS spectra was 1 $\times$ 10⁶ charges in the 375-1500 m/z range with a maximum injection time of 50 ms and resolution of 60,000 at 200 m/z. Isolation of precursors was performed with 1.4 m/z. Precursors were fragmented by higher-energy C-trap dissociation (HCD) with a normalized collision energy of 30 eV. MS/MS scans were acquired at a resolution of 15,000 at 200 m/z with an AGC target value of 100,000 and maximum injection time of 110 ms and dynamic exclusion for 15 s.

Raw data files were processed using Proteome Discoverer (PD) version 2.4.1.15 (Thermo Scientific). For each of the TMT experiments, raw files from all fractions were merged and searched with the SEQUEST HT search engine with a Mouse UniProt protein database downloaded on 2019/12/13 (92,249 entries). cysteine carbamidomethylation was specified as fixed modifications, while Methionine oxidation, acetylation of the protein N-terminus, TMTpro (K) and TMT6plex/TMTpro (N-term) and phosphorylation (STY) were set as variable modification. The precursor and fragment mass tolerances were 10 ppm and 0.6 Da respectively. A maximum of two trypsin missed cleavages were permitted. Searches used a reversed sequence decoy strategy to control peptide false discovery rate (FDR) and 1% FDR was set as threshold for identification.

Viral transductions

Lentivirus suspensions were produced by transfection of lentiviral vector with second generation packaging constructs psPAX2 and pMD2.G (Addgene plasmids 12260 & 12259) into 70–80% confluent HEK293T cells using Lipofectamine 2000. Viral particles were collected, filtered through 0.45 μm sterile filters, and incubated with the cells of interest for 24 h with 8 $\mu\text{g}/\text{mL}$ of polybrene. Cells were recovered in complete medium overnight before addition of selection media including hygromycin B (Life Technologies, Cat #: 10687-010), puromycin (Sigma Aldrich), G418 (Thermo Fisher, Cat #: 10-131-035), or blasticidin (Thermo Fisher, Cat #: R21001).

For KP LUAD-derived cells, HEK293 Freestyle (HEK293F) cells were cultured in Dulbecco's Modified Eagle's (DME) high glucose medium (Media Preparation Core, MSKCC, Cat #: 52100047), supplemented with 10% fetal bovine serum (FBS) (Sigma Aldrich, Cat #: F2442), 2 mM L-glutamine (Thermo Fisher Scientific) and 100 units/mL of penicillin/streptomycin (Thermo Fisher Scientific). The viral media was then concentrated using a Sorvall WX90+ Ultracentrifuge using a SureSpin 632 rotor spinning at $>130,000$ rcf for 2 h at 4°C. The supernatant was then discarded, and the viral pellet was incubated at 4°C overnight before being separated into aliquots for future use. The virus was titrated in vitro using Green-Go reporter cells.⁷⁰

qRT-PCR analysis

RNA was extracted from cells using the RNeasy Mini Kit (Qiagen, Cat #: 74106). 1 μg total RNA was used as a template for cDNA synthesis using the Transcriptor First Strand cDNA Synthesis Kit (Roche, Cat #: 04379012001). Relative gene expression was determined by quantitative PCR on the ViiA 7 Real-Time PCR System (Life Technologies) using SYBR Green assays (Life Technologies). Relative mRNA levels were calculated by the $2^{-\Delta\Delta C_t}$ method using murine *Gapdh* and human *GAPDH* as internal. qRT-PCR primers are listed in Table S4.

RNA sequencing and data analysis

After RiboGreen quantification and quality control by Agilent BioAnalyzer, 500 ng of total RNA underwent polyA selection and TruSeq library preparation according to instructions provided by TruSeq Stranded mRNA LT Kit (Illumina, Cat #: RS-122-2102), with 8 cycles of PCR. Samples were barcoded and run on a NovaSeq 6000 in a PE100 run, using the NovaSeq 6000 S4 Reagent Kit (200 Cycles) (Illumina). An average of 35 million paired reads was generated per sample. Ribosomal reads represented 0.8–1.6% of the total reads generated and the percent of mRNA bases averaged 92%.

RNA sequencing reads were 3' trimmed for base quality 15 and adapter sequences using version 0.4.5 of TrimGalore (https://www.bioinformatics.babraham.ac.uk/projects/trim_galore), and then aligned to mouse assembly mm10 with STAR v2.7.10b using default parameters. Data quality and transcript coverage were assessed using the tool CollectRNASeqMetrics from Picard (<http://broadinstitute.github.io/picard/>). Read count tables were generated with HTSeq v0.9.1. Normalization and expression dynamics were evaluated with DESeq2 using the default parameters and outliers were assessed by sample grouping in principal component analysis. Gene set enrichment analysis (GSEA, <http://software.broadinstitute.org/gsea>) was run against MSigDB v6 using the pre-ranked option and $\log_2$ fold change for pairwise comparisons.

ChIP-seq

1×10^7 cells were crosslinked at room temperature for 10 min by swirling with 1% formaldehyde (Sigma Aldrich) followed by addition of 125 mM glycine to quench the reaction. After washing the cells three times with PBS, cells were resuspended with ChIP lysis buffer (50 mM HEPES/KOH pH 7.5, 150 mM NaCl, 0.1% Na-deoxycholate, 1% Triton X-100, 1 mM EDTA, 0.1% SDS) supplemented with protease inhibitors (Roche, cOmplete, mini, EDTA-free protease inhibitor tablets, Cat #: 11836170001) and phosphatase inhibitors (Thermo Scientific, Halt Phosphatase Inhibitor Cocktail, Cat #: 78427, 1:1000). Cell lysates in 1 mL of ChIP lysis buffer were sonicated for sheared chromatin to be 200–500 bp. Supernatants cleared by centrifugation were incubated with the designated antibody at 4°C overnight on a rotator. Dynabeads™ Protein G were blocked with 2.5 mg/mL UltraPure™ BSA (Thermo Scientific, Cat #: AM2616) at 4°C overnight. The beads were then washed three times with ChIP buffer (50 mM Tris-HCl pH 8.0, 150 mM NaCl, 1 mM EDTA, 1% Triton X-100) and incubated with the chromatin-antibody samples at 4°C for 1–2 h. These samples were washed 6 times with high salt ChIP buffer (20 mM Tris, pH 8.0, 500 mM NaCl, 2 mM EDTA, 1% of Triton X-100, 0.1% of SDS) followed by two washings with Tris-EDTA (TE) buffer. Chromatin complexes were eluted twice in TE with 1% ultrapure SDS (Thermo Fisher Scientific) at 65°C for 15 min, and reverse-crosslinked with RNase A (Thermo Fisher Scientific, Cat #: EN0531) for 4 h or overnight, followed by incubation with Proteinase K solution (Thermo Fisher Scientific, Cat #: EO0491) for 1 h at 45°C. DNA was purified using QIAquick PCR Purification Kit (Qiagen). For library construction and sequencing, immunoprecipitated DNA was quantified by PicoGreen and the size was evaluated by Agilent BioAnalyzer. Illumina sequencing libraries were prepared using the KAPA HTP Library Preparation Kit (Kapa Biosystems, Cat #: KK8234) according to the manufacturer's instructions with 0.02–5 ng input DNA and 8–14 cycles of PCR. Barcoded libraries were run on the NovaSeq 6000 in a PE100 run, using the NovaSeq 6000 S2 or S4 Reagent Kit (200 or 300 Cycles) (Illumina). An average of 36 million paired reads were generated per sample.

ChIP-qPCR analysis

For ChIP-qPCR analysis, DNA samples immunoprecipitated following the ChIP protocol were analyzed by qRT-PCR, and the amplification product was expressed as percentage of the input. The ChIP-qPCR primer pairs for the indicated genes are listed below. ChIP-qPCR primers are listed in [Table S4](#).

CUT&RUN

CUT&RUN samples were prepared with CUTANA™ ChIC/CUT&RUN Kit (Epicpyher, Cat #: 14-1048). For each CUT&RUN reaction, cell pellets (5×10^5 cells) were washed twice with wash buffer (20 mM HEPES-KOH, pH 8.0; 150 mM NaCl; 0.5 mM spermidine; and protease inhibitor (Roche), followed by resuspension with wash buffer. Cells in wash buffer were incubated with activated Concanavalin-A beads for 10 min at room temperature. Beads were resuspended with immunoprecipitation buffer (Wash buffer; 0.01% digitonin; 2 mM EDTA) and incubated with 0.5 μ g of designated antibody at 4°C overnight on a nutator. Antibodies included anti-H3K4me3 antibody (Epicpyher, Cat #: 13-0041), anti-H3K4me1 antibody (Epicpyher, Cat #: 13-0057), anti-H3K27Ac antibody (Active Motif, Cat #: 39133), anti-H3K27me3 antibody (Active Motif, Cat #: 39157), and anti-RNA polymerase II antibody (Active Motif, Cat #: 39097). The following day, beads were washed twice with digitonin buffer (wash buffer including 0.01% digitonin) and incubated with CUTANA pAG-MNase for 10 min at room temperature, followed by wash twice with digitonin buffer. pAG-MNase was activated by incubation with 1 μ L of 100 mM CaCl₂ for 2 h at 4°C on nutator. The reaction was terminated by incubation with stop buffer for 10 min at 37 °C in thermocycler. CUT&RUN enriched DNA was purified using CUTANA DNA purification kit (Epicpyher, Cat #: 14-0050). For library construction and sequencing, Immunoprecipitated DNA was quantified by PicoGreen and the size was evaluated by Agilent BioAnalyzer. Illumina sequencing libraries were prepared using the KAPA HTP Library Preparation Kit according to the manufacturer's instructions with 0.02-5 ng input DNA and 8-14 PCR cycles. Barcoded libraries were run on the NovaSeq 6000 in a PE100 run, using the NovaSeq 6000 S2 or S4 Reagent Kit (200 or 300 cycles) (Illumina). An average of 36 million paired reads were generated per sample.

ATAC-seq

For Omni-ATAC, samples containing 5×10^4 cells were washed twice with cold ATAC-RSB (10 mM Tris-HCl pH 7.5, 10 mM NaCl, 3 mM MgCl₂) and resuspended with ice-cold ATAC-lysis buffer (10 mM Tris-HCl pH 7.4, 10 mM NaCl, 3 mM MgCl₂, 0.1% Nonidet P-40, 0.1% Tween-20, 0.01% digitonin), followed by incubation on ice for 3 min. After washing the cells with cold ATAC-wash buffer (10 mM Tris-HCl pH 7.4, 10 mM NaCl, 3 mM MgCl₂, 0.1% Tween-20), cells were resuspended with 50 μ L of transposition mix (10 mM Tris-HCl pH 7.5, 5 mM MgCl₂, 10% dimethyl formamide, 100 nM Nextera TDE1 (Illumina, Cat #: FC-121-1030), 0.01% Digitonin, 0.1% Tween-20) and incubated at 37°C for 30 min in a thermomixer with 1000 rpm mixing. DNA was purified using a QIAquick PCR Purification Kit (Qiagen). ATAC-seq libraries were prepared using the NEBNext High-Fidelity 2 $\times$ PCR Master Mix (NEB, Cat #: M0541) as previously described.⁷³ Libraries were purified by double-sided size selection with AMPure XP beads (Beckman Coulter) and assessed using a Bioanalyzer High-Sensitivity DNA Analysis kit (Agilent, Cat #: 5067-4626).

For sequencing, after PicoGreen quantification and quality control by Agilent BioAnalyzer, libraries were pooled for equimolar run on a NextSeq 2000 in a PE50 run, using the NextSeq 1000/2000 P² Reagents (100 Cycles) (Illumina). The loading concentration was 1 nM and a 2% spike-in of PhiX was added to the run to increase diversity and for quality control purposes. The run yielded on average 5×10^7 reads per sample.

MNase-seq

Samples containing 1×10^7 cells were washed twice with cold PBS and cell pellets were resuspended with cold MNase-RSB (10 mM Tris-HCl, pH 7.5, 10 mM NaCl, 1 mM MgCl₂). After incubation on ice for 10 min, 10% Nonidet P-40 was added to make a final 0.5% Nonidet P-40 and cells were homogenously mixed by inverting. Nuclei were pelleted at 1300 $\times$ g for 5 min and resuspended with MNase-digestion buffer (10 mM Tris-HCl pH 8.0, 15 mM NaCl, 60 mM KCl, 1 mM CaCl₂, 250 mM sucrose, 0.5 mM dithiothreitol, DTT). Nuclei were incubated with varying concentrations of MNase to optimize the fragmentation (~60% mononucleosome, ~20% dinucleosome, and ~10% trinucleosome). We treated nuclei with 100 U of MNase (Thermo Fisher Scientific, Cat #: 88216). After incubation for 5 min at 37°C with mixing, stop solution (1% SDS, 100 mM NaHCO₃, 20 mM EDTA) was added and incubated at 65°C for 90 min. DNA fragments were incubated with Tris-HCl pH 7.5, Proteinase K and Glycoblue (Thermo Fisher Scientific, Cat #: AM9515) for 16 h at 65°C. Mononucleosomal DNA (~147 bp) was extracted using QIAquick Gel Extraction Kit (Qiagen).

For library construction and sequencing, Immunoprecipitated DNA was quantified by PicoGreen and the size was evaluated by Agilent BioAnalyzer. Illumina sequencing libraries were prepared using the KAPA HTP Library Preparation Kit according to the manufacturer's instructions with 0.02-5 ng input DNA and 8-14 cycles of PCR. Barcoded libraries were run on the NovaSeq 6000 in a PE50 run, using the NovaSeq 6000 S2 or S4 Reagent Kit (200 or 300 Cycles) (Illumina). An average of 5×10^7 paired reads were generated per sample.

MNase-qPCR analysis

For MNase-qPCR analysis, mononucleosomal DNA was analyzed by qRT-PCR, and the amplification product was expressed as percentage of the input. MNase-qPCR primer pairs for the indicated genes are listed in [Table S4](#).

DRIPc-seq

DRIPc-seq samples were prepared as described.⁴⁹ Briefly, DNA was extracted by treating cell samples with SDS and Proteinase K at 37°C overnight, followed by phenol-chloroform isoamyl alcohol extraction using phase lock gel tubes (VWR, Cat #: 10847-800) followed by ethanol precipitation. Extracted DNA was digested with restriction enzyme cocktail (BsrGI (NEB, Cat #: R3575S), EcoRI (NEB, Cat #: R3101S), HindIII (NEB, Cat #: R3104S), SspI (NEB, Cat #: R3132S), XbaI (NEB, Cat #: R0145S)) in NEB buffer 2 with 1 mM spermidine (Sigma Aldrich, Cat #: 05292-1ML-F) and bovine serum albumin (BSA) (NEB, Cat #: B9001) at 37°C overnight. Digested DNAs were purified by phenol-chloroform isoamyl alcohol (Thermo Fisher Scientific, Cat #: 15593031) extraction using phase lock gel tube followed by ethanol precipitation. 10 µg of digested DNA were incubated with RNase H (NEB, Cat #: M0297S) for 6 h at 37°C in RNase H buffer. 8 µg of DNA/RNA hybrids treated with RNase H and 8 µg of DNA/RNA hybrids from five separate tubes without RNase H were immunoprecipitated with S9.6 antibodies (Millipore Sigma, Cat #: MABE1095) at 4°C overnight on rotator. Dynabeads™ Protein G beads were blocked with 2.5 mg/mL UltraPure™ BSA (Thermo Scientific) at 4°C overnight. The beads were then washed twice with binding buffer (10 mM NaPO₄ pH 7.0, 140 mM NaCl and 0.05% Triton X-100, 10 mM Tris-HCl pH 8 and 1 mM EDTA) and incubated with the DNA/RNA-antibody samples at 4°C for 2 h. The beads were washed three times with binding buffer, followed by incubation with 300 µL of elution buffer and 7 µL of proteinase K at 55°C for 45 min with mixing. After incubation, DNA/RNA hybrids were separated by magnetic rack and subjected to Phenol-Chloroform Isoamyl alcohol extraction using phase lock gel tubes followed by ethanol precipitation.

Recovered DNA/RNA hybrids were incubated with DNase I (NEB, Cat #: M0303S) at 37°C for 45 min and DNase I was then inactivated by addition of 1 µL of 0.5 M EDTA, pH 8.0 and incubation at 75°C for 15 min. Ethanol-precipitated RNA strands were subjected to reverse transcription with IScript reverse transcription supermix (BioRad, Cat #: 1708840) per manufacturer's instructions. cDNA samples were cleaned by AMPure XP beads to recover DNA fragments 200-500 bp. Eluted fragments were subjected to second strand synthesis with dUTP followed by AMPure XP bead purification. Samples were subjected to sonication using a Diagenode Bioruptor with 15 cycles, end repair (NEB, Cat #: E6050), dATP-tailing (NEB, Cat #: M0212S), and adaptor ligation with TruSeq Single indexes (Illumina, Cat #: 20015960). Libraries then cleared of uracil-containing DNA strands using uracil N-glycosylase (Thermo Fisher Scientific, Cat #: N8080096), amplified, and purified with AMPure XP beads. After PicoGreen quantification and quality control by Agilent TapeStation, libraries were pooled for an equimolar run on a NovaSeq 6000 in a PE100 run, using the NovaSeq 6000 S4 Reagent Kit (200 Cycles) (Illumina). The loading concentration was adjusted to 0.5nM, and a 1% spike-in of PhiX was added to the run for quality control purposes. The run yielded on average 10⁸ reads per sample.

Epigenome analysis

ATAC-seq, MNase-seq, ChIP-seq, and CUT&RUN sequencing reads were trimmed and filtered for quality and adapter content using version 0.4.5 of TrimGalore (https://www.bioinformatics.babraham.ac.uk/projects/trim_galore), with a quality setting of 15 and running cutadapt (v1.15) and FastQC (v0.11.5). Reads were aligned to mouse assembly mm10 with version 2.3.4.1 of bowtie2 (<http://bowtie-bio.sourceforge.net/bowtie2/index.shtml>) and were deduplicated using MarkDuplicates in version 2.16.0 of Picard Tools. To ascertain regions of chromatin accessibility, MACS2 (<https://github.com/taoliu/MACS>) was used with a p-value setting of 0.001. ChIP and CUT&RUN target enrichment also used MACS2 with the same parameters and scored against matched input or IgG as the control, respectively. The BEDTools suite v2.29.2 (<http://bedtools.readthedocs.io>) was used to create normalized read density profiles. A global peak atlas was created by first removing blacklisted regions (<http://mitra.stanford.edu/kundaje/akundaje/release/blacklists/mm10-mouse/mm10.blacklist.bed.gz>) then merging all peaks within 500 bp for ChIP and CUT&RUN or taking 500 bp windows around peak summits for ATAC and counting reads with featureCounts (v1.6.1). DESeq2 was used to normalize read density (median ratio method) and to calculate differential enrichment for all pairwise contrasts. Peak-gene associations were created by assigning all intragenic peaks to that gene, and otherwise using linear genomic distance to transcription start site. Peak intersections were calculated using bedtools and intersectBed with 1 bp overlap. Gene set enrichment analysis (GSEA, <http://software.broadinstitute.org/gsea>) was performed with the pre-ranked option and default parameters, where each gene was assigned the single peak with the largest (in magnitude) log₂ fold change associated with it. Motif signatures were obtained using Homer v4.5 (<http://homer.ucsd.edu>). Profile and tornado plots were created using deepTools v3.3.0 by running computeMatrix and plotHeatmap on normalized bigwigs with average signal sampled in 25 bp windows and flanking region defined by the surrounding 3 kb. DRIPc data was processed as described.⁴⁹ Briefly, samtools (v1.8) and awk were used to separate positive and negative strand data into bedGraph files using the BEDTools script bedtoolsgenomecov and scaled to the number of mapped reads in each sample.

DNA affinity precipitation

DNA affinity precipitation was used to test the DNA binding activity of RREB1. 393T3 cells were transduced with a vector encoding HA-tagged RREB1(1-1291) or HA-tagged RREB1(1-346). Cell samples were collected and lysed in buffer containing 0.5% Nonidet P-40, 100 mM EDTA, and 100 mM Tris-HCl, pH 8.0. Cell lysates were incubated with biotin-labeled probes and poly(dI-dC) at 4°C overnight on a rotator. Lysates were then incubated with streptavidin at 4°C for 1-2 h on a rotator. Streptavidin beads were then washed three times with lysis buffer and eluted in NuPAGE LDS (lithium dodecyl sulfate) sample buffer (LDS at pH 8.5 with SERVA Blue G250 and phenol red) (Thermo Fisher Scientific, Cat #: NP0007) supplemented with NuPAGE Sample Reducing Agent

(500 mM dithiothreitol) (Thermo Fisher Scientific, Cat #: NP0009). Eluted proteins were subjected to western immunoblot analysis. Synthetic biotin-labeled dsDNA oligonucleotide probes used for DNA precipitation assays are listed in [Table S4](#).

Immunoblotting and Immunoprecipitation

Cells were washed twice with PBS and lysed using RIPA buffer (EMD Millipore, Cat #: 20-188) supplemented with protease inhibitor cocktail and phosphatase inhibitor cocktail. Protein concentration was measured using the BCA system (Pierce). Cell lysates were then mixed with NuPAGE LDS sample buffer supplemented with NuPAGE Sample Reducing Agent and heated at 95°C for 10 min. Equal amounts of protein were separated by NuPAGE Novex 4-12% Bis-Tris gels (Thermo Scientific) using 1x MOPS SDS running buffer (Thermo Scientific, Cat #: NP0001), and transferred to nitrocellulose membranes after electrophoresis. Membranes were blocked with 1:1 Odyssey blocking buffer (Tris-buffered saline) (LICOR Biosciences, Cat #: 927-60001) and PBS + 0.1% Tween (PBST) for 1 h, incubated overnight with primary antibodies, washed 4x in PBST, incubated 1 h with secondary antibody (LI-COR Biosciences) in Odyssey Blocking Buffer, and washed 4x in PBST. Signal is detected with the 680 and 800 channels of the Odyssey CLx imager.

Anti-HA-tag antibody (Cell Signaling Technology, Cat #: 3724S, 1:2,000), anti-SMAD2/3 antibody (Cell Signaling Technology, Cat #: 8685S, 1:2,000), anti-SMAD4 antibody (Cell Signaling Technology, Cat #: 46535S, 1:1,000), anti-DHX9 antibody (Abcam, Cat #: ab26271, 1:2,000), anti-INO80 antibody (Proteintech, Cat #: 18810-1-AP, 1:1,000), anti-CBP antibody (Cell Signaling Technology, Cat #: 7389S, 1:1,000), anti-Flag antibody (Sigma Aldrich, Cat #: F1804, 1:2,000), anti-RREB1 antibody (Santa Cruz Biotechnology, Cat #: sc-515901, 1:300), anti-FRA1 antibody (Santa Cruz Biotechnology, Cat #: sc-28310, 1:500), anti-JUNB antibody (Cell Signaling Technology, Cat #: 3753S, 1:1,000), anti-RBBP5 antibody (Bethyl Laboratories, Cat #: A300-109A, 1:1,000), anti-KDM6B antibody (Cell Signaling Technology, Cat #: 3457S, 1:1,000), anti-HMGA2 antibody (Cell Signaling Technology, Cat #: 5269S, 1:1,000), anti-TGFBR2 antibody (Proteintech, Cat #: 66636-1-Ig, 1:1,000), anti-SNAIL antibody (Cell Signaling Technology, Cat #: 3879S, 1:1,000), anti-ZEB1 antibody (Proteintech, Cat #: 21544-1-AP, 1:500), anti-HAS2 antibody (Thermo Scientific, Cat #: PA5-25593, 1:500), anti-PDGFB antibody (Novus Biologicals, Cat #: NBP1-58279, 1:500), anti-IL11 antibody (Thermo Scientific, Cat #: PA5-95982, 1:500), anti-RuvBL1 antibody (Proteintech, Cat #: 10210-2-AP, 1:1,000), anti-RuvBL2 antibody (Cell Signaling Technology, Cat #: 8959S, 1:1,000), anti-ACR8 antibody (Proteintech, Cat #: 17334-1-AP, 1:1,000), anti-ACTL6A antibody (Proteintech, Cat #: 10341-1-AP, 1:1,000), anti-GAPDH antibody (Cell Signaling Technology, Cat #: 2118S, 1:3,000), and anti- α -Tubulin antibody (Sigma Aldrich, Cat #: T6074, 1:10,000).

For immunoprecipitation (IP), cells were wash with ice cold PBS twice and incubated with IP buffer (25 mM Tris-HCl pH 7.5, 150 mM NaCl, 1 mM EDTA, 1% Nonidet P-40, 5% Glycerol) for 20 min on ice. Cell lysates were collected and immunoprecipitated at 4°C overnight on rotator with designated antibodies. Dynabeads™ Protein G were blocked with 2.5 mg/mL UltraPure™ BSA (Thermo Fisher Scientific) at 4°C overnight. Dynabeads were washed three times with IP buffer and incubated with the immunoprecipitation samples at 4°C for 1-2 h. The beads were then washed 5 times with IP buffer, and protein was eluted from beads in 2X LDS sample buffer at 70°C under shaking for 10 min. Protein samples were separated by using a magnetic rack and subjected to immunoblotting.

Anti-HA-tag antibody (Cell Signaling Technology, Cat #: 3724S), anti-SMAD2/3 antibody (Cell Signaling Technology, Cat #: 8685S), anti-DHX9 antibody (Proteintech, Cat #: 17721-1-AP), anti-INO80 antibody (Proteintech, Cat #: 18810-1-AP), anti-anti-CBP antibody (Cell Signaling Technology, Cat #: 7389S), and anti-Flag antibody (Sigma Aldrich, Cat #: F1804).

Immunohistochemistry and Immunofluorescence

Tissue blocks were prepared from tissues fixed in 4% paraformaldehyde overnight and dehydrated in 70% ethanol. Paraffin embedded sections were deparaffinized using Histo-Clear (National Diagnostics, Cat #: HS-200) followed by rehydration using 100-70% ethanol. Endogenous peroxidase activity was quenched with 1% H₂O₂ at RT for 10 min. Antigen retrieval was performed in a steamer for 30 min in prewarmed citrate-based (Vector Laboratories, Cat #: H-3300) retrieval solution. For enzymatic detection, tissue sections were placed in 2.5% normal horse serum (Vector Laboratories, Cat #: S-2012) and 1% BSA in Tris-buffered saline (TBS) solution, followed by and overnight incubation with the designated primary antibody in blocking solution. ImmPRESS HRP Anti-Rabbit Ig (Vector Laboratories, Cat #: MP-7401) and ImmPACT DAB Peroxidase (Vector Laboratories, Cat #: SK-4105) were used for detection. For fluorescent detection, tissue sections were placed in 10% normal goat serum (Life Technologies, Cat #: 50062Z) or 10% normal donkey serum (Sigma Aldrich, Cat #: D9663-10ML) with 1% BSA in TBS. After incubation with primary antibodies diluted in blocking solution, sections were incubated in fluorophore conjugated secondary antibodies for 2 h, followed by three washes and staining with nuclear dye Hoechst 33342 (Thermo Fisher Scientific, Cat #: H3570). Detection was followed by dehydration of tissue in 70-100% ethanol and HistoClear, followed by mounting with Vectashield mounting medium (enzymatic detection) (Vector Laboratories, Cat #: H-1000-10) or ProLong diamond antifade mountant (fluorescent detection) (Life Technology, Cat #: P36970).

For immunofluorescence from culture cells, cells were plated on glass coverslips were fixed for 10 min at room temperature in 2% paraformaldehyde (393T3 cells) or for 20 min at room temperature in 4% paraformaldehyde (tumoroids). For permeabilization, samples were treated with PBS with 0.1% Triton-X 100 (PBSTr) for 30 min (393T3 cells) or with PBS with 0.05% Triton-X 100 for 10 min (393T3 cells), followed by a 1 h block in PBSTr supplemented with 2.5% normal goat serum and 1% BSA. Samples were incubated with primary antibodies in blocking solution, followed by three washes with PBSTr. After 1 h of incubation with secondary antibody,

samples washed three times with PBSTr, incubated in Hoechst solution (Thermo Fisher Scientific) for 5 min, and mounted with ProLong diamond antifade mountant (Life Technology).

Flow cytometry

Cells were prepared in 2% heat-inactivated FBS in PBS. DAPI (final concentration 1 $\mu\text{g/ml}$) was added to each sample to identify dead cells and cells were analyzed using a BD LSR Fortessa II Analyzer at the Flow Cytometry Core Facility at MSKCC. Doublets were excluded and DAPI⁻/GFP⁺ cells were evaluated for mCherry⁺ signal.

RNA fluorescence in situ hybridization

4 μm FFPE mouse tissue sections were mounted on Superfrost Plus microscope slides (Fisher). Samples were loaded into Leica Bond RX, baked at 60°C for 30 min, dewaxed with Bond Dewax Solution (Leica, Cat #: AR9222), and pretreated with EDTA-based epitope retrieval ER2 solution (Leica, Cat #: AR9640) at 95°C for 15 min (no proteolytic retrieval). The sections were incubated with probes mPdgfb-C1 (Advanced Cell Diagnostics Bio, Cat #: 424658), mHas2-C2 (Advanced Cell Diagnostics Bio, Cat #: 465178-C2), and mI11-C3 (Advanced Cell Diagnostics Bio, Cat #: 552468-C3) at 42°C for 2 h. The hybridized probes were detected using RNA-scope LS Multiplex Reagent Kit (Advanced Cell Diagnostics Bio, Cat #: 322800) according to manufacturer's instructions. Slides were then incubated with Alexa Fluor 488 tyramide (Life Technologies, Cat #: B40953), CF 594 Tyramid (Biotium, Cat #: 92174), and Alexa Fluor 647 (Life Technologies, Cat #: B40958) for 20 min at room temperature. Slides were washed in PBS and incubated in 5 $\mu\text{g/mL}$ 4',6-diamidino-2-phenylindole (DAPI) in PBS for 5 min, rinsed in PBS, and mounted in Calbiochem Mowiol 4–88 (Fisher Scientific, Cat #: 47-590-4100GM).

In situ proximity ligation assay

In situ PLA was conducted employing the Duolink *In Situ* Reagents from Olink Bioscience (Sigma Aldrich, DUO92008-100RXN), following manufacturer's instruction. Cell cultures were plated on glass coverslips and fixed for 10 min at room temperature in 3% paraformaldehyde solution. Following fixation, cells were blocked with Duolink blocking solution and incubated overnight with two primary antibodies derived from distinct host species (one mouse and one rabbit antibody). The samples were then incubated with Duolink *In Situ* PLA Probe Anti-mouse PLUS (Sigma Aldrich, Cat #: DUO92001-100RXN) and anti-rabbit minus PLA probes (Sigma Aldrich, Cat #: DUO92005-100RXN). Probes included unique DNA strands to facilitate the hybridization of introduced oligonucleotides. After ligation and amplification, coverslips were mounted with ProLong diamond antifade mountant (Life Technology).

Protein expression and purification

RREB1 ZF1-5 (human, Uniprot, Q92766, aa 60-264 aa) was cloned in pOPINF using a synthesized DNA template with optimized codons for bacterial expression (Thermo Fisher Scientific). The sequence was confirmed by DNA sequencing (GATC Biotech). The protein was expressed in *E. coli* B834(DE3) strain (Merck KGaA, Cat #: 69041), fused to an N-terminal His-tag with the 3C peptidase cleavage site. Cells were grown at 37 °C in Terrific Broth (CondaLab, Cat #:1246) and induced with IPTG (NZYTech, Lda, Cat #: MB02603, 0.5 mM) at an OD₆₀₀ of 0.8 for three hours. Bacterial cultures were centrifuged and cells were lysed at 4 °C (EmulsiFlex-C5, Avestin) in 50 mM Tris (Cymit Quimica, SL, Cat #: AN-AG00HEKS), 400 mM NaCl (Merck KGaA, Cat #: S9625), 40 mM imidazole (Merck KGaA, Cat #: A1073.1000), 1 mM TCEP (ChemoSapiens S. L., Cat #: AB121644) and Tween 20 (Panreac Quimica, S.L.U., Cat #: A4974.0500) 0.2% V/V pH 8 at 25 °C in the presence of lysozyme (Merck KGaA, Cat #: L6876) and DNase I (Merck KGaA, Cat #: 11284932001). Supernatants containing the soluble proteins were purified using HiTrap HP 5 mL column (Cytiva) and eluted by an imidazole gradient, using a NGC™ Quest 10 Plus Chromatography System (Bio-Rad). Fractions containing the protein of interest were pooled, and cleaved with 3C protease overnight at 4 °C. Finally, the protein was purified by size exclusion chromatography on a HiLoad™ 16/600 Superdex™ 75 pg (in 20 mM Tris, 100 mM NaCl and 2mM TCEP at pH 7.5 at 25 °C and kept at –80 °C.

Electrophoretic mobility shift assay (EMSA)

Each pair of EMSA probes was prepared by annealing the Cy5-DNA strand with its complementary non-labeled strand (CondaLab). DNAs were mixed at equimolar concentrations (3 mM) in 20 mM Tris (Cymit Quimica, SL, Cat #: AN-AG00HEKS) pH 7.0 at 25 °C and 100 mM NaCl (Merck KGaA, Cat #: S9625), heated at 90 °C for 3 min and cooled down to room temperature for 2 h.

Protein and DNA were incubated for 30 min at 4 °C in 10 μL of binding buffer (100 mM Tris, 10% glycerol, Merck KGaA, Cat #: 49767I) using a fixed concentration of the duplex DNAs (7.5 nM) and increasing amounts of the protein. Electrophoresis was performed in native 6% polyacrylamide gels (1.5 mm thick), prepared with 40% Acrylamide/Bis Solution, 19:1, (Bio-Rad, Cat #: 1610144). The gels were run for 30 min in TG buffer at 150 V at 4 °C and exposed to a Typhoon imager (GE Healthcare).

SMAD4 MH2 domain M1 model generation

The M1 model was generated using ColabFold v1.5.5⁷⁴ and the MH2 domain (starting at residue 272 and modifications as described in the text). The initial loop (272 to 320) is maintained as in the MH2 WT structure (Protein ensemble Database (PED): PED00194, model1).^{53,75} UCSF Chimera was used to prepare figures.⁷⁶

QUANTIFICATION AND STATISTICAL ANALYSIS

Images were analyzed using Fiji software (NIH). Statistical analyses and sample sizes are given in each figure or figure legend. Statistical tests were performed in the GraphPad Prism software. *P* values were computed by two-tailed Mann-Whitney U test and t-test. **P* < 0.05; ***P* < 0.01; ****P* < 0.001; *****P* < 0.0001.

Supplemental figures

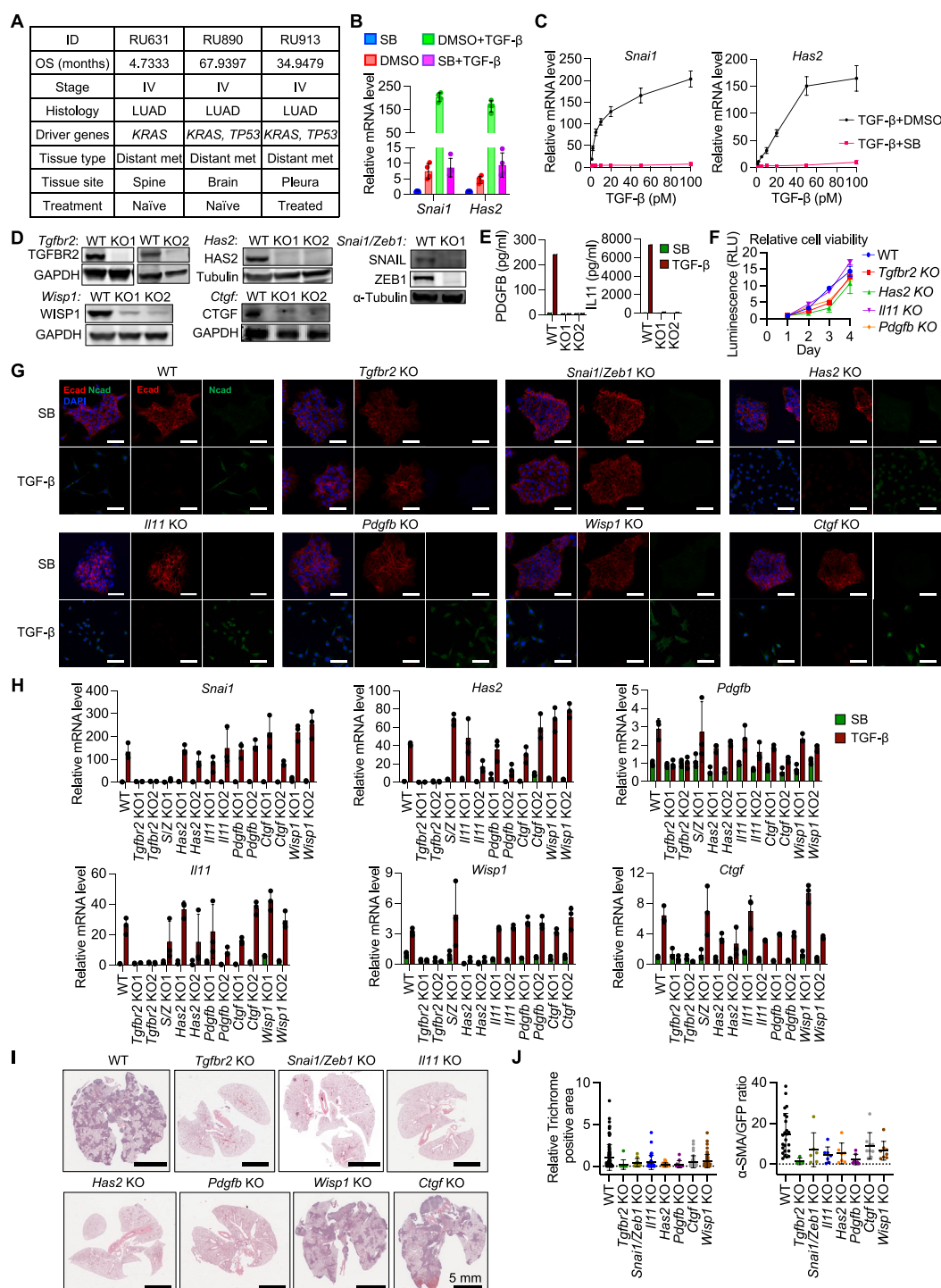


Figure S1. The EMT and fibrogenic branches TGF- β response are required for metastasis, related to Figure 1

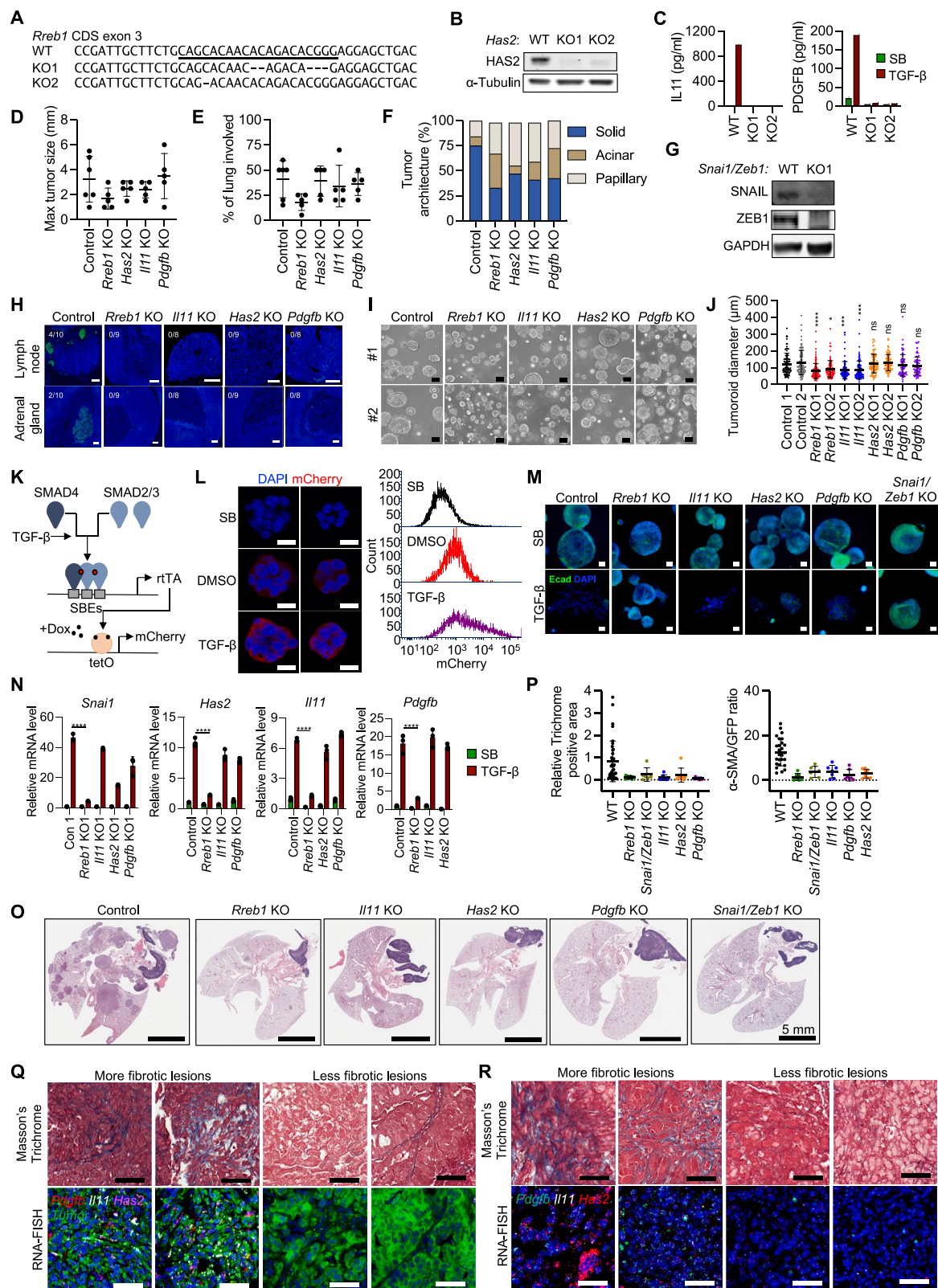
(A) Clinical annotation of the LUAD patient-derived xenografts used to generate tumoroids.

(B) *Snai1* and *Has2* mRNA levels in 393T3 cells incubated with DMSO (vehicle), SB, TGF- β + DMSO, or TGF- β + SB for 1.5 h. Data are mean $\pm$ SD; $n = 4$.

(C) *Snai1* and *Has2* mRNA levels in 393T3 cells incubated with varying concentrations of TGF- β , plus DMSO or SB for 1.5 h. Data are mean $\pm$ SD; $n = 4$.

(legend continued on next page)

- (D) Lysates from WT, *Tgfb2*^{-/-}, *Has2*^{-/-}, *Ctgf*^{-/-}, *Wisp1*^{-/-}, or *Snai1/Zeb1*-KO 393T3 cells were subjected to western immunoblotting using the antibodies indicated on the left of each panel.
- (E) PDGFB (left) and IL11 (right) quantification in cell culture supernatants from WT, *Pdgfb*-KO (left), and *Il11*-KO (right) 393T3 cells incubated with SB or TGF- β for 24 h.
- (F) Growth kinetics of WT, *Tgfb2*^{-/-}, *Il11*^{-/-}, *Has2*^{-/-}, and *Pdgfb*-KO cells as determined by CellTiter-Glo luminescent cell viability assay. *RLU*, relative light unit.
- (G) Representative E-cadherin (Ecad) and N-cadherin (Ncad) IF and DAPI staining of WT, *Tgfb2*^{-/-}, *Snai1/Zeb1*^{-/-}, *Has2*^{-/-}, *Pdgfb*^{-/-}, *Il11*^{-/-}, *Wisp1*^{-/-}, and *Ctgf*-KO 393T3 cells incubated with SB or TGF- β for 48 h. Scale bars, 50 μ m.
- (H) *Snai1*, *Has2*, *Il11*, *Pdgfb*, *Wisp1*, and *Ctgf* mRNA levels in the same series of 393T3 derivatives incubated with SB or TGF- β for 1.5 h. Data are mean $\pm$ SD; *n* = 3. *S/Z*, *Snai1/Zeb1*.
- (I) Representative H&E staining of lungs from mice 21 days after tail-vein inoculation of WT, *Tgfb2*^{-/-}, *Il11*^{-/-}, *Has2*^{-/-}, *Pdgfb*^{-/-}, *Ctgf*^{-/-}, *Wisp1*^{-/-}, and *Snai1/Zeb1*-KO 393T3 cells. Scale bars, 5 mm.
- (J) Quantification of trichrome staining (left) and α -SMA staining (right) in lung metastatic colonies from the experiments in [Figure 1D](#).



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Figure S2. Fibrogenic factors drive pulmonary metastasis in a LUAD GEMM, related to Figure 1

- (A) sgRNA targeting *Rreb1* CDS exon 3 and mutant *Rreb1* genomic sequences in cells derived from two independent tumors.
- (B) Tumoroids derived from control and *Has2*-KO KP primary lung tumors were subjected to western immunoblotting with the antibodies indicated on the left.
- (C) IL11 (left) and PDGFB (right) quantification in the cell culture supernatants from control, *Il11*-KO (left), and *Pdgfb*-KO (right) KP tumoroids treated with SB or TGF- β for 48 h.
- (D and E) Maximal size of primary lung tumors (D) and proportion of lung section areas occupied by tumor (E) in control, *Rreb1*-, *Has2*-, *Il11*-, or *Pdgfb*-KO KP GEMMs.
- (F) Tissue histology of primary tumors from control, *Rreb1*-, *Has2*-, *Il11*-, or *Pdgfb*-KO KP GEMMs.
- (G) Lysates from WT or *Snai1/Zeb1*-KO KP tumoroids generated by CRISPR engineering were subjected to western immunoblotting with the antibodies indicated on the left.
- (H) Representative images of mediastinal lymph node and adrenal gland sections from control, *Rreb1*-, *Il11*-, *Has2*-, or *Pdgfb*-KO mice subjected to GFP IF to detect metastatic lesions. Scale bars, 100 μ m.
- (I and J) Representative phase contrast images of KP tumoroids cultured in Matrigel for 6 days (I) and diameter of the tumoroids (J). Scale bars, 100 μ m. Data are mean $\pm$ SD; one-way ANOVA; ns: not significant, * $p < 0.05$, *** $p < 0.001$, **** $p < 0.0001$.
- (K) Schematic representation of the TGF- β and SMAD-dependent transcriptional reporter. *SBE*, SMAD-binding element; *TRE*, tetracycline-responsive promoter element; *Dox*, doxycycline.
- (L) KP LUAD tumoroids transduced with the TGF- β /SMAD reporter were grown in Matrigel for 2 days and incubated with 2.5 μ M SB, vehicle (DMSO), or 100 pM TGF- β for 6 h. Images of reporter mCherry expression and quantification are shown. Scale bars, 20 μ m.
- (M) Representative E-cadherin IF and DAPI staining of control, *Rreb1*-, *Has2*-, *Il11*-, and *Pdgfb*-KO KP tumoroids grown in Matrigel for 4 days and incubated with SB or TGF- β for 48 h. Scale bars, 100 μ m.
- (N) *Snai1*, *Has2*, *Il11*, and *Pdgfb* mRNA levels in control, *Rreb1*-, *Il11*-, *Has2*-, and *Pdgfb*-KO KP tumoroids grown in Matrigel for 4 days and incubated with SB or TGF- β for 4 h. Data are mean $\pm$ SD; $n = 3$; two-way ANOVA; **** $p < 0.0001$.
- (O) Representative H&E staining of lung tissue sections from mice 56 days after tail-vein inoculation of control, *Rreb1*-, *Il11*-, *Has2*-, *Pdgfb*-, or *Snai1/Zeb1*-KO KP tumoroids. Scale bars, 5 mm.
- (P) Quantification of trichrome staining (left) and α -SMA staining (right) in lung metastatic colonies from the experiments in Figure 1H.
- (Q and R) Representative Masson's trichrome staining and RNA-FISH of mouse lung tissue sections after tail-vein inoculation of KP tumoroids (Q) or 393T3 cells (R). Scale bars, 50 μ m.

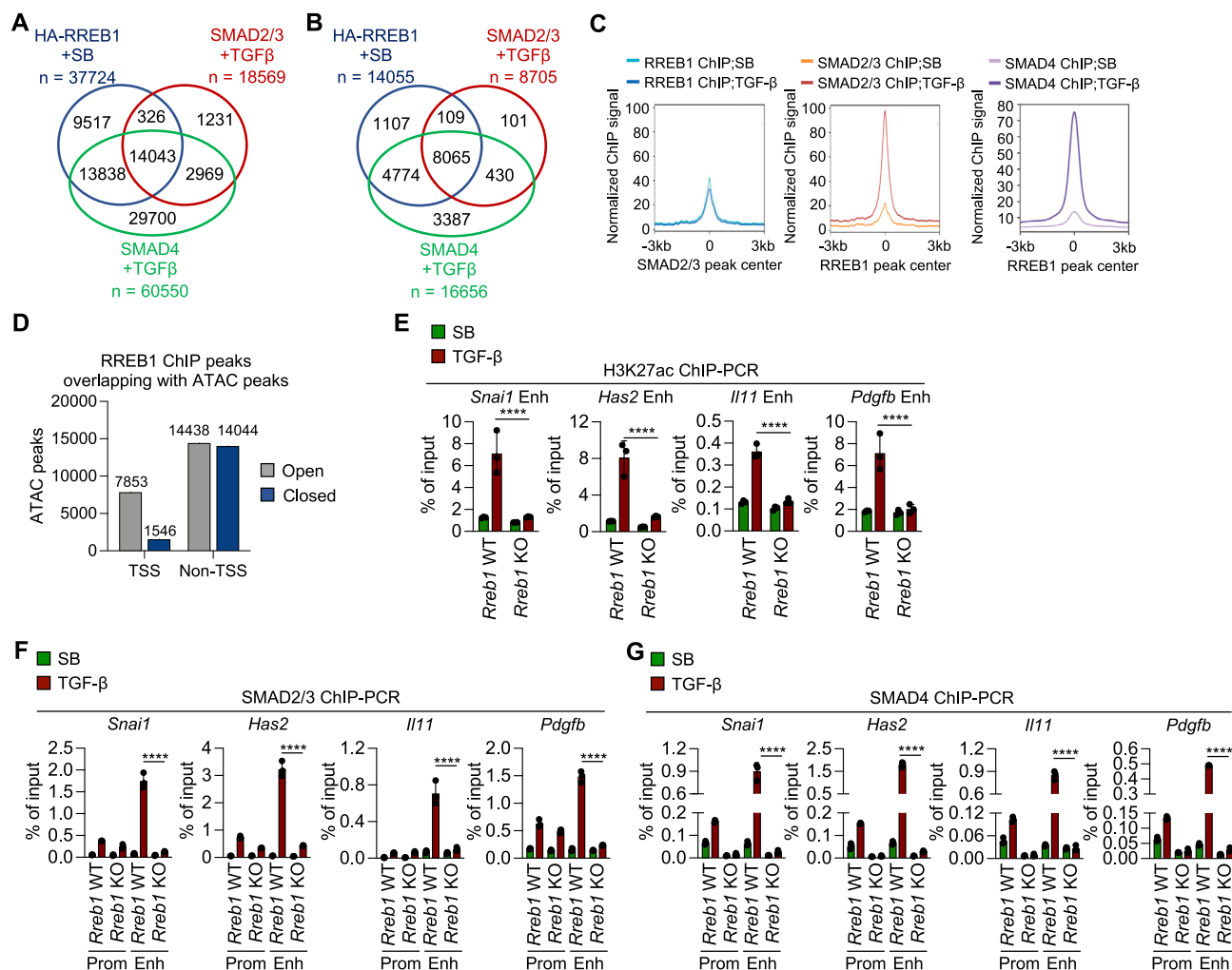


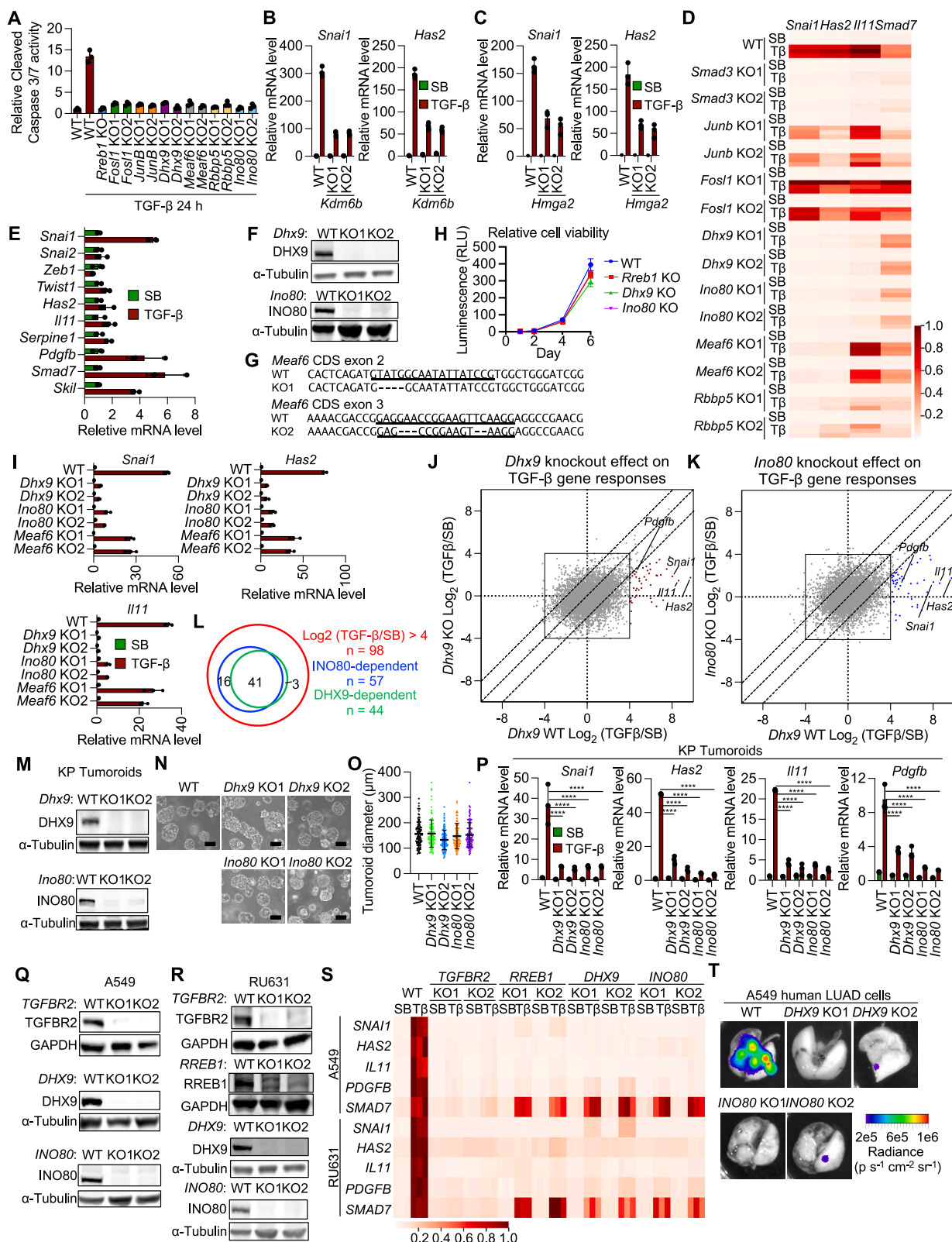
Figure S3. RREB1 enables TGF-β-induced binding of SMADs to primed enhancers, related to Figure 2

(A and B) Venn diagram showing the overlap between HA-RREB1, SMAD2/3, and SMAD4 genome-wide peaks (A) and the annotated genes in these peaks (B) based on ChIP-seq analysis of 393T3 cells expressing HA-RREB1 incubated with SB or TGF-β for 1.5 h.

(C) Metaplots of HA-RREB1 (left), SMAD2/3 (middle), and SMAD4 (right) ChIP-seq peak summits relative to summits of SMAD2/3 in TGF-β-treated cells (left) or HA-RREB1 in SB-treated cells (middle and right) based on analysis of Figure 2E data.

(D) Composite plot analysis representing HA-RREB1 ChIP-seq peaks (based on data in Figures 2D and 2E) and their overlap with ATAC-seq peaks in SB-treated 393T3 cells. TSS, transcription start site.

(E–G) ChIP-PCR analysis of H3K27ac (E), SMAD2/3 (F), and SMAD4 (G) on the primed enhancers of the indicated genes in WT or *Rreb1*-KO 393T3 cells treated with SB or TGF-β for 1.5 h. Data are mean ± SD; n = 3; two-way ANOVA; ****p < 0.0001. Enh, enhancer; Prom, promoter.



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Figure S4. Identification of essential RREB1 cofactors, related to Figure 3

(A) WT, *Rreb1*^{-/-}, *Fosl1*^{-/-}, *JunB*^{-/-}, *Dhx9*^{-/-}, *Meaf6*^{-/-}, *Rbbp5*^{-/-}, and *Ino80*-KO SMAD4-restored PDAC cells were incubated with the AKT inhibitor MK220612 (2.5 μ M) and SB505124 (2.5 μ M) or TGF- β (100 pM) for 24 h to trigger EMT-dependent apoptosis.¹⁰ Cleaved caspase 3/7 activity was then determined. Data are mean $\pm$ SD. *n* = 3.

(B and C) *Snai1* and *Has2* mRNA levels in SMAD4-restored PDAC cells expressing shRNAs against *Kdm6b* (B) or *Hmga2* (C) and incubated with SB or TGF- β for 1.5 h. Data are mean $\pm$ SD. *n* = 3.

(D) Heat map representation of *Snai1*, *Has2*, *Il11*, and *Smad7* mRNA levels in SMAD4-restored PDAC cells treated with SB or TGF- β (T β) for 1.5 h. Data are mean $\pm$ SD. *n* = 3.

(E) Cells were isolated from *Pdx1*-Cre; *Kras*^{LSL-G12D}; *p53*^{R172H/wt} PDAC tumors in mice and grown as tumoroids in Matrigel for 4 days. Tumoroids were incubated with SB or TGF- β for 4 h, and the mRNA levels for the indicated genes were determined. Data are mean $\pm$ SD. *n* = 4.

(F) WT, *Dhx9*^{-/-}, and *Ino80*-KO 393T3 cell lysates were subjected to western immunoblotting with the indicated antibodies.

(G) sgRNA targeting *Meaf6* CDS exon 2 and exon 3, and mutant *Meaf6* genomic sequences in cells derived from two independent tumors (bottom).

(H) Growth kinetics of WT, *Rreb1*^{-/-}, *Dhx9*^{-/-}, and *Ino80*-KO 393T3 cells as determined by CellTiter-Glo luminescent cell viability assay. *RLU*, relative light units.

(I) *Snai1*, *Has2*, and *Il11* mRNA levels in WT, *Dhx9*^{-/-}, *Ino80*^{-/-}, and *Meaf6*-KO 393T3 cells incubated with SB or TGF- β for 1.5 h. Data are mean $\pm$ SD. *n* = 3.

(J and K) TGF- β -dependent change in the mRNA levels of individual genes in WT (x axis) and *Dhx9*-KO 393T3 cells (y axis) (J) or WT (x axis) and *Dhx9*-KO (y axis) 393T3 cells (K), as determined by RNA-seq analysis from Figure 3G. Off-diagonal TGF- β -responsive genes of interest are highlighted.

(L) Venn diagram showing the overlap between DHX9- and INO80-dependent TGF- β -responsive genes ($\log_2$ -transformed fold change ≥ 4 ; false discovery rate [FDR] ≤ 0.05) based on the RNA-seq datasets from the experiments Figures S5J and S5K.

(M) WT, *Dhx9*^{-/-}, and *Ino80*-KO KP tumoroid lysates were subjected to western immunoblotting analysis with the indicated antibodies.

(N and O) Representative phase contrast microscopy images (N) and diameter (O) of KP tumoroids cultured in Matrigel for 6 days. Scale bars, 100 μ m. Data are mean $\pm$ SD.

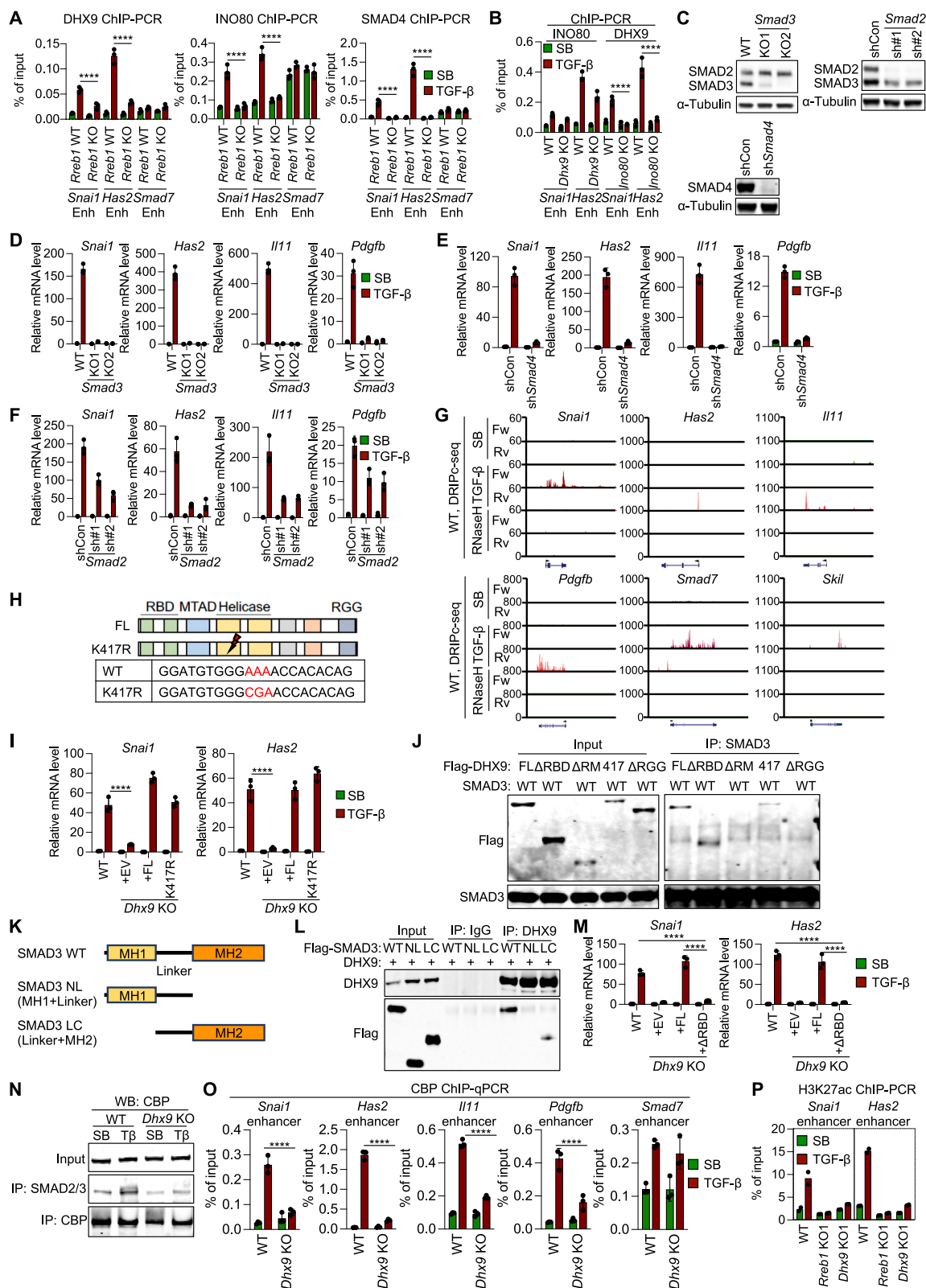
(P) *Snai1*, *Has2*, *Il11*, *Pdgfb*, and *Smad7* mRNA levels in WT, *Dhx9*^{-/-}, and *Ino80*-KO KP tumoroids grown in Matrigel for 4 days and treated with SB or TGF- β for 2 h. Data are mean $\pm$ SD; *n* = 3; two-way ANOVA; *****p* < 0.0001.

(Q) WT, *DHX9*^{-/-}, and *INO80*-KO A549 cell lysates were subjected to western immunoblotting with the indicated antibodies.

(R) WT, *TGFBR2*^{-/-}, *DHX9*^{-/-}, and *INO80*-KO RU631 PDX-derived tumoroids were subjected to western immunoblotting with the antibodies indicated on the left.

(S) Heat map representation of *SNAI1*, *HAS2*, *IL11*, *PDGFB*, and *SMAD7* mRNA levels in WT, *TGFBR2*^{-/-}, *RREB1*^{-/-}, *DHX9*^{-/-}, and *INO80*-KO A549 human cells treated with SB or TGF- β (T β) for 1.5 h and RU631 PDX-derived tumoroids grown in Matrigel for 2 days and treated with SB or TGF- β (T β) for 2 h.

(T) Representative *ex vivo* BLI images corresponding to athymic mice 63 days after tail-vein inoculation of WT, *DHX9*^{-/-}, or *INO80*-knockout A549 cells in Figure 3R.



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Figure S5. DHX9, INO80, and SMAD requirements for fibrogenic EMT gene responses, related to Figure 4

- (A) ChIP-PCR analysis of DHX9, INO80, and SMAD4 binding to *Snai1*, *Has2*, and *Smad7* enhancers in WT and *Rreb1*-KO 393T3 cells treated with SB or TGF- β for 1.5 h. Data are mean $\pm$ SD; $n = 3$; two-way ANOVA; **** $p < 0.0001$.
- (B) ChIP-PCR analysis of DHX9 and INO80 binding to *Snai1* and *Has2* enhancers in WT, *Dhx9*-, and *Ino80*-KO 393T3 cells. Data are mean $\pm$ SD; $n = 3$; two-way ANOVA; **** $p < 0.0001$.
- (C) 393T3 cells expressing control shRNA (shCon), *Smad4* shRNA (sh*Smad4*), or *Smad2* shRNA (sh*Smad2*) were subjected to western immunoblotting with the indicated antibodies.
- (D–F) *Snai1*, *Has2*, *Il11*, and *Pdgfb* mRNA levels in WT and *Smad3*-KO 393T3 cells (D), in 393T3 cells expressing control or *Smad4*-targeting shRNA (sh*Smad4*) (E), or *Smad2*-targeting shRNA (F) after treatment with SB or TGF- β for 1.5 h.
- (G) DRIPc-seq tracks in *Snai1*, *Has2*, *Il11*, *Pdgfb*, *Smad7*, and *Skil* loci in SB- or TGF- β -treated WT and *Rreb1*-KO cells. Genomic DNA extracted from WT cells treated with TGF- β was incubated with RNaseH as a negative control. *Fw*, forward strand; *Rv*, reverse strand.
- (H) Schematic of DHX9 domain structure and mutations.
- (I) *Snai1* and *Has2* mRNA levels in WT or *Dhx9*-KO 393T3 cells expressing empty vector (EV), full-length DHX9 (FL), or helicase-activity mutant (K417R) and treated with SB or TGF- β for 1.5 h. Data are mean $\pm$ SD; $n = 3$; two-way ANOVA; **** $p < 0.0001$.
- (J) HEK293T cells expressing SMAD3 and various FLAG-DHX9 mutants were lysed and immunoprecipitated with anti-SMAD3 antibodies. Immune complexes were subjected to western immunoblotting with the antibodies indicated on the left.
- (K) Schematic of SMAD3 mutants.
- (L) Lysates of HEK293T cells expressing DHX9 and various FLAG-SMAD3 mutants were immunoprecipitated (IP) with anti-DHX9 antibodies. Immune complexes were subjected to western immunoblotting with the indicated antibodies.
- (M) *Snai1* and *Has2* mRNA levels in 393T3 cells expressing empty vector (EV), WT-DHX9 (FL), or DHX9- Δ RBD (Δ RBD) and treated with SB or TGF- β for 1.5 h. Data are mean $\pm$ SD; $n = 3$; two-way ANOVA; **** $p < 0.0001$.
- (N) WT or *Dhx9*-KO 393T3 cells were incubated with SB or TGF- β (T β) for 1.5 h, lysed, and immunoprecipitated with the antibodies indicated on the left. The immune complexes were subjected to western immunoblotting with the antibodies indicated at the top.
- (O) ChIP-PCR analysis of CBP enrichment at the enhancers of *Snai1*, *Has2*, *Il11*, *Pdgfb*, and *Smad7* in WT or *Dhx9*-KO 393T3 cells. Data are mean $\pm$ SD; $n = 3$; two-way ANOVA; **** $p < 0.0001$.
- (P) ChIP-PCR analysis of H3K27ac enrichment to the enhancers of *Snai1* and *Has2* in WT, *Rreb1*-, and *Dhx9*-KO 393T3 cells treated with SB or TGF- β for 1.5 h. Data are mean; $n = 2$.

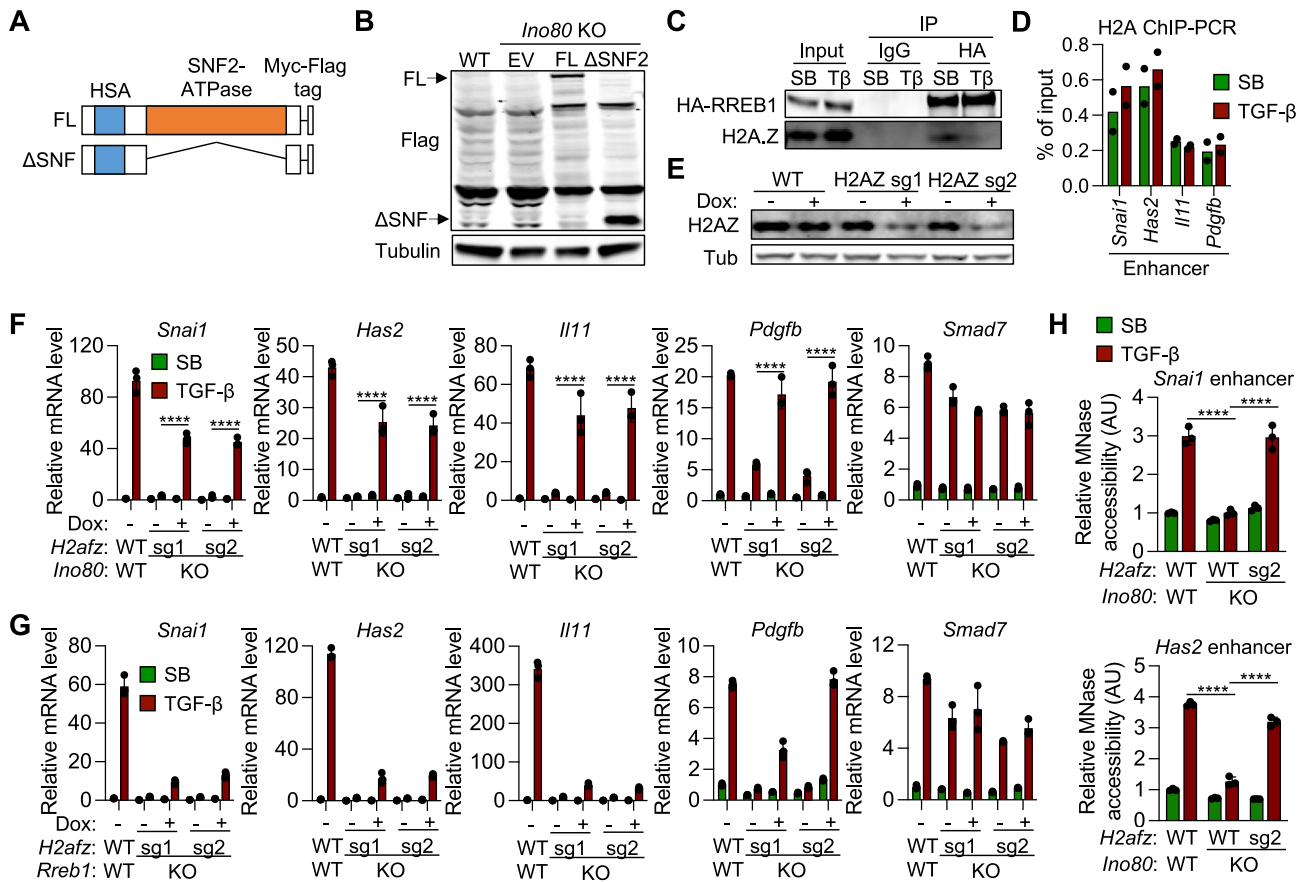


Figure S6. INO80-dependent chromatin remodeling at RREB1-primed enhancers, related to Figure 6

(A) Schematic of INO80 domain structure and mutant.

(B) WT and *Ino80*-KO 393T3 cells expressing recombinant INO80 mutants were subjected to western immunoblotting with the indicated antibodies.

(C) WT 393T3 cells expressing HA-RREB1 were treated with SB or TGF- β , lysed, and immunoprecipitated with anti-HA antibodies. Immune complexes were subjected to western immunoblotting with the indicated antibodies.

(D) ChIP-PCR analysis of H2A enrichment to the enhancers of *Snai1*, *Has2*, *Il11*, and *Pdgfb* in 393T3 cells treated with SB or TGF- β for 1.5 h. Data are mean; $n = 2$.

(E) WT and inducible *H2afz*-KO 393T3 cells were treated with doxycycline for 24 h and subjected to western immunoblotting with the indicated antibodies.

(F and G) *Ino80*-KO (F) and *Rreb1*-KO (G) 393T3 cells were transduced with two independent sgRNAs against *H2afz* for CRISPR interference and further engineered to express doxycycline (Dox)-inducible Cas9. mRNA levels of *Snai1*, *Has2*, *Il11*, *Pdgfb*, and *Smad7* were analyzed after incubation of the cells with SB or TGF- β for 1.5 h. Data are mean $\pm$ SD; $n = 3$; two-way ANOVA; **** $p < 0.0001$.

(H) Relative MNase accessibility determined by qPCR using MNase-digested DNA extracted from 393T3 cells from Figure 6G. Data are mean $\pm$ SD; $n = 3$; one-way ANOVA; **** $p < 0.0001$.

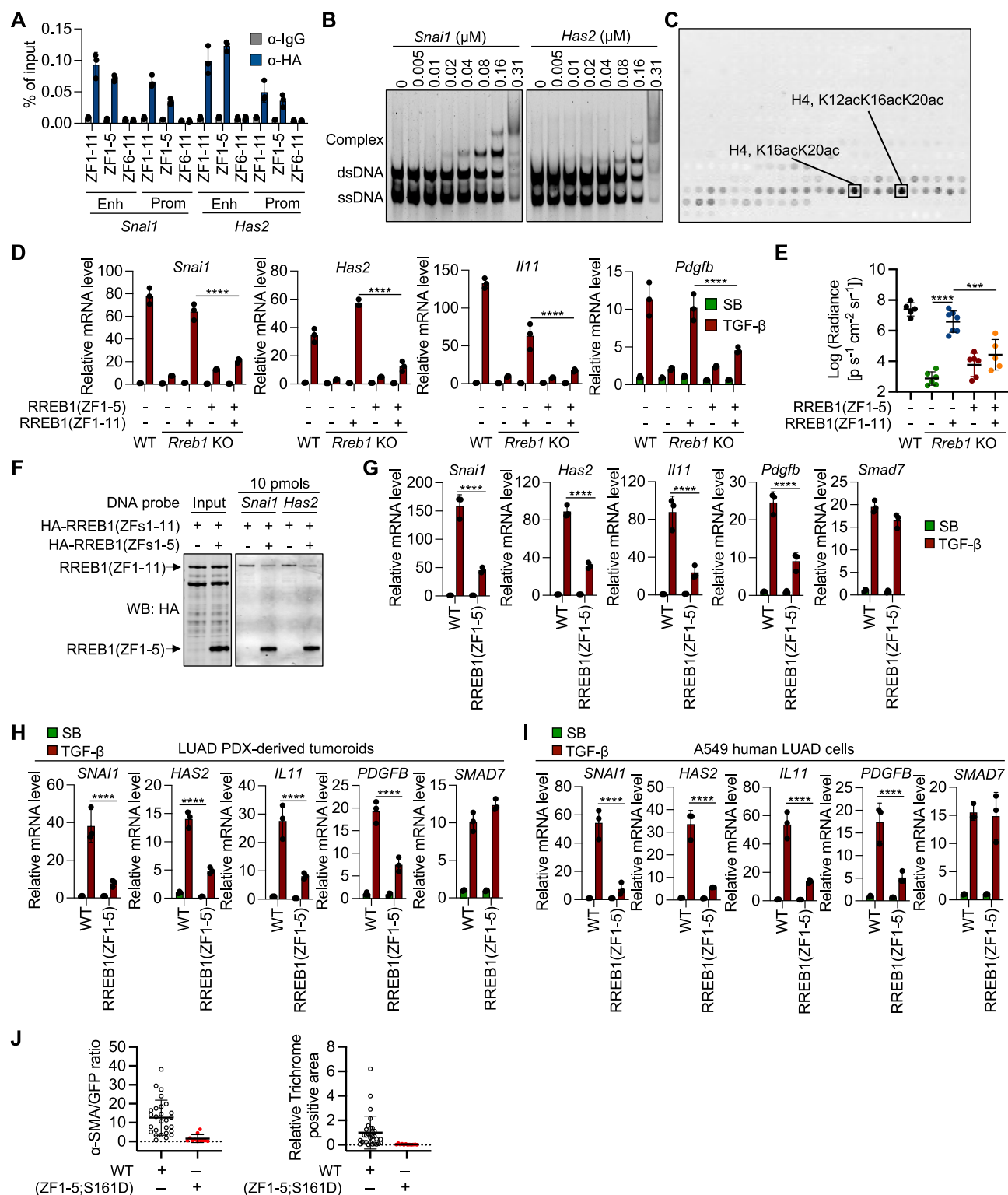


Figure S7. Effect of an RREB1 inhibitor on fibrogenic EMT gene responses and metastasis, related to Figure 7

(A) ChIP-PCR analysis of HA-RREB1(ZF1-11) and HA-RREB1(ZF1-5) binding to the indicated regulatory loci of *Snai1* and *Has2* in *Rreb1*-KO 393T3 cells. *Enh*, enhancer; *Prom*, promoter.

(B) Titration of *Snai1* and *Has2* RREs probes (Table S4) with recombinant RREB1(ZF1-5) followed by EMSA.

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(C) Binding of recombinant RREB1(ZF1-5) to a histone peptide array ([Table S3](#)).

(D) WT and *Rreb1*-KO 393T3 cells expressing HA-RREB1(ZF1-11) HA-RREB1(ZF1-5) were treated with SB or TGF- β for 1.5 h. mRNA levels of *Snai1*, *Has2*, *Il11*, *Pdgfb*, and *Smad7* were then determined. Data are mean $\pm$ SD; $n = 3$; two-way ANOVA; **** $p < 0.0001$.

(E) Lung metastasis burden in mice 21 days after inoculation of WT and *Rreb1*-KO 393T3 cells expressing HA-RREB1(ZF1-11) or HA-RREB1(ZF1-5), as quantified by firefly luciferase BLI. Data are mean $\pm$ SD; $n = 5$ –6 mice per group. Two-way ANOVA; *** $p < 0.001$; **** $p < 0.0001$.

(F) *Rreb1*-KO 393T3 cells were transduced with a HA-RREB1(ZF1-11)-vector alone or together with a HA-RREB1(ZF1-5) vector. HA-RREB1 binding to dsDNA *Snai1* or *Has2* promoter oligonucleotide probes was determined by DNA affinity precipitation assays.

(G–I) *Snai1*, *Has2*, *Il11*, *Pdgfb*, and *Smad7* mRNA levels in WT or HA-RREB1(ZF1-5)-expressing 393T3 mouse LUAD cells (G), human LUAD PDX-derived tumoroids (H), and A549 human LUAD cells (I). 393T3 and A549 cell monolayers were treated with SB or TGF- β for 1.5 h (G and I). LUAD PDX-derived tumoroids were grown in Matrigel for 2 days and treated with SB or TGF- β for 2 h after culture (H). Data are mean $\pm$ SD; $n = 3$; two-way ANOVA; **** $p < 0.0001$.

(J) Quantification of trichrome staining (left) and α -SMA staining (right) in lung metastatic colonies from the experiments in [Figure 7H](#).